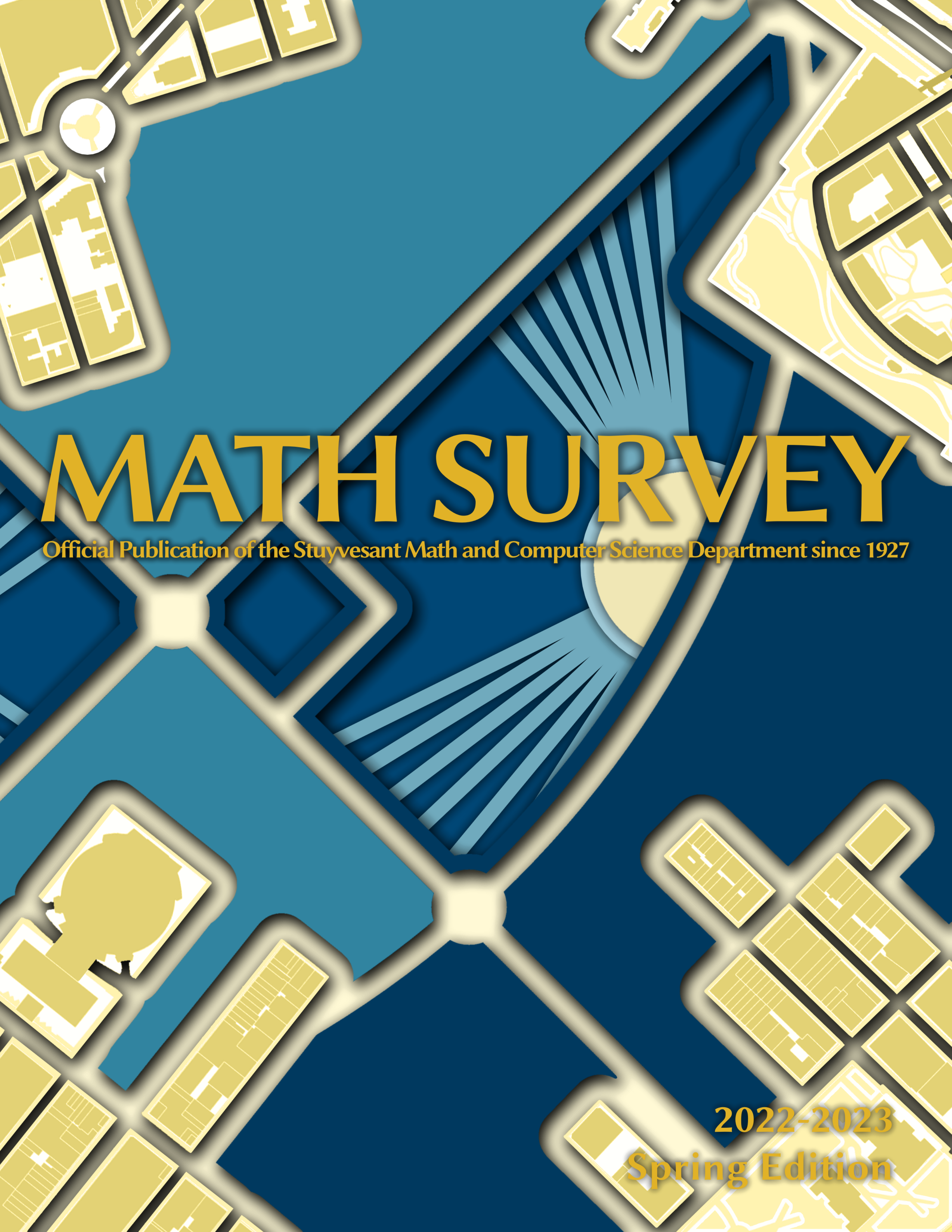

MATH SURVEY

Official Publication of the Stuyvesant Math and Computer Science Department since 1927

2022-2023

Spring Edition

Contents

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Stuyvesant Math Survey

2022-2023 Winter Edition

The Official Mathematics and Computer Science Publication
of the Stuyvesant High School Math Department since 1927

Special thanks to Milan Haiman (Class of 2019), Mr. Gary Rubinstein, Mr. Stan Kats, and Mr. Thomas Lin for their support for this edition.

Visit https://stuymathsurvey.com for the online version.
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Editor's Note

We are thrilled to share with you the *2022-2023 Spring Edition of Math Survey*, marking our second edition of the year. Starting with this edition, we are joining forces with the *Math and CS Research*, enhancing and expanding the publication endeavors of Stuyvesant's Math and Computer Science department.

First and foremost, I would like to thank Mr. Rubinstein for his invaluable guidance and for contributing a thoughtful foreword. I would also like to extend my gratitude to the Parents' Association for their generous funding for our printed copies. Finally, thank you to all the staff members who have dedicated their time and effort to make this edition a success. Congratulations to the graduating members, and best of luck to the next year's team as you continue the wonderful work!

Rin Fukuoka
June 2023

Foreword

Mr. Gary Rubinstein has been teaching mathematics at Stuyvesant for 20 years. He is also a dedicated math author, having published "Descartes' Method for Constructing Roots of Polynomials with 'Simple' Curves" in the Convergence, Mathematical Association of America's online journal. In addition, he has written the Barron's Regents Exams and Answers for Algebra I and Algebra II.

It's a great time to be a math student. Back when I was in high school in the late 1980s, there were only two sources to learn from: the teacher and the textbook. I remember my own math teachers very well. They were hard workers and they had a concrete goal that we would ace the Regents exams, but it didn't leave a lot of time for more inspirational and intriguing mathematics. Had there been the Internet and YouTube and access to math journals, I wonder if I would have utilized these great resources.

One of the courses I teach is called Math Research. Over the years, it has become mainly an elective that some freshmen take when they want extra math but don't want to compete on the math team. The course culminates in students writing a research paper in which they learn some interesting math topic and try to extend that topic a little with their own original ideas. Over the years, while grading these papers, I have found myself doing my own research on the topics the students chose. To grade accurately, I needed to fully understand the topic. In these studies I did as part of my grading process, I have learned some of the most interesting mathematics I have ever known.

I once had a student who wanted to write about the math of Astronomy. As modern Astronomy requires a lot of Calculus, I told this student to go back to ancient Astronomy. It turned out that many cultures throughout history have pondered the mystery of the glowing orbs in the night sky. This led to the invention of Trigonometry and, more specifically, spherical Trigonometry. I had no idea that such a thing existed: trigonometry where the sides of the triangles were on the surface of a sphere, and the triangle was a bit like the base of a pyramid where the vertex of the pyramid was at the center of the sphere.

I have some books about spherical Trigonometry that I pick up and read from time to time. There aren't many books published about it nowadays since it is no longer a hot topic. But it is fun to peer into the past and to learn something that few people alive today, even among people who love math, are interested in. It's like being in a world where Shakespeare was not a household name, and I discover an obscure British playwright and dive into these plays, thinking what a shame it is that he is not well known.

There's too much math to master it all. But for any topic, if you desire, you can dive into it and see where it leads. In fact, that might lead you to even more topics and mathematical connections.

And that's what I think the purpose of the *Math Survey* is. It is an opportunity for students writers to follow a mathematical interest and see where it leads. And hopefully the students who read the *Math Survey* are inspired to do a little of their own research on the topics and to follow the path that interests them.

Gary Rubinstein
Math Teacher
June 2023

Wrap Around Relay

Mikayla Lin

1 Instructions

Use previous answers to solve each question.

2 Problems

1. Let A be the answer to problem 3. Compute $A^2 - 40A + 388$?

2. Let B be the answer to problem 1. Suppose there are B people at a dinner party. Each person comes with a friend, and they insist on sitting next to each other. How many ways are there to seat the B people at a round table if rotations are considered identical?

3. Let C be the answer to problem 2. How many positive integers less than 100 are multiples of C?

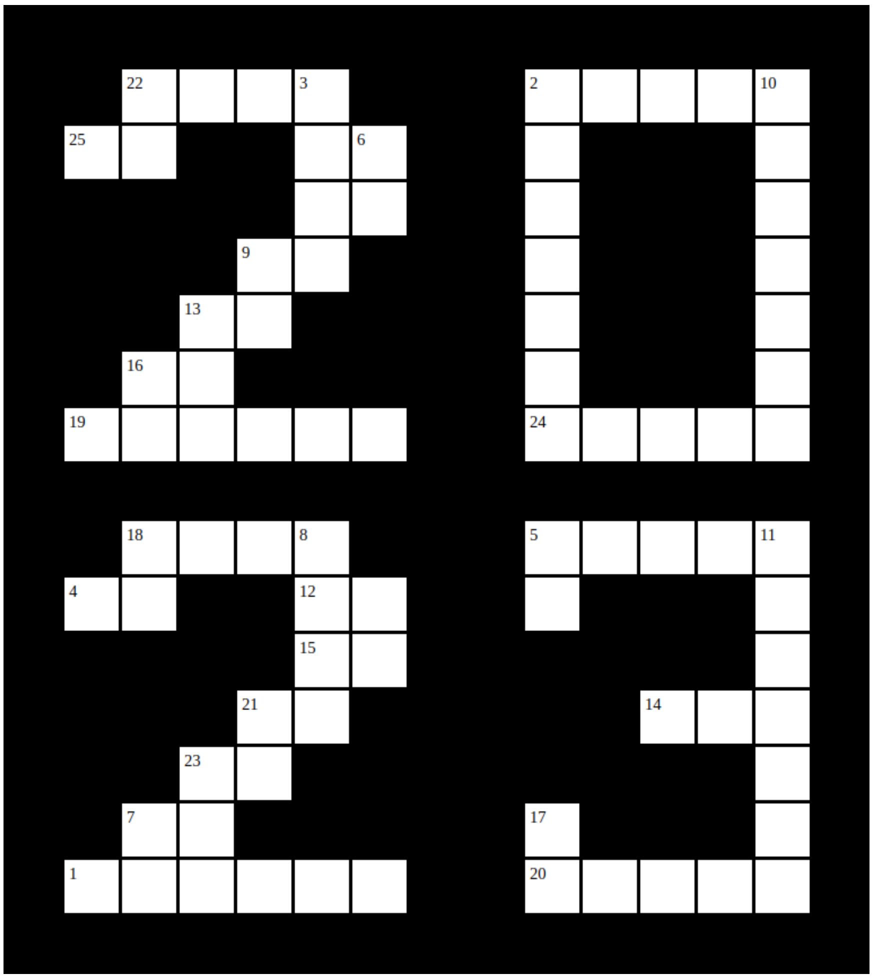
Find (A, B, C).

The solution is on page 5.

Number Crossword

Amanda Louie

1 Puzzle



1.1 Introduction + Rules

This crossword contains clues that have been written by various Math Survey members and friends who have been kind enough to contribute to this puzzle. Their names are credited next to the clues that they have written, thank you so much to everyone who submitted a clue!
Rules: This puzzle works like a standard crossword, only the answers are numbers. Answer the clues for across and down to fill in the puzzle!

Across

1. $(2023 \div 17) - 90)(2023 + 2628)$

2. $(2023 - 2020)(2023 + 4120)$

4. Let S be the sum of the digits of 33^{22}. Given that:

   - The first 10 digits of 33^{22} are 2, 5, 5, 4, 5, 0, 4, 5, 3, 0 respectively,
   - The 27^{th} through 33^{rd} digits are 0, 4, 9, 5, 1, 4, 8 respectively,
   - Digits 17 and 18 are 7 and 4,
   - The average of all the remaining even-placed digits except for the last is an integer,
   - The difference of the remaining odd-placed digits and the remaining even-placed digits except for the last is a multiple of 5,

   Find the product of $\frac{S-1}{8}$ and the smallest factor of S other than 1.
   Credit: Patrick Was

5. $(2023 \times 36) + 1195$

7. This number is the sum of the first 6 triangular numbers (numbers where given a triangular number of objects, you can arrange those objects into an equilateral triangle).

12. $\int_{-12}^{12} \sqrt[3]{x - \lfloor x \rfloor}\, dx = ?$
    Credit: Zawad Dewan

14. This number is a three digit palindromic number equal to the product of the sum of its digits and ten more than the product of its digits.

15. Compute the number of factors of $196^2 + 578^2$.
    Credit: Julia Kozak

18. $((2023 \div 119) + 20) \times 77$

19. $(2023 \times 243) + 716$

20. $((2023 + 75) \times 40$

21. Let n be the number of ordered triples of integers (x,y,z) are there such that $x^2 + y^2 + z^2 = 34$. Compute $2n - 5$.
    Credit: Kathleen Zhang

22. $(2023 - 1940) \times 42$

23. $(2023 \div 17) - 45$

24. Rin is deciding how to auction a piece of jewelry to Elie and Josiah. Her first option is to limit both bidders to one bid hidden from each other, and have the winner pay the loser's bid for the jewelry. Her second option is to let each bidder bid multiple times, but force both bidders to pay their own final bids, even though only the winner will get the jewelry. Elie and Josiah value the jewelry at \$127 and \$151 respectively, and both bring \$10 more than they value their jewelry at. Both bidders decide to submit bids, bid optimally in integer amounts

and don't communicate with each other. If Elie bids first, what is the product of Josiah's final bids in each of Rin's potential auction plans?
Credit: Lee Eisenberg

25. $(2023 \div 119) + 2$

Down

2. $((2023 \times 375) + 106) \times 2$
3. $(2023 \times 3) + 245$
5. Find

$$\lim_{n\to\infty} \frac{\displaystyle\sum_{k=1}^{n} (.99)^k}{\displaystyle\sum_{k=1}^{n} \left(\frac{2k+2}{k^2+2k} - \frac{2k+4}{k^2+4k+3} \right)}$$

   Credit: Ethan Sharma

6. $(2023 \div 119) + 70$
8. $(2023 \times 4) + (20230 \div 17) - 151$
9. Find the largest prime under 100 that is the sum of another prime and a positive integer to the fifth power.
   Credit: Daphne Qin
10. $(4046 \times 2023) + (2023 \times 23^2) + (2023 \times 23 \times 6) + (2023 \times 19) + 82$
11. $2023^2 + -(2023 \times 460) + 1871$
13. $2023 - 1761$
16. $(2023 \div 17) - 60$
17. Let the sequence N be

$$\frac{2}{2 \times 3} + \frac{2}{3 \times 8} + \frac{2}{4 \times 15} + \frac{2}{5 \times 24} + \cdots\cdots$$

   . The sum of the first nth terms of the sequence N is $\frac{189}{390}$. Compute the value of n.
   Credit: Lucia Liu

18. Find the smallest two-digit number that is divisible by the number of primes less than the number.
    Credit: Mr. Kats

The solution is on page 6.

Solution to Wrap Around Relay

Mikayla Lin

1 Solution

Let's first write each variable in terms of the other. Question 1 is pretty straightforward, the expression is already in terms of A. However, we can complete the square to simplify the expression a little more.

$$B = A^2 - 40A + 388 = (A-20)^2 - 12$$

Question 2 is a combo problem. Each person wants to sit with their friend, so we can just treat them as $\frac{B}{2}$ pairs with $\frac{B}{2}$ seats. To account for rotations, we can fix one pair at a certain position and then choose the places for the $\frac{B}{2} - 1$ remaining pairs. Thus there are $(\frac{B}{2} - 1)!$ ways to seat them. However, within each pair, we could have person 1 then person 2, or person 2 then person 1. So we have 2 choices for each pair, or a multiplier of $2^{\frac{B}{2}}$.

$$C = (\tfrac{B}{2} - 1)! 2^{\frac{B}{2}}$$

Question 3 tells us that A is however many multiples of C are less than 100.

$$A = \lfloor \tfrac{99}{C} \rfloor$$

Lets look at question 2 first, as its the most restrictive. From the problem statement, we know that B must be a positive even integer. Now lets trace that back to question 1. From our completing the square, B is positive when $(A-20)^2 > 12$. Question 3 tells us that A is a positive integer, so we can conclude that: $A > 20 + \sqrt{12}$ meaning $A > 23$. More than that for B to be an even number, $(A-20)^2$ must have the same parity as 12, so A has to be even as well. This means A can be 24, 26, 28, ...

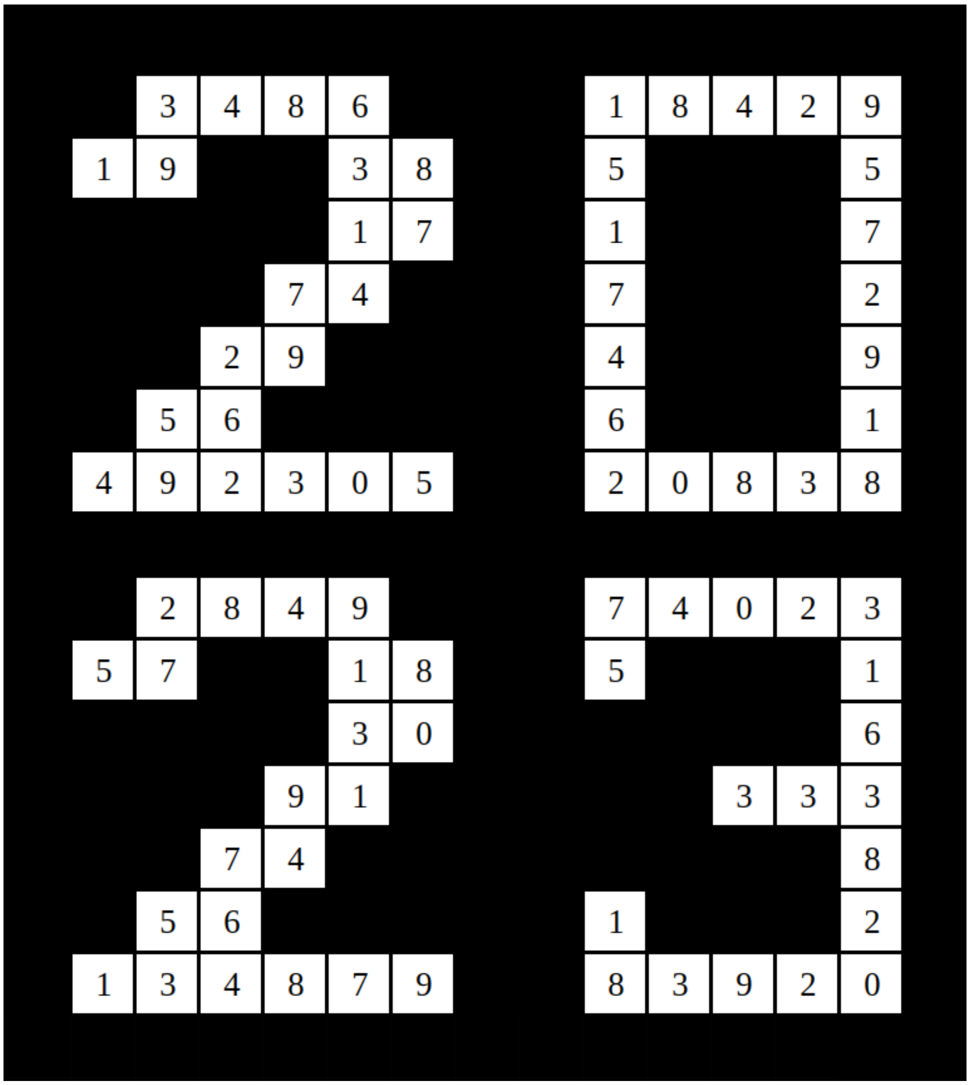
Looking at the expression for A: $A = \lfloor \frac{99}{C} \rfloor$ tells us that C could be at most 4, as 5 or any number greater would give A < 20.

All that's left is to try out different values of C. The only value of C that produces an even A is 4. Plugging in the numbers you would get that $A = \lfloor \frac{99}{4} \rfloor = 24$, $B = (24-20)^2 - 12 = 4$, and $C = (\frac{4}{2} - 1)! 2^{\frac{4}{2}} = 1! \cdot 4 = 4$, which is what we set C to be. This confirms that $C = 4$ works as a solution and gets you that $\boxed{\text{(A, B, C) = (24, 4, 4)}}$.

Solutions to Number Crossword

Amanda Louie

1 Puzzle



Across

1. $(2023 \div 17) - 90)(2023 + 2628) =$ **134879**

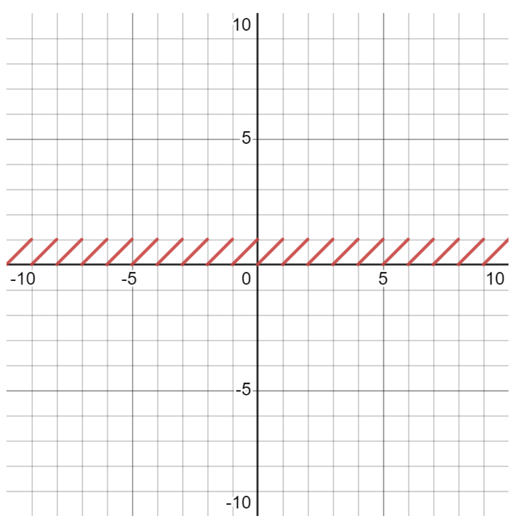
2. $(2023 - 2020)(2023 + 4120) = \mathbf{18429}$

4. We can find the number of digits in 33^{22} using $1 + \lfloor 22 \log 33 \rfloor$, which yields 34.
Because $11|33^{22}$ and $9|33^{22}$, we can apply their divisibility rules to find the sum of the remaining values. The last digit must be 9 because the ones digit of powers of 3 cycles every 4, and dividing 22 by 4 leaves a remainder of 2. Using these digits, we can find that the current alternating sum is -4, assuming we subtract the odd digits from the even ones, and that the current total sum is 84. There are 14 values remaining. We can express the remaining digits as $a, b, c, \ldots n$, where $A = a + b + \ldots + n$ and $B = -a + b - \ldots + n$. Based on the divisibility rules, we can write the following systems:
$A \equiv 6 \mod 9$
$B \equiv 4 \mod 11$ and $B \equiv 0 \mod 5$
It is also given that the average of all the remaining even numbered integers is an integer, so $b + d + \cdots + n \cong 0 \mod 7$. So, $\frac{A+B}{2}$ must be divisible by 7. The smallest value that satisfies B is 15, so that should be the first case tried. Cycling through some values for A that work, it is seen that $A = 69$ and $B = 15$.
$84 + 69 = 153$
$\frac{153-1}{8} \cdot 3$
$19 \cdot 3$
57
Credit: Patrick Was

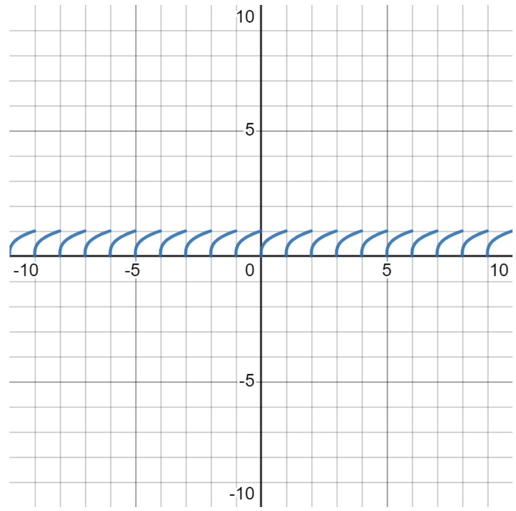
5. $(2023 \times 36) + 1195 = \mathbf{74023}$

7. $\mathbf{56} = 1 + 3 + 6 + 10 + 15 + 21$

12. A nice approach to solving integrals is by looking at the area under the curve and above the x-axis. Let's first consider the graph of $x - \lfloor x \rfloor$:



Notice that each line segment can be represented by $y = x$ where $0 \leq x \leq 1$. Now, by taking the cube root of this equation, we have $\sqrt[3]{x - \lfloor x \rfloor}$:



Again, we see that each little curve can be represented by the equation $y = x^{\frac{1}{3}}$ where $0 \leq x \leq 1$. We have $12 + 12 = 24$ of these curves, so $24 \cdot \int_0^1 \sqrt[3]{x}\, dx = 24 \cdot \frac{3}{4} = 18$.
Credit: Zawad Dewan

14. **333**$= (3 + 3 + 3)(10 + (3 \times 3 \times 3))$

15. We can first notice that $196^2 + 578^2 = 14^4 + 4 \cdot 17^4$. Then, by Sophie Germain's Identity, we can factor the expression as $(14^2 + 2 \cdot 14 \cdot 17 + 2 \cdot 17^2)(14^2 - 2 \cdot 14 \cdot 17 + 2 \cdot 17^2) = 4 \cdot 625 \cdot 149$. This has prime factorization $2^2 \cdot 5^4 \cdot 149$, thus the number of factors is $3 \cdot 5 \cdot 2 = \mathbf{30}$.
Credit: Julia Kozak

18. $((2023 \div 119) + 20) \times 77 = \mathbf{2849}$

19. $(2023 \times 243) + 716 = \mathbf{492305}$

20. $((2023 + 75) \times 40 = \mathbf{83920}$

21. We have to represent 34 as as the sum of squares. The only squares that we can use are 0, 1, 4, 6, 16, and 25. There are only two sets of sums that work: 0, 25, 9 and 16, 9, 9. In the case of 0, 25, 9, there are 3! ways to choose the order and $2^2 = 4$ ways to choose the sign of 5 and 3. In total there are $6 \times 4 = 24$ ways. In the case of 16, 9, 9, there are 3 ways to choose the order and $2^3 = 8$ ways to choose the sign. In total there are $3 \times 8 = 24$ ways. Adding these up, there are $24 + 24 = 48$ ways $= n$. Thus, $2n - 5 = 2 \times 48 - 5 = 96 - 5 = \mathbf{91}$.
Credit: Kathleen Zhang

22. $(2023 - 1940) \times 42 = \mathbf{3486}$

23. $(2023 \div 17) - 45 = \mathbf{74}$

24. In the first auction plan, both Josiah and Elie will want to bid exactly what they value the jewelry at. That way, they are guaranteed to pay less than their bid regardless of the outcome. Josiah wins here by bidding \$151. In the second auction plan, whoever has the lower bid at the end will be in danger of losing money without getting anything. This means that both bidders will continue bidding more and more to avoid losing money, even going past what they value the jewelry at. Thus Elie bids all the way up to \$137, his highest possible bid,

making Josiah's next and final bid \$138. Multiplying these two bids gives $151 \times 138 =$ **20838**.
Credit: Lee Eisenberg

25. $(2023 \div 119) + 2 =$ **19**

Down

2. $((2023 \times 375) + 106) \times 2 =$ **1517462**

3. $(2023 \times 3) + 245 =$ **6314**

5. First, the numerator is an infinite geometric series with $|r| < 1$, so we can use the formula for infinite geometric series:
$$\frac{1}{1 - .99} = \frac{1}{.01} = 100$$
Let's try to manipulate the summation in the denominator. We can factor the denominators and notice a potential partial fraction decomposition:
$$\sum_{k=1}^{n} \left(\frac{2k+2}{k(k+2)} - \frac{2k+4}{(k+1)(k+3)} \right)$$
We want to factor the first term into some $\frac{A}{k} + \frac{B}{k+2}$ and the second term into some $\frac{C}{k+1} + \frac{D}{k+3}$. Combining each pair of fractions we get $\frac{A(k+2)+Bk}{k(k+2)}$ and $\frac{C(k+1)+D(k+3)}{(k+1)(k+3)}$. We want to match up the coefficients of k and the constants, so we can see that $A = 1$ due to the constant term so $B = 1$ to get a $2k$. We solve $C + D = 2$ and $C + 3D = 4$ to get that $C = 1$ and $D = 1$ as well. So, this simplifies to
$$\sum_{k=1}^{n} \left(\frac{1}{k} - \frac{1}{k+1} + \frac{1}{k+2} - \frac{1}{k-3} \right)$$
We can see that this telescopes. Let's look at the first few terms:
$$\begin{array}{l} 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} \\ \quad\; \frac{1}{2} - \frac{1}{3} + \frac{1}{4} - \frac{1}{5} \\ \qquad\;\; \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} \\ \qquad\qquad \frac{1}{4} - \frac{1}{5} + \frac{1}{6} - \frac{1}{7} \end{array}$$
And so on. All terms after $\frac{1}{4}$ will completely cancel since there will be 4 instances of each with alternating signs. All we have left are 1, $\frac{1}{2}$, and $\frac{1}{3}$. The $\frac{1}{2}$ cancels, and we are left with $1 + \frac{1}{3} = \frac{4}{3}$. The sum is equal to $\frac{4}{3}$, so
$$\lim_{n\to\infty} \frac{\sum_{k=1}^{n} (.99)^k}{\sum_{k=1}^{n} \left(\frac{2k+2}{k^2+2k} - \frac{2k+4}{k^2+4k+3} \right)} = \frac{100}{\frac{4}{3}} = 100\frac{3}{4} = \mathbf{75}.$$
Credit: Ethan Sharma

6. $(2023 \div 119) + 70 = \mathbf{87}$

8. $(2023 \times 4) + (20230 \div 17) - 151 = \mathbf{9131}$

9. Since the number raised to the fifth power is positive, we know that our only options are $1^5 = 1$ or $2^5 = 32$. $3^5 = 243$ is too large.
Let's say we choose 1^5, which is an odd number. For the sum to be prime, our only possible scenario is $1^5 + 2 = 3$. We can be fairly certain, though, that the answer is bigger than 3. Let's try using 2^5 to get bigger solutions.
$2^5 = 32$. Starting from the largest prime less than 100 and going down:

$$97 - 32 = 65$$

$$89 - 32 = 57$$

$$83 - 32 = 51$$

$$79 - 32 = 47$$

47 is prime. Thus, our answer is **79**.
Credit: Daphne Qin

10. It's reasonable to infer that the terms are supposed to somehow cancel each other out. Therefore, we should actively be looking for a way to rewrite each term in this sequence as a difference between two expressions.
Notice that the sequence N can be rewritten as

$$\frac{2}{2(2^2-1)} + \frac{2}{3(3^2-1)} + \frac{2}{4(4^2-1)} + \frac{2}{5(5^2-1)} + \cdots\cdots$$

in which every term can be represented as

$$\frac{2}{a(a^2-1)}$$

for some number a. This expression can then be written as the following:

$$\frac{2}{a(a+1)(a-1)}$$

Notice that (a + 1) - (a - 1) = 2. Therefore, we can rewrite the expression as:

$$\frac{(a+1)-(a-1)}{a(a+1)(a-1)} = \frac{(a+1)}{a(a+1)(a-1)} - \frac{(a-1)}{a(a+1)(a-1)}$$

$$= \frac{1}{(a-1)\times a} - \frac{1}{a\times(a+1)}$$

So, the sequence N can be rewritten as

$$\left(\frac{1}{1\times 2} - \frac{1}{2\times 3}\right) + \left(\frac{1}{2\times 3} - \frac{1}{3\times 4}\right) + \left(\frac{1}{3\times 4} - \frac{1}{4\times 5}\right) + \cdots\cdots$$

It's important to note that for any nth term of the sequence, the a value is not equal to n but $n+1$. The sum of the first nth terms of the sequence N can thus be written as

$$\frac{1}{1 \times 2} - \frac{1}{(n+1) \times (n+2)} = \frac{189}{380}$$

$$\frac{1}{(n+1) \times (n+2)} = \frac{1}{380}$$

The value of n is **18**.
Credit: Lucia Liu

18. Start by listing out the first few primes: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29... Now, we can start by testing the smallest two-digit integer and go up from there.
There are four primes less than 10 and 11. 4 divides neither.
There are five primes less than 12 and 13. 5 divides neither.
There are six primes less than 14, 15, 16, and 17. 6 divides none of them.
There are seven primes less than 18 and 19. 7 divides neither.
There are eight primes less than 20, 21, and 22. 8 divides none of them.
Now there are nine primes less than 23, 24, 25, 26 and 27, and 28. 9 divides **27**.
Credit: Mr. Kats

Cross Sections of Real Life: Conics

Ethan Sharma

1 Introduction

Little Timmy throws a baseball to his friend 10 feet away. It would be standard to model this trajectory with a parabola. We can say this by making the simplification that the only accelerating force that acts on the ball is Earth's gravity.

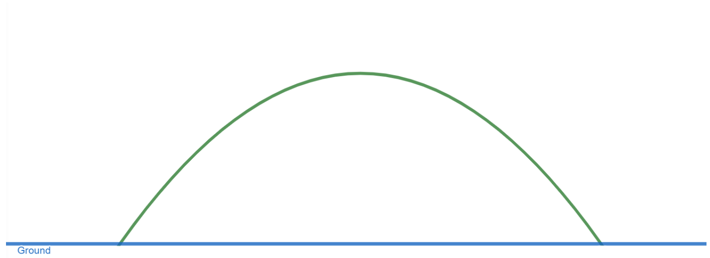


Figure 1: The baseball's path

Now, this works fine when we only want to see the ball's path from Timmy to his friend. But our parabolic path will keep going. Ignoring the physical body of the Earth, this is what our parabola models for the path of the ball after Timmy throws it:

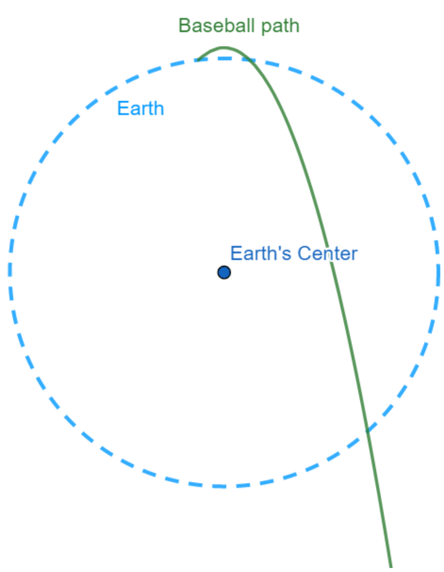


Figure 2: Baseball's path (not to scale)

Yeah...Something's off. This baseball is getting launched out of Earth's orbit into space, somehow reaching escape velocity, with only the Earth's gravity accelerating it?? Of course, that isn't possible. Earth's gravity pulls mass towards the center, so this baseball can't just go flying in this parabolic path.

Here we find the mistake that led to this issue: We take the Earth's gravity to be a constant accelerating force downwards; however if we look from this perspective, the accelerating force of gravity will be changing direction and magnitude depending on how far the ball is from the center. The parabolic approximation is good enough when the distance from the Earth's center barely changes, like we have in Timmy's example, but that does not apply to the general case.

Consider a satellite in orbit. We know that the reason it revolves around Earth is due to Earth's gravitational pull, and it revolves *around* the earth, in a somewhat circular path. This gives us a hint as to what should be happening to the ball: It won't go in the same direction forever, instead the ball will be pulled towards the center of the Earth. Now, let's ignore the other problems presented by the physical body of the Earth and just say that Little Timmy throws his baseball high enough to get into Earth's orbit.

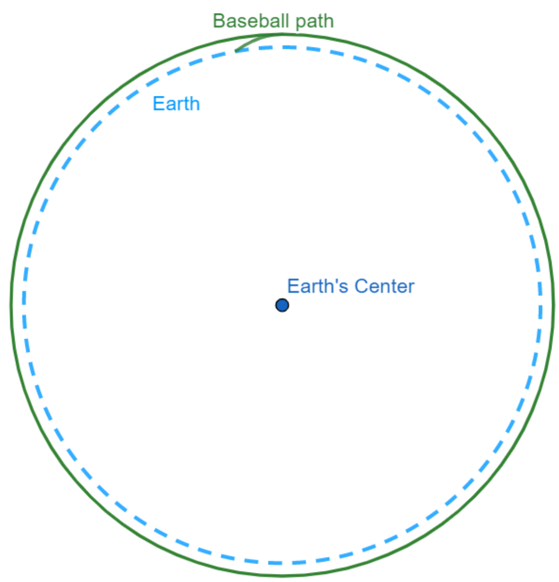


Figure 3: Baseball's orbit (not to scale)

Now, even though Timmy is still just throwing the ball, the path is now somewhat circular. *Somewhat.* The Earth isn't perfectly spherical, however, as the circumferences around the poles and around the equator are different. So, the ball's orbit won't exactly be circular, but **elliptical**. In fact, the path in Figure 2 should be elliptical as well.

We like to simplify curved surfaces into circular pieces because dealing with circles is a lot easier than dealing with other curves. A circle, with its infinite symmetries, represents a perfect, ideal object. This perfection is near impossible to recreate in real life, but it is often useful if we want to approximate values.

So what gives? Parabolic paths are actually elliptical? Ellipses are basically circles? Parabolas are actually circles? These objects seem to have little in common, but yes, they are all connected. Parabolas, circles, and everything in between can all be classified as **conic sections.**

2 Definitions

2.1 Sections of a Cone

The most direct definition of conic sections is the set of curves formed by the intersection of a plane and a double cone, or two right cones on top of one another. This "double cone" can also be seen as the solid of revolution of a line $y = mx + b$ about the y-axis.

Figure 4: $z^2 = x^2 + y^2$ (Note these cones have no "base.")

We can obtain all four types of conic sections depending on the plane that intersects with the double cone. The cross section is a circle if the plane is parallel to a base of the cones. An ellipse is formed when the plane intersects one of the cones four times. A circle is sometimes considered a special case of an ellipse. A parabola is formed when the slope of the plane is equal to the slope of a line containing the sides of the cones. Finally, a hyperbola is formed when the plane intersects each cone two times.

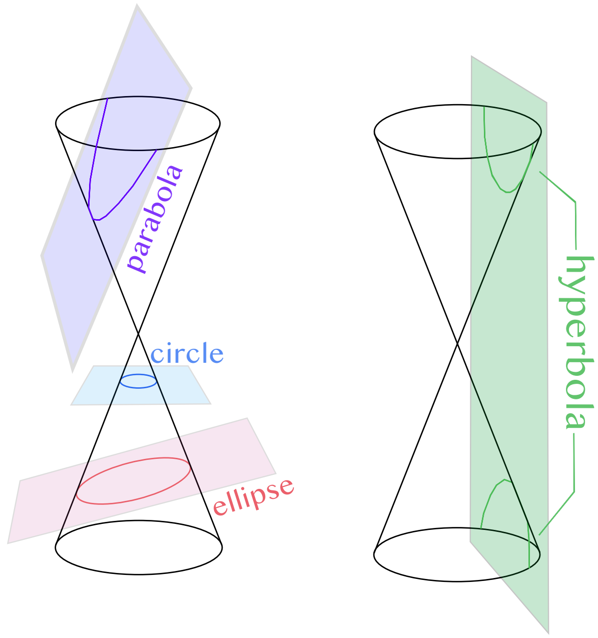


Figure 5: All four types of conic sections

2.2 Eccentricity

All of the conic sections are also connected by the idea of **eccentricity**. The eccentricity of a conic section is a quantity that describes how much it "deviates" from a circle.

An eccentricity of 0 means the figure is a circle. (A circle doesn't deviate at all from being a circle.) Between 0 and 1, the figure is an ellipse. When the eccentricity equals 1, the figure is a parabola, and when the eccentricity is greater than 1, the figure is a hyperbola.

Given a focus and directrix (a point and a line), eccentricity is defined to be the ratio between the distance from any point on the conic section to the focus and the distance from that point to the directrix. So, any conic section can be formed given a focus, directrix, and eccentricity.

2.3 General Form

The general form for all conic sections is:

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$$

From this equation we can see that, generally, any polynomial with degree less than or equal to 2 (in x or y) is a conic section. Certain cases for the coefficients of this equation will yield different conic sections, as we will discuss in later sections.

3 Circles

The most well-known definition of a circle is its locus definition: the set of all points equidistant from a given point (the center) on a two-dimensional plane. How does this line up with our definition of conic sections?

We've defined an eccentricity of 0 to be a circle. This also means, by the focus-directrix definition of eccentricity, the ratio of the distance from a point on the circle to some focus and the distance from a point on the circle to some directrix is 0. So where is the focus and directrix for any circle?

Well, if we considered some line l and a point P, where would the ratio of those distances be 0? Let X be any point that can be on our circle. Looking at our definition for eccentricity,

$$e = \frac{d(X, P)}{d(X, l)}$$

The only way $e = 0$ is possible is when $d(X, P) = 0$. The only point X such that $d(X, P) = 0$ is P, so that means our circle is only the point P! This doesn't quite work with what we already know about circles, so what's wrong?

The issue is the directrix. As long as $d(X, l)$ is finite, the figure will be a degenerate circle. So, let's consider,

$$\lim_{d(X,l)\to\infty} e = \lim_{d(X,l)\to\infty} \frac{d(X, P)}{d(X, l)} = 0.$$

This is true as long as $d(X, P)$ is finite. So when the directrix is a line infinitely far away from the circle, all points that are a finite distance from the focus are on the circle? That isn't right either, which is why we need a radius. P, the focus, is the center of the circle, so a radius would just be a constant $d(X, P)$.

So for a non-degenerate circle, the directrix is a line at infinity and the focus is the center of the circle. The directrix ends up being insignificant in this case, because it would be a line at infinity for any circle, so it doesn't help at all. A center and a radius identifies a circle, as we have eccentricity ($e = 0$), a focus (the center), and a directrix (insignificant). The radius is a byproduct of the fact that we can't determine anything about $d(X, P)$ when the directrix is at infinity except that it is finite.

3.1 General Form

In the general form for conic sections, the figure is a circle if $A = C$ and $B = 0$. This can be seen by experimentation, for example looking at $2(x - 5)^2 + 2(y - 6)^2 = 25$, we can see that it is a circle centered at (5,6) with radius $\frac{5}{\sqrt{2}}$, and it has $A = C$ and D, E, and F terms. However, there is no B term, which makes sense because an xy term would come from some form of $(ax + by)^2$ which would not result in a circle.

4 Ellipses

Ellipses are often described as squished circles, because that is exactly what they are. If you take a circle, grab it at diametrically opposite points, and pull on it, you have created an ellipse.

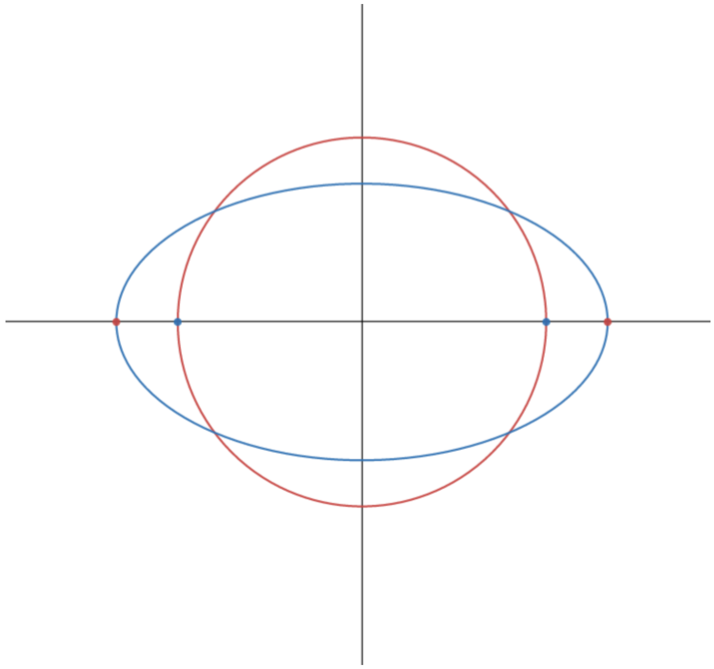


Figure 6: Circle pulled from blue points to red points forms ellipse.

4.1 Foci

An ellipse is most commonly defined as the set of points such that the sum of the distance from two points to a point on the figure is constant. With this definition, an ellipse has two foci, and the sum of the distances from each foci to a point on the ellipse is a constant value. Here we can see the similarity to a circle: A circle is the set of points a constant distance away from the center (a focus), and an ellipse is the set of points such that the distances from its two foci sum to a constant value, which we call the **focal sum**. With our pulling on the circle example, we can imagine the focus of the circle gets pulled apart into two foci. The ellipse still has a center, though, and it's at the midpoint of the line segment connecting the foci.

4.2 Anatomy

The general form for a horizontally-oriented ellipse centered at (h, k) is:

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$$

If $a > b$, then a and b are called the **semimajor** and **semiminor** axes of the ellipse, which are each half of the **major** and **minor** axes of the ellipse. The major axis can be thought of as the line connecting the points farthest apart on the ellipse, and these points are called the **vertices** of the ellipse. The minor axis is the line connecting the points on the ellipse closest to each other. In an ellipse centered at the origin, like in Figure 6,

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

The minor axis lies on the y-axis, and the major axis lies on the x-axis, as long as $a > b$. If $b > a$, then the ellipse is vertically-oriented and the major and minor axes are on the y-axis and

x-axis respectively. In Figure 6, the red points are the vertices of the ellipse. The foci in an ellipse of this form can be found using a and b, utilizing the focal sum.

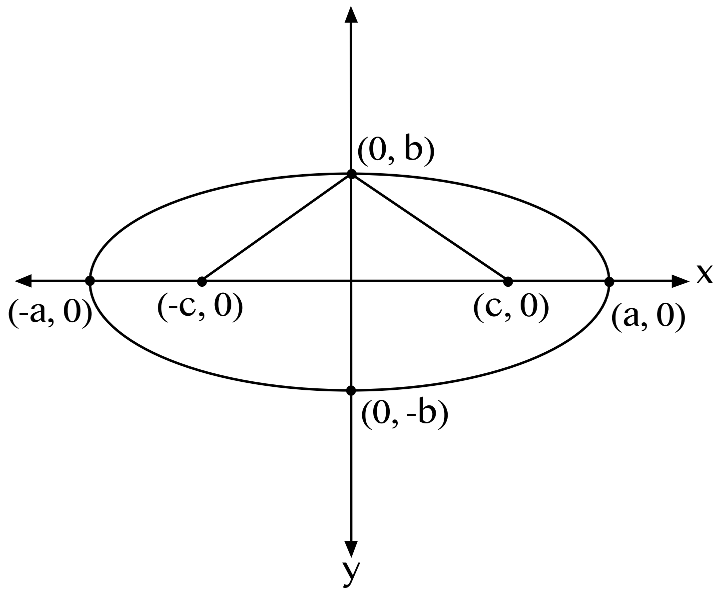


Figure 7: Anatomy of ellipse

The distance from the center of the ellipse to either focus is c. We can find c as follows:

- The focal sum to the point $(a, 0)$ is $(a - c) + (a + c) = 2a$.
- Consider the triangle formed by $(0, b)$, $(-c, 0)$ and $(c, 0)$. This triangle is isosceles since the length of the line segments connecting $(0, b)$ to the foci can both be expressed as $\sqrt{c^2 + b^2}$.
- The focal sum $2a$ gives us $2a = 2\sqrt{c^2 + b^2}$, which yields $a^2 = b^2 + c^2$.

So now, given an ellipse in its general form, we are able to find its foci.

4.3 Focus-Directrix

For an ellipse, the eccentricity is between 0 and 1. To say this, there must be some way an ellipse can be defined with a focus and directrix. And there is! For a horizontally oriented ellipse centered at the origin, consider some vertical line $x = k$. We know that this line can't intersect the ellipse, so let it be to the right of the ellipse.

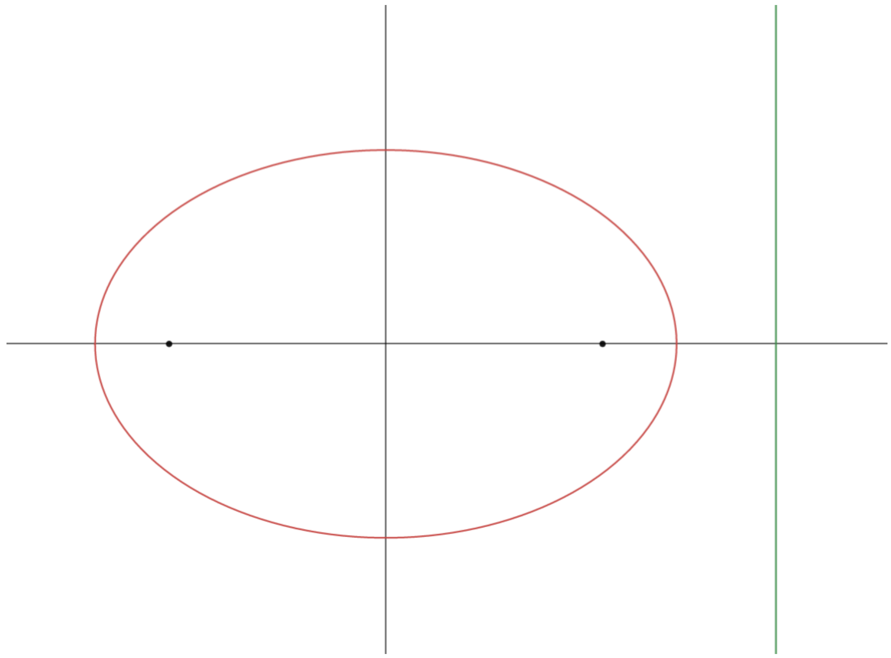


Figure 8: Ellipse, with foci at black points and line $x = k$

We know the ratio of the distance from a point on the ellipse to the focus to the distance from that point on the ellipse to the directrix is the eccentricity. Consider $(a, 0)$, the vertex of the ellipse on the right. The distance from this vertex to the focus is $a - c$, and the distance from this focus to the directrix is $k - a$. We can write

$$e = \frac{a-c}{k-a}$$

Now, consider the eccentricity at (-a, 0):

$$e = \frac{a+c}{k+a}$$

These values are equivalent as they are both the eccentricity of the ellipse, so we can solve for k:

$$\begin{aligned}
\frac{a+c}{k+a} &= \frac{a-c}{k-a} \\
(a+c)(k-a) &= (a-c)(k+a) \\
ak - a^2 + ck - ac &= ak + a^2 - ck - ac \\
2ck &= 2a^2 \\
k &= \frac{a^2}{c}
\end{aligned}$$

Now we have found the equation for the directrix in terms of a and c.

4.4 Eccentricity, Part 2

Eccentricity can also be defined based on the conic section itself. For a circle, it's 0, and for a parabola, it's 1. For an ellipse, however, the eccentricity is always between 0 and 1, and can be defined as

$$e = \frac{c}{a},$$

where a is the semimajor axis and c is the distance from the center to a focus. We can prove this new definition for eccentricity using our equation for the directrix. Consider $(a, 0)$ again: The distance from $(a, 0)$ to the focus is $a - c$, and the distance from $(a, 0)$ to the directrix is $\frac{a^2}{c} - a$. Simplifying, we get:

$$\frac{a - c}{\frac{a^2}{c} - a} = \frac{ac - c^2}{a^2 - ac}$$
$$\frac{ac - c^2}{a^2 - ac} = \frac{c(a - c)}{a(a - c)}$$
$$e = \frac{c}{a}$$

This makes sense in the context of the "pulling on a circle" example: Since the focus of the circle is also the center, $c = 0$, so the eccentricity of a circle is 0. Since $0 < \frac{c}{a} < 1$, $c < a$. So, as long as $c < a$, or the semimajor axis is greater than the distance from the focus to the center, $e < 1$, so the figure is an ellipse. If $c > a$, then the focus is forced outside of the ellipse, and this becomes a hyperbola and $e > 1$ (although c is defined differently for hyperbolas, which will be discussed later). Notice that our derivation breaks when $c = a$, which is when the figure will be a parabola, not an ellipse (it doesn't make sense for the focus to be on a vertex of the ellipse).

We can also express the equation for the directrix in terms of eccentricity.

$$k = \frac{a^2}{c}$$

We can simplify this with $e = \frac{c}{a}$:

$$k = \frac{a}{e}$$

Now we've shown that the directrix of a horizontally-oriented ellipse centered at the origin is at $x = \frac{a}{e}$. Note that this allows us to use one focus with the directrix to form the ellipse.

4.5 General Form

In the general form for a conic, the figure will be an ellipse if $A \neq C$ and $B = 0$. This makes sense since an ellipse is a circle if $a = b$, when the coefficients of x^2 and y^2 are equal, but when $a \neq b$, the figure is an ellipse. Ellipses can, however, have a nonzero B term, and this is the case in ellipses that are not horizontally or vertically oriented.

5 Parabolas

As mentioned in the introduction, parabolas are often used for estimating projectile motion. This is because, when assuming the acceleration of an object is constant (usually from gravity), the function for position of that object will be a quadratic. The general form for a parabola is

$$y = ax^2 + bx + c.$$

The focus-directrix definition for a parabola is fairly well known as the set of points equidistant from a focus and directrix. From this definition, it is easy to see that the eccentricity for parabolas is 1. The standard form for a parabola arises from this definition, which is

$$4p(y-k) = (x-h)^2,$$

where p is the distance from the vertex to the directrix (or the focus). Both of these formulas are for vertically-oriented parabolas; horizontally-oriented parabolas can be obtained by switching $(x-h)$ and $(y-k)$.

5.1 Eccentricity

Let's make more sense of our previous definition of eccentricity ($e = \frac{c}{a}$) for a parabola. Consider an ellipse with a fixed focus F and a directrix l to the right of the focus. Right now, $0 < e < 1$, and we want e to approach 1. We will do this by increasing c and a by some amount x, which can be represented by moving the center of the ellipse x units. Notice that, when moving the center, the other focus of the ellipse also changes such that it is c away from the new center. Consider

$$\lim_{x \to \infty} \frac{a+x}{c+x} = 1$$

This is true since a and c in this equation are constant, and we increase it by some value x. This can be visualized with the following figures:

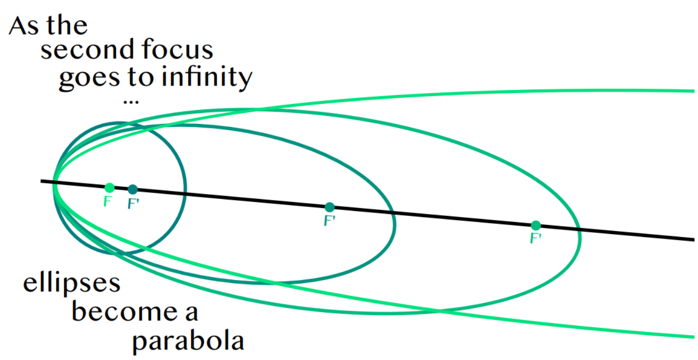


Figure 9: Parabola from ellipse

It can be observed that a parabola is essentially an ellipse with a second focus "at infinity." In the case of projectile motion on Earth, as seen in the introduction, the Earth's center is so far away from the ball's path on the Earth's surface, so instead of needing to model the path with an ellipse, it can be estimated with a parabola.

5.2 General Form

Recall the General form for a conic section:

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$$

For a parabola, in the general form of a conic, if $B = 0$, either $A = 0$ or $C = 0$, so there can only be one squared term. We can also have a nonzero B term with a parabola if we rotate the figure, just like with an ellipse. If $B = 0$ and there is an x^2 term, then there must be a y term to be a parabola. Otherwise, the figure would be two vertical lines. For example, $x^2 + 1 = 0$ will be the lines $x = -1$ and $x = 1$, but $x^2 - y = 0$ is just the parabola $y = x^2$. We have to set a restriction on the B term since the B term causes weird stuff to happen, for example something like $x^2 + xy + 1 = 0$ ends up being a hyperbola.

5.3 The xy term

The B term of the general form for conics creates a few issues when trying to categorize conics based on coefficients of the terms. Notice that we don't usually have this term in general forms for specific conics such as $y = ax^2 + bx + c$ or $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$, as the x and y terms are usually separate. So where does the xy term come from? I've mentioned that there can be a B term if the figure is rotated, or just not specifically vertically or horizontally oriented.

Let's see this with an example: Take the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$:

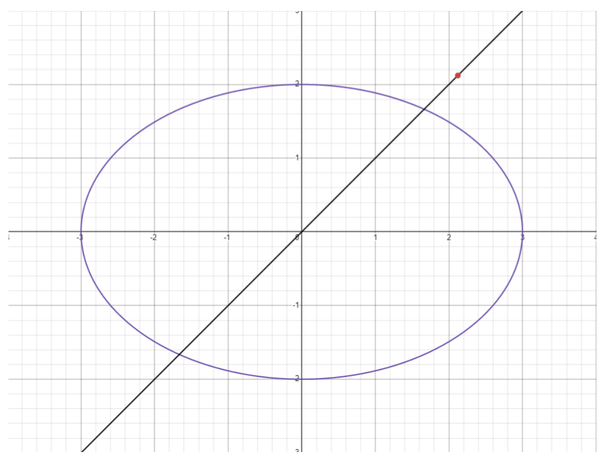


Figure 10: $\frac{x^2}{9} + \frac{y^2}{4} = 1$

Let's rotate the ellipse 45° counterclockwise. Looking at the point (3,0), since rotation is a rigid motion, after rotating this point should still be 3 units away from the origin. Instead of forming an angle of 0° with the x-axis, however, after rotation, the point would lie on a line that creates a 45° angle with the origin. We can find the coordinates of this point by considering an isosceles right triangle with hypotenuse 3, which will have a side length of $\frac{3}{\sqrt{2}}$. So, by rotating by 45°, we have transformed the point (3,0) to $(\frac{3}{\sqrt{2}}, \frac{3}{\sqrt{2}})$. Obviously I chose the point and angle so that the rotation would be nice, but this can be generalized. We can consider a rotation as doing a 'change of variables' that rotates the xy plane to some uv plane about the origin by some angle θ, and the following relation is true:

$$u = x\cos\theta - y\sin\theta$$
$$v = y\cos\theta + x\sin\theta$$

We can see that $\frac{3}{\sqrt{2}} = 3\cos 45° - 0\sin 45°$ and $\frac{3}{\sqrt{2}} = 0\cos 45° + 3\sin 45°$, so this works for our example. Just for fun, let's prove this:

Dealing with rotation in the rectangular plane is annoying, so let's use polar coordinates.

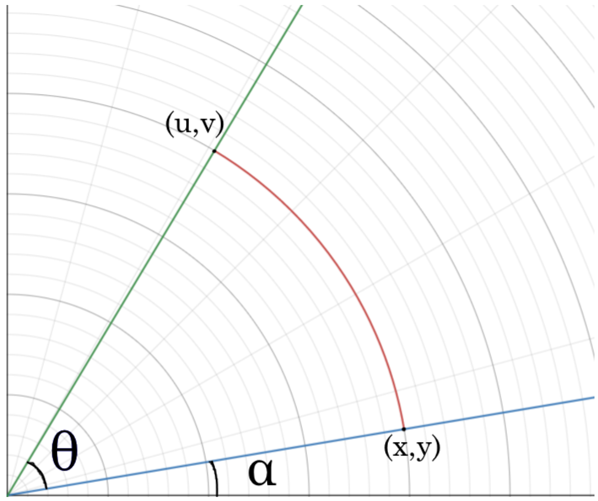


Figure 11: Polar Coordinate System (The points are in rectangular form)

Since we are rotating about the origin, the distance from (0,0) to (x,y) is the same as the distance from (0,0) to (u,v). Let r denote that common distance. Denoting α and θ as they are on Figure 11, we have the following equations:

$$x = r\cos\alpha \tag{1}$$
$$y = r\sin\alpha \tag{2}$$
$$u = r\cos(\alpha+\theta) \tag{3}$$
$$v = r\sin(\alpha+\theta) \tag{4}$$

We can apply the sin and cos addition formulas to (3) and (4) to get:

$$u = r\cos\alpha\cos\theta - r\sin\alpha\sin\theta$$
$$v = r\sin\alpha\cos\theta + r\sin\theta\cos\alpha$$

Plugging in (1) and (2), we get:

$$u = x\cos\theta - y\sin\theta$$
$$v = y\cos\theta + x\sin\theta$$

So we have shown that we can rotate the xy plane to some uv plane by any angle θ using these formulas. Let's see what happens when we use this with the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ rotated by 45°. Our rotated ellipse is only rotated in the xy plane; it is vertically oriented in the uv plane (since the uv plane is a rotation of the xy plane). So, in the uv plane, we are dealing with $\frac{u^2}{9} + \frac{v^2}{4} = 1$, and we can substitute the above equations to find this in the xy plane:

$$\frac{u^2}{9} + \frac{v^2}{4} = 1$$
$$u = x\cos\theta - y\sin\theta$$
$$v = y\cos\theta + x\sin\theta$$
$$\frac{(x\cos\theta - y\sin\theta)^2}{9} + \frac{(y\cos\theta + x\sin\theta)^2}{4} = 1$$
$$\frac{x^2(\cos\theta)^2 - 2xy\cos\theta\sin\theta + y^2(\sin\theta)^2}{9} + \frac{y^2(\cos\theta)^2 + 2xy\cos\theta\sin\theta + x^2(\sin\theta)^2}{4} = 1$$
$$4x^2(\cos\theta)^2 - 8xy\cos\theta\sin\theta + 4y^2(\sin\theta)^2 + 9y^2(\cos\theta)^2 + 18xy\cos\theta\sin\theta + 9x^2(\sin\theta)^2 = 36$$
$$4x^2\cos\theta^2 + 9x^2\sin\theta^2 + 4y^2\sin\theta + 9y^2\cos\theta^2 + 10xy\cos\theta\sin\theta = 36$$
$$4x^2((\cos\theta)^2 + (\sin\theta)^2) + 5x^2(\sin\theta)^2 + 4y^2((\sin\theta)^2 + (\cos\theta)^2) + 5y^2(\cos\theta)^2 + 10xy\cos\theta\sin\theta = 36$$
$$4x^2 + 4y^2 + 5x^2(\sin\theta)^2 + 10xy\cos\theta\sin\theta + 5y^2(\cos\theta)^2 = 36$$

Or, expressed in the general form:

$$(4 + 5(\sin\theta)^2)x^2 + (4 + 5(\cos\theta)^2)y^2 + (10\cos\theta\sin\theta)xy - 36 = 0$$

Now, finally, let's plug in 45°:

$$(4 + 5(\sin 45^\circ)^2)x^2 + (4 + 5(\cos 45^\circ)^2)y^2 + (10\cos 45^\circ\sin 45^\circ)xy - 36 = 0$$
$$(4 + \frac{5}{2})x^2 + (4 + \frac{5}{2})y^2 + (\frac{10}{2})xy - 36 = 0$$
$$\frac{9}{2}x^2 + \frac{9}{2}y^2 + 5xy - 36 = 0$$

As we can see, rotated conic sections create these xy terms (and are often pretty ugly too). This is why it becomes difficult to generalize conics by the general form, since the equations for any specific conic can take many different forms. We can also see here that $A = C$, so we need a restriction on B when making the claim in 4.2.

6 Hyperbolas

Hyperbolas are the least appreciated conic section. Circles and parabolas are the most known, and ellipses are fairly well known, but hyperbolas are more obscure. Hyperbolas can be thought of as "inverted" ellipses.

Hyperbolas have a very similar definition to ellipses. An ellipse is the set of points such that the sum of the distances from the foci are a constant value, the focal sum. A hyperbola is the set of points such that the absolute value of the difference of the distances from two points is constant. From this definition, the general form for a hyperbola emerges:

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

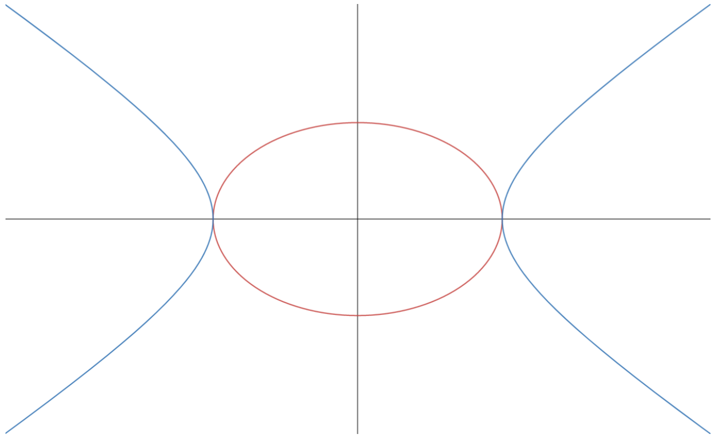


Figure 12: $\frac{x^2}{9} + \frac{y^2}{4} = 1$ in red, $\frac{x^2}{9} - \frac{y^2}{4} = 1$ in blue

6.1 Anatomy

The anatomy of a hyperbola is similar to an ellipse as well, with some key differences. Figure 12 is a diagram with the equations $\frac{x^2}{9} + \frac{y^2}{4} = 1$ and $\frac{x^2}{9} - \frac{y^2}{4} = 1$.

Notice that a and b are the same for the ellipse and the hyperbola. The major axis of the ellipse is also called the transverse axis of the hyperbola, and the minor axis of the ellipse is also called the conjugate axis of the hyperbola. The major, minor, transverse, and conjugate axes are all axes of symmetry for each figure. They have the same vertices, but the foci are at different places, and thus e is a different value.

The value of c, the distance from the center to the focus, is different in hyperbolas and ellipses with relation to a and b. We showed that $a^2 = b^2 + c^2$ or $c^2 = b^2 - a^2$ for ellipses, and it can be similarly shown that $c^2 = a^2 + b^2$ for hyperbolas. Now it makes sense that eccentricity is greater than 1 for hyperbolas, since $e = \frac{c}{a} = \frac{\sqrt{a^2+b^2}}{a} = \sqrt{\frac{a^2+b^2}{a^2}} = \sqrt{1 + \frac{b^2}{a^2}} > 1$.

We can now use the anatomy to derive the general form for a hyperbola. By definition, $|d_2 - d_1|$ is constant. Consider the focal difference at point (a,0) (Refer to Figure 13):

- The focal difference to the point $(a, 0)$ is $(2a + c) - (c) = 2a$.
- By the pythagorean theorem, $d_2 = \sqrt{y^2 + (c + x)^2}$ and $d_1 = \sqrt{y^2 + (x - c)^2}$
- We now have $\sqrt{y^2 + (c + x)^2} - \sqrt{y^2 + (x - c)^2} = 2a$.
- Simplifying, we get $x^2(c^2 - a^2) - a^2y^2 = a^2(c^2 - a^2)$
- Plugging in $b^2 = c^2 - a^2$, we get the general form for a hyperbola:

$$b^2x^2 - a^2y^2 = a^2b^2$$
$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

Just as parabolas can be thought of as ellipses that “break” as the focus goes to infinity, hyperbolas can be thought of as ellipses that go “around” infinity, with the second focus of the ellipse coming back from negative infinity.

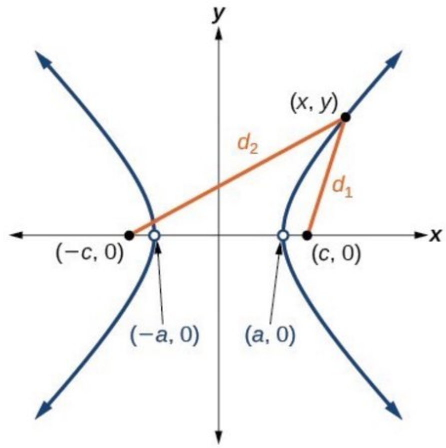


Figure 13: Hyperbola anatomy

6.2 Linear Asymptotes

One cool aspect of hyperbolas are that they have linear asymptotes. We can see this by manipulating the general form:

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$
$$\frac{y^2}{b^2} = \frac{x^2}{a^2} - 1$$
$$y^2 = \frac{b^2x^2}{a^2} - b^2$$
$$y^2 = \frac{b^2}{a^2}(x^2 - a^2)$$
$$y = \pm\frac{b}{a}\sqrt{x^2 - a^2}$$

To analyze end behavior, we can observe that as x gets arbitrarily large, it will dominate the square root term, so for large x, this is approximately:

$$y = \pm\frac{b}{a}\sqrt{x^2}$$
$$y = \pm\frac{b}{a}x.$$

With this result we have shown that as x approaches infinity, hyperbolas asymptotically approach the lines $y = \pm\frac{b}{a}x$.

7 Real World Applications

You might be thinking, okay, all conics are connected, so what? Looking past the math of conic sections, they have applicable uses in real life. I've already mentioned projectile motion, but it is very useful to make the approximation from an ellipse to parabola for ease of calculation.

7.1 Orbits

Newton's laws of universal gravitation can be used to show that the path of all orbits are conic sections. Kepler's first law also states that all planets move in elliptical orbits. For example, the Earth's orbit around the sun and the Moon's orbit around the Earth are elliptical. Satellite orbit around the Earth can be either circular or elliptical. Parabolic and hyperbolic paths occur when the object in orbit is at or past escape velocity. When the object is exactly at escape velocity, the path is parabolic. If the object is travelling faster than escape velocity, like a comet, it will follow a hyperbolic path. The linear asymptotes of a hyperbola represent the path of the comet as it becomes arbitrarily far away from the sun; as the sun has less and less gravitational pull on it, the comet will travel in a straight line (as long as no other forces act on it).

7.2 Reflective Properties

Each conic section has its own reflective property that makes it useful to us. I won't prove them here, but feel free to try them on your own.

Parabolas have the property that any line originating from the focus will be reflected at a path perpendicular to the directrix. The opposite is true as well, so this is useful for sending out signals as well as collecting signals. Paraboloids are 3D objects that are just solids of revolutions of a parabola. These are used in real life for satellite dishes and car headlights. This property makes paraboloid shapes very useful since they will work even if a signal is very weak, as long as the signal is pointed towards the dish.

Figure 14: Satellite dish collecting signal at focus of circular paraboloid

Ellipses have the property that any path originating from one focus will be reflected off of the ellipse to the other focus. We can see the connection with the parabola reflective property here, since the parabola's second focus is essentially at infinity, so the reflected lines are parallel lines which meet "at infinity." Furthermore, signals sent from one focus at the same time will reach the second focus at the same time. This, again, is used to send signals, as well as some obscure areas such as lithotripsy.

Hyperbolas have the property that any path directed at one focus will be deflected by the hyperbola towards the other focus. This is similar to the parabola in that it can take paths pointed at one place and redirect it to another. This is used in radio direction (between two towers) and is useful for mirrors in telescopes to reflect light towards the eyepiece.

7.3 Hyperbolic Geometry

Hyperbolas also have interesting connections to non-Euclidean geometry. A figure called the hyperbolic paraboloid was one of the first used to investigate non-Euclidean geometric properties. It also looks like a potato chip(see Figure 15).

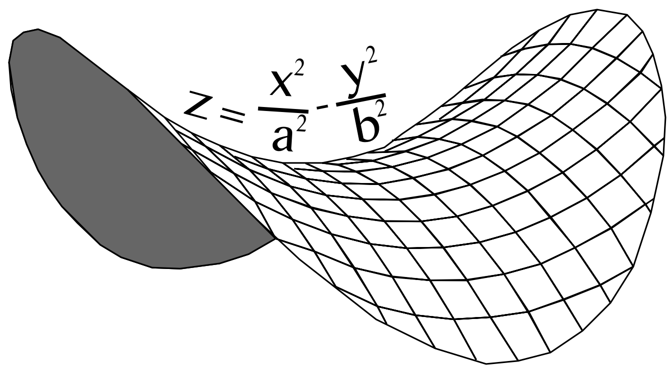


Figure 15: Hyperbolic paraboloid

Hyperbolic geometry is very useful in the studies of special and general relativity and descriptions of spacetime. It also has many other interesting practical uses, such as modelling human vision or in cameras.

8 Conclusion

Conic sections may just seem like some obscure topic in your math class, but they can essentially be looked at as the curves of nature. They can be used to describe the very fabric of our universe and for our own technological advances. Conics may be challenging to work with mathematically, but seeing how they are all connected through eccentricity and of all things, a cone, makes it a fascinating subject.

Bitwise Operators and its Nifty Applications

Jun Hong Wang

1 Introduction

Imagine these two scenarios: you are writing math-heavy code for a small, hardware limited computer, like a Raspberry Pi, or it's a very old computer with fewer optimizations. By using modern programming norms, we could write code that works on a modern computer, but these are older, smaller computers with fewer resources to work with. What can we do? How did programmers do this type of math back then?

2 Bitwise Operators

A bitwise operator is a normal operator, but instead of working with strings or base 10 numbers, they operate purely on bits and binary representations of strings and numbers. Although there aren't bitwise mathematical operators like addition or subtraction, these operations can be created by using bitwise logical operators (AND, OR, NOT, XOR) and logical shifts (<<, >>, exact representation varies by language). To understand how these operators work, we must convert base 10 numbers into their binary equivalents.

2.1 NOT

NOT takes one argument, and flips the bits. All bits set to 1 are set to 0, and all bits set to 0 are set to 1. Notice that this takes one argument, which makes NOT a unary operator. This isn't the same as boolean not, which is used for values like true and false.

In signed integers, where the leftmost bit represents the sign of the integer, a NOT will change the sign of the number. Keep in mind that for these examples, all of the integers are positive, and no bits represent the sign.

NOT 14 = 17

	2^4	2^3	2^2	2^1	2^0
14	0	1	1	1	0
17	1	0	0	0	1

2.2 AND

AND takes two arguments, and compares each bit using a logical and operator. Only if both bits are 1 will the result be 1.

14 AND 24 = 8

	2^4	2^3	2^2	2^1	2^0
14	0	1	1	1	0
24	1	1	0	0	0
8	0	1	0	0	0

2.3 OR

OR also takes two arguments, and compares each bit using a logical or operator. Only if both bits are 0 will the result be 0.

14 OR 24 = 30

	2^4	2^3	2^2	2^1	2^0
14	0	1	1	1	0
24	1	1	0	0	0
30	1	1	1	1	0

2.4 XOR

XOR takes two arguments, and is like OR, except 1 is only returned when only one of the two arguments has 1 in that bit.

14 XOR 24 = 22

	2^4	2^3	2^2	2^1	2^0
14	0	1	1	1	0
24	1	1	0	0	0
22	1	0	1	1	0

2.5 Left Shift

Left shift moves each bit to the left, and sets the new rightmost bit to a 0. The 2^3 bit becomes the 2^4 bit, and the new 2^0 bit is set to 0. It will take the form $a << b$, and the result should be $a * 2^b$.

$14 << 1 = 28$

	2^4	2^3	2^2	2^1	2^0
14	0	1	1	1	0
28	1	1	1	0	0

2.6 Right Shift

Right shift moves each bit to the left, and sets the new leftmost bit to a 0. The 2^4 bit becomes the 2^3 bit and so on, and the rightmost bit, the 2^0 bit, gets cut off.

$24 >> 1 = 12$

	2^4	2^3	2^2	2^1	2^0
14	0	1	1	1	0
22	0	1	1	0	0

3 Why They Are Important

Most programmers won't run into these operators unless they go very deep into programming or into certain niches, so why should the average programmer care? After all, the normal operators we've been taught work perfectly fine. However, on some legacy machines or resource limited computers, there is a benefit to using bitwise operators. On old computers, bitwise operators are faster, and also use less resources and power.

4 Uses

Uses: A normal implementation of an isEven function which takes an integer input and determines whether the input is even and would use %2 (modulus 2, or remainder after dividing by 2), but one interesting implementation uses logical shifts.

Pseudocode:
input n
$m = n >> 1$
$m = m << 1$
return $m == n$

This works because if a number is shifted to the right, the rightmost bit is removed, and when the number is shifted back to the left, a 0 gets added. In even numbers, the rightmost bit is always a 0, so when the shifts happen, the end result is the same as the input, so the function will return true.

Another use is addition. The numbers below are in binary, not base 10.

$1 + 1 = 10$, the 2^0 bit of the output is the 2^0 bits XOR each other, while the 2^1 bit of the output is the 2^0 bit AND each other. The XOR part is the sum, while the AND is the carry.

Pseudocode:
a and b as input
while $b! = 0$
carry = a AND b
$a = a$ XOR b
$b =$ carry $<< 1$
return a

This might look like a bunch of nonsense, but this is roughly what the circuit for addition looks like, which is why historically, this was used instead of base 10 addition. Here, the sum is stored inside a, and a left shifted carry is stored in b. Ultimately, b will become 0, and the values of the other variables will not change after this, so a, the sum, is returned.

Here, $0b$ specifies that this is the binary representation of a number.

There are also other uses, like cryptography, or bitcoin, where maximum efficiency for calculations is required. Not only that, but by combining some bitwise operators, you can use them for security purposes.

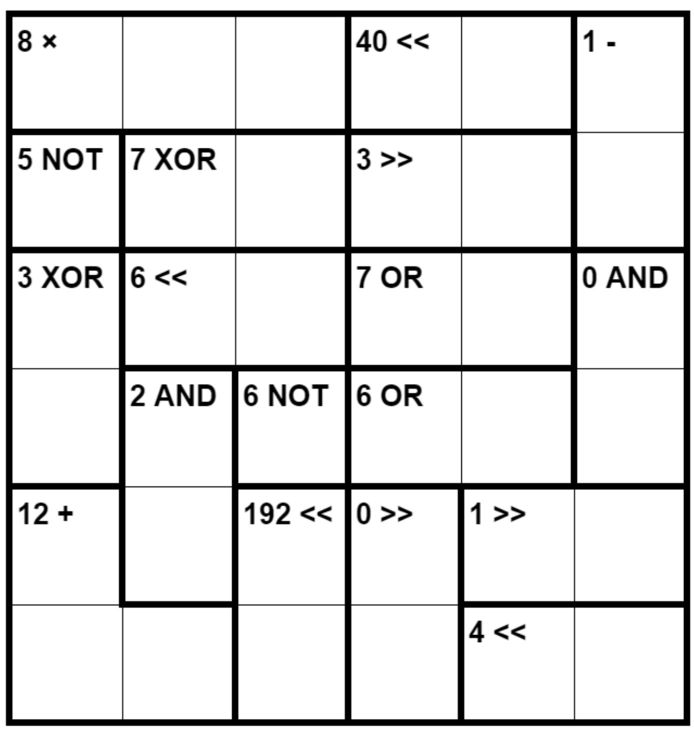
```
>>> bitwiseAdd(6,10)
 0b10 carry
 0b1100 a
 0b100 b
 0b100 carry
 0b1000 a
 0b1000 b
 0b1000 carry
 0b0 a
 0b10000 b
 0b0 carry
 0b10000 a
 0b0 b
16
```

5 Bitwise Kenken

In this kenken puzzle, the numbers in each square will be represented in binary. Each row and column must contain numbers from 1_{10} to 6_{10}, or 001_2 to 110_2 exactly once. Applying the operation on each of the bolded group of squares must result in the number (in base 10) to the left of the operation.

For example, look at *5 NOT* on the leftmost column. $5_{10} = 101_2$, and **010** NOT = 101.

For all numbers, assume that they are 3-bit numbers (only 3 binary digits). For bitwise operators, the inputs may be in any order.



5.1 Solution

References

"Bitwise Operators in Java." GeeksforGeeks. April 17, 2023.
https://www.geeksforgeeks.org/bitwise-operators-in-java/

Kodichath, Sreedev. "Real World Uses of Bitwise Operators." Tarka Labs Blog. September 3, 2020.
https://blog.tarkalabs.com/real-world-uses-of-bitwise-operators-c41429df507f

Narein, Adith. "Addition using Bitwise Operations". OpenGenus IQ.
https://iq.opengenus.org/addition-using-bitwise-operations/.

Numbers Are Based

Elijah White

1 Introduction

When we do math, we usually do not think about what number base we are using. Because we are taught in base-10 from a young age, we think of the decimal system as the default, even though this is not inherently true. The primary reason that we humans use base-10 is because we have 10 fingers. There are a vast number of different systems of number representation that have their own benefits and detriments. Of these, there are many types: positive and negative integer bases, non-real bases, and irrational bases are some examples.

1.1 How Number Bases Work

We use number bases to express the value of numbers. In positional number systems, we do this by writing strings of symbols called digits. Different digits have different place values. These place values depend on the base and are defined to be powers of the base.

The value of a number represented in a certain number base is found by summing the products of each digit and its respective place value. For example, 123.45 (base-10) is $1(10^2) + 2(10^1) + 3(10^0) + 4(10^{-1}) + 5(10^{-2})$.

The radix point is a dot that separates the integer place values from the fractional ones, e.g. the decimal point in base-10. For any base b, on the left of the radix point, we start with the $b^0 = 1$ place value. To the left of that is the $b^1 = b$ place value, then the b^2 place value, etc. To the right of the radix point, there is the b^{-1} place value, the b^{-2} place value, etc.

In addition to having place value, digits are restricted to certain positive integers, and for a reason. If we had a non-integer digit p in a place value b^k, we could instead represent the value pb^k as $\lfloor p \rfloor b^k + b(p - \lfloor p \rfloor)b^{k-1}$, and $0 < b(p - \lfloor p \rfloor)b^{k-1} < b^k$, i.e. we could have the digit in the place value b^k be $\lfloor p \rfloor$ and express the rest of the value in smaller place values. For positive bases, we do not need negative digits because, to express negative numbers, we instead use a negative sign that applies to all of the digits. For a natural number base b, digits are limited to being integers 0 to $b - 1$ (we will explain why later). For bases that exceed 10, we use other symbols, like letters, for digits. For example, base-16 uses 0 to 9 and a to f.

However, this is not the only way to represent numbers. Number systems that use bases as previously described are called positional number systems, and a digit is a type of numeral, which is any symbol used to represent a number. They are called positional because the position of a numeral determines its value. Most of the numeral systems that we use today, such as decimal, are positional number systems. Other types of number systems include additive, multiplicative, and ciphered numeral systems. They also can have bases, but the bases have a different significance than they do in positional systems.

1.2 A Brief History of Positional Number Systems

The first civilization to develop a positional number system, and just a number system in general, that we know of were the Sumerians. They used the sexagesimal, or base-60, system. They used symbols for 1 and 10 to construct digits 1 to 59. For example, 47 would be four of the 10 symbols and seven of the 1 symbols written together. They could then express numbers in base-60 by placing these digits horizontally from right to left in ascending order of place value, like we do in decimal. So, even though they used base-60, their system had elements of base-10 in regards to how they constructed digits. Also, at first, they did not have a symbol for 0 and instead left an empty space for any place value that did not contribute any value to the target number. Circa 300 B.C.E., however, the Babylonians, who adopted the Sumerians' system, invented a symbol for 0 as a placeholder to replace this empty space (but it was only a placeholder, they did not use the symbol for the actual number 0). Base-60 is still prevalent today: there are 60 seconds in a minute, 60 minutes in an hour, and $360 = 60 \cdot 6$ degrees in a full rotation.

The Mayans had a similar system to that of the Sumerians. They used a base of 20, and they had symbols for 5 and 1 with which they constructed their digits 1 through 19. They placed these from bottom to top in ascending order of place value. After the Babylonians, around 400 C.E., they too independently invented 0 as a placeholder for place values that were empty.

The Chinese numeral system, which was invented sometime in the 2nd millennium B.C.E., is a multiplicative one. This means that each numeral consists of two symbols, when denoting the place value, and one denoting the numerical value. For example, they would use the symbols for hundreds and the symbol for 4 together to represent 400. So, a multiplicative system is similar to a positional one, but each digit consists of two numerals instead of one.

Finally, the Hindu-Arabic numeral system, which evolved from Brahmi and Gwalior numerals, was implemented and developed by Indian mathematicians by the 7th century. Its numerals replaced Roman numerals in Europe around the 12th or 13th century and later Chinese numerals in China, and they are the ones we use today.

1.3 Some Notation and Terminology

We denote what base we are using with a subscript that is the base in base-10; e.g. 57_{13} is in base-13_{10}. Assume that all numbers and arithmetic are in base-10 unless specified otherwise.

A number system (b, D) is that which uses base-b and the set D as digits. Like the base alone, the system used to express a number is denoted by a subscript: $a_{(b,D)}$ is a in base-b using digits D.

We denote the digit in the place value b^k of a number in base-b as $(d_k)_b$, or sometimes with another letter like c_k, where $k \in \mathbb{Z}$. So, we say that the value of a digit $(d_k)_b$ is $d_k b^k$ (we will always express the base, b, in base-10). A number a can be represented in base-b in base-b positional notation by $(d_s d_{s-1} \dots d_2 d_1 d_0 . d_{-1} d_{-2} \dots d_{t+1} d_t)_b$, where $a = \sum_{k=t}^{s}(d_k b^k)$. If a digit consists of multiple terms or characters, it will be enclosed by parentheses and possibly separated from other digits with a space for clarity.

Note that if an equation is true in any base, it is true in all bases, as changing the base only changes how we represent values, not the values themselves.

When doing arithmetic, we "carry" numbers over to certain place values and "borrow" from certain place values. Let us just quickly make the definitions of these processes clear and consistent. Let a be the result of an arithmetic operation in base-b with the set of digits $\{d_k\}$, and let b^h be the place value that we are currently working in. When we say that we carry a number n over p places, we increase d_{h+p} by n and decrease d_h by nb^p, which preserves the value of the number because we change its value by $nb^{h+p} - nb^p(b^h) = 0$. When we borrow n from p places away, we decrease d_{h+p} by n and increase d_h by nb^p, which works because we change the value of the number by $-nb^{h+p} + nb^p(b^h) = 0$.

We say that a complex number can be expressed in a number system if that number can be represented in the number system as either, for real bases, as the sum of a real and imaginary part with no other operations, or, for non-real bases, as one number without the use of any operations.

So, we can think of a number system (b, D) as "complete" if every number can be expressed in base-b with the set of digits D in exactly one "useful" unique way. We say "useful" here because by one unique way, we mean "one unique way" like it is in the decimal system, as every number technically has two unique representations in decimal, e.g. ten is both 10 and $9.\overline{9}$.

We can think of a system as "under-complete" if it lacks the ability to express at least one number in base-b with digits D. We can think of it as "over-complete" if it can express at least one number in base-b with digits D in more than one unique way.

Let D_b denote a set of digits in base-b such that the system (b, D_b) is as complete as possible, and/or just the standard set of digits that we use for base-b.

Let $[n]$ denote the set of nonnegative integers less than n.

2 Positive Integer Bases

For a positive integer base b, $D_b = [b]$, which is why base-10 uses digits 0 to 9.

In base-b, we need the digit $b - 1$ because, for example, without it, it would not be possible to express $b - 1$ as a single number. For example, we need 9 in base-10 because without 9, we could not express 9 in base-10 without the use of operations. We would inconveniently need to write 9 as $10 - 1$ or $8 + 1$ or something similar. So, the system $(b, [b - 1])$ would be under-complete. Similarly, we need the digits $b - 2$, $b - 3$, $\ldots$, and 0.

We do not need the digit b because, if we had $(d_k)_b = b$ for any k, the value of d_k would be $b(b^k) = b^{k+1}$. Therefore, $b = 10_b$, and instead of $d_k = b$, we can have $d_{k+1} = 1$. So, the system $(b, [b + 1])$ is over-complete.

Extending this, we do not any digit $(d_k)_b \geq b$, which we will put in the form $qb + r$ $(q, r \in \mathbb{N})$ where $r < b$, because then the value of d_k would be $(qb + r)b^k = qb^{k+1} + rb^k$ (we could also reason that $qb + r = (qr)_b$). So instead of $d_k = qb + r$, we could have $d_{k+1} = q$ and $d_k = r$. If $q \geq b$, then we could repeat this process with $d_{k+1} = q$.

Now, does D_b *have* to be $[b]$? Not necessarily. We could instead have $D_b = \{[b] - n\}$ for an integer n where $0 < n < b$, which would include negative digits. For example, we could use $[10] - 4$ or the integers -4 to 5 (inclusive) for base-10. Since the value of $(d_k)_{(b,[b])}$ is

$$d_k b^k = (b + d_k - b)b^k = 1(b^{k+1}) + (d_k - b)b^k,$$

to express the value of $(d_k)_{(b,[b])}$ in $(b, [b]-n)$ where $d_k \geq b-n \notin [b]-n$, we can have $(d_{k+1})_{(b,[b]-n)} = 1$ and $(d_k)_{(b,[b]-n)} = (d_k)_{(b,[b])} - b < 0$. More simply put, $(d_k)_{(b,[b])} = (1\ (d_k - b))_{(b,[b]-n)}$. For example, if we wanted to write 76 in the system $(10, [10] - 4)$, i.e. using the digits -4 to 6:

$$76_{(10,[10])} = 1(10^2) - 2(10^1) - 4(10^0), \text{ which is } (1\ (-2)\ (-4))_{(10,[10]-4)}.$$

However, the standard is $D_b = [b]$ because nonnegative digits are simpler to work with.

To count in positive bases other than 10, instead of converting every number from base-10 to base-b, you can count through base-10 and only record numbers that only use the digits $0, 1, \ldots b-1$, or, better yet, think of the numbers that only use these digits in increasing order from 1. For example, to count from 1 in base-3:

$$1, 2, 10, 11, 12, 20, 21, 22, 100, 101, 102, 110, 111, 112, 120, 121 \ldots$$

Also, unary, or base-1, only uses one digit, 0. To make unary at least somewhwat usable, we can think of each "0" representing a 1 because 1 to any integer power is 1, and so unary is essentially a system of tally marks.

2.1 Conversion

The easiest way to convert a number a from base-b to base-10 is to sum the values of the digits of a using base-10 arithmetic, which is what we are used to:

$$a_b = (d_s \ldots d_2 d_1 d_0 . d_{-1} d_{-2} \ldots d_t)_b \Rightarrow a = \sum_{k=t}^{s} (d_k b^k)$$

We will discuss ways of converting from base-10 to other bases and between two non-decimal bases. There are several methods to convert a from base-10 into base-b, where $b \in \mathbb{N}$. One such method is the following process:

1. Find the largest power of b, b^N, such that $b^N \leq a$ (use base-10).
2. Let d_N (the digit in the place value b^N) be $\lfloor \frac{a}{b^N} \rfloor$, and let R_N be the remainder of $\frac{a}{b^N}$ (again, use base-10 for this arithmetic).
3. Let $d_{N-1} = \lfloor \frac{R_N}{b^{N-1}} \rfloor$, and let R_{N-1} be the remainder of $\frac{R_N}{b^{N-1}}$ (use base-10).
4. Repeat step 3 with d_{N-2} and R_{N-2}, d_{N-3} and R_{N-3}, etc. n times until

$$a - \sum_{k=N-n+1}^{N} (d_k b^k) = 0 \text{ , i.e. } R_{N-n+1} = 0.$$

5. If $N - n + 1 > 0$, which means that we have finished expressing all of a's value before reaching all the way to the place value b^0, concatenate $N - n$ zeros at the end of the result from the previous steps.

In other words, starting at the largest power of b less than or equal to a, we repeatedly find the largest multiple of a power of b, $d_k b^k$, less than or equal to the remaining amount of a, make d_k the digit for the b^k place value, and subtract $d_k b^k$ from a.

This power method starts by finding the digit in the largest place value. There is another method, though, that starts at the smallest place value. Let a be the number we are converting and we will start at the place value b^s (use base-10 for the arithmetic of every step):

1. Let d_s be the remainder of $\frac{a}{b^{s+1}}$, divided by b^s, and let $R_0 = a - d_s b^s$.
2. Let d_{s+1} be the remainder of $\frac{R_0}{b^{s+2}}$, divided by b^{s+1}, and let $R_1 = R_0 - d_{s+1} b^{s+1}$.
3. Let d_{s+2} be the remainder of $\frac{R_1}{b^{s+3}}$, divided by b^{s+2}, and let $R_2 = R_1 - d_{s+2} b^{s+2}$.
4. Continue this process with d_3 and R_3, d_4 and R_4, etc. n times until

$$a - \sum_{k=s}^{n-1} (d_k b^k) = 0 \text{ , i.e. } R_{n-1} = 0$$

Said in English, we are finding the remainder of the number divided by the next place value, which is the value that we need to express in the current place value (since it is smaller than the next place value and cannot be expressed in any larger place values), and then setting the digit equal to that divided by the current place value. Then, we subtract that digit's value from the number and do the same with the next place value. For example, if we wanted to convert 14 into base-3, we would first find the remainder of 14 divided by 3, which is 2, to find the value that we need to express in the ones digit because everything else can be expressed in larger place values, like those of 3 and 9.

This method only works well for real bases, though, as calculating remainders is different when dealing with imaginary numbers.

If a is an integer, and we do not already know the best place value to start at, it is usually the easiest to start with $s = 0$. However, if the number we are converting is not an integer, it might be difficult to find the place value to start at. If we start at the wrong place value, this method will not work because we are assuming that the values of all d_k for $k < s$ are 0.

We can use this to convert non-integer values by guessing and checking with different starting bases, but this might not work well if the radix representation of a number in the other base does not terminate. If this is the case, we can use division to convert it by writing it as a fraction, converting the numerator and denominator, and then performing division. To get an approximation in another base when the number is irrational, we can use a base-10 approximation for it and then convert that. So, if we wanted an approximation of π in base-7, we could convert 3.14. Note that since the representation of the number π does not use a positional number system (we just use the symbol π), π, and other numbers that we use special characters for, mean the same thing in all bases, kind of like 0.

As an example, say we want to convert 197 into base-12. Using the power method:

1. The greatest power of 12 that does not exceed 197 is $12^2 = 144$.
2. d_2 is $\lfloor \frac{197}{144} \rfloor$, which is 1.
3. d_1 is $\lfloor \frac{197-1(144)}{12} \rfloor = \lfloor \frac{53}{12} \rfloor$, which is 4.
4. d_0 is $\lfloor \frac{53-4(12)}{1} \rfloor$, which is 5.
5. So, $197 = 145_{12}$.

Using the remainder method:

1. d_0 is the remainder of $\frac{197}{12}$, which is 5.
2. d_1 is the remainder of $\frac{197-5}{144} = \frac{192}{144}$, which is 48, divided by 12, which is 4.
3. d_2 is the remainder of $\frac{192-4(12)}{12^3}$, which is 144, divided by 144, which is 1.
4. So, like before, $197 = 145_{12}$.

We can use a very similar method to convert directly from any base-c to any other base b. The only difference when converting from a non-decimal base c is that we have to do arithmetic in base-c, like how we did all arithmetic in base-10 when converting from base-10. We would also have to first convert the base b itself, which is given in base-10, into base-c. However, it is most likely faster to convert from base-c to base-10 and then from base-10 to base-b, as arithmetic in other bases takes much longer and is much more difficult for us.

Even so, there is a method of grouping for converting between bases that are powers of each other. To convert a number a from base-b to base-b^h where h is a positive integer, we can take the digits of a_b h at a time starting at the radix point to the left and right. We can then convert each of those groups of h digits into single digits in base-b^h and concatenate all the results in order of how the groups of digits were formed to get a_{b^h}, including the radix point. The reason why this works is a consequence of the fact that $(n-1)(n^0 + n^1 + n^2 + \cdots + n^{k-1} + n^k) = n^{k+1} - 1$. Rewriting that for this context:

$$(b-1)(b^0 + b^1 + b^2 + \cdots + b^{h-2} + b^{h-1}) = b^h - 1.$$

Since $(b-1)(b^0 + b^1 + b^2 + \cdots + b^{h-2} + b^{h-1})$ is the maximum value that the groups of digits can obtain (since $b-1$ is the largest digit possible in base-b), the range of possible values that the groups of digits produce is the same as the range of digits of b^h. For example, the maximum possible value of such a group of 3 digits in base-4 is $3(4^0 + 4^1 + 4^2) = 63 = 4^3 - 1$, which is equal to the maximum possible value of a digit in base-64.

So, as an example of this method of conversion, 12032.2_4 is equal to $6e.2_{64}$ because $12_4 = 6_{64}$, $032_4 = e_{64}$, and $2_4 = 2_{64}$.

To convert from base-b^h to base-b, reverse this process by converting each digit in base-b^h into a sequence of h digits in base-b.

2.2 Arithmetic

To do arithmetic in different bases, we can just convert the operands into base-10, do the arithmetic in base-10, and convert the result back into the original base. However, this is not particularly interesting, and we will discuss how to do arithmetic in different bases without converting to base-10.

Let the kth place value be the one that we are working in, and let m_k and n_k be the digits of the operands in the kth place value. Assume that m_k is a digit of the minuend and n_k is a digit of the subtrahend for subtraction.

When doing arithmetic in different bases, there are two main differences: when to and what to carry and borrow.

Addition is very similar for all positive integer bases. For addition in a positive integer base b, we carry a 1 over one place when $m_k + n_k \geq b$, since $b(b^k) = b^{k+1}$.

Subtraction is also very similar for all positive integer bases. We borrow b from one place away when $m_k - n_k \leq 0$. For example, $21_6 - 5_6 = 12_6$.

Multiplication is also consistent. When $m_k n_k \geq b$, we carry the bs digit of $m_k n_k$ over 1 place (like how we would carry the 10s digit of the product over 1 place). So, one-digit times one-digit multiplication works differently in every base. For example, $3_6 \times 4_6 = 20_6$.

The only difference in division is the process of determining how many of the divisor "go in to" the specified digits in the dividend.

Doing operations in non-decimal bases (especially multiplication and division) can be difficult since we do not have non-decimal multiplication tables memorized, so having some sort of multiplication table can help with these operations, like this one for base-12:

x	1	2	3	4	5	6	7	8	9	χ	ε	10
1	1	2	3	4	5	6	7	8	9	χ	ε	10
2	2	4	6	8	χ	10	12	14	16	18	1χ	20
3	3	6	9	10	13	16	19	20	23	26	29	30
4	4	8	10	14	18	20	24	28	30	34	38	40
5	5	χ	13	18	21	26	2ε	34	39	42	47	50
6	6	10	16	20	26	30	36	40	46	50	56	60
7	7	12	19	24	2ε	36	41	48	53	5χ	65	70
8	8	14	20	28	34	40	48	54	60	68	74	80
9	9	16	23	30	39	46	53	60	69	76	83	90
χ	χ	18	26	34	42	50	5χ	68	76	84	92	χ0
ε	ε	1χ	29	38	47	56	65	74	83	92	χ1	ε0
10	10	20	30	40	50	60	70	80	90	χ0	ε0	100

Figure 1: Duodecimal times table

where χ is the digit for 10 and ϵ is the digit for 11.

As mentioned previously, we can use division to convert non-integer values. For example, if we wanted to convert $\frac{1}{7}$ into base-12, we would convert the numerator and denominator into base-12, which gives us the same $\frac{1}{7}$, and then we would perform long division (assume all arithmetic and numbers are in base-12):

$$
\begin{array}{rll}
 & 0.\overline{186\chi} & \\
7 & \overline{1.0000} & \\
 & \underline{-7} & \text{7 goes into 10: 1 time} \\
 & 50 & 10 - 7 = 5 \\
 & \underline{-48} & \text{7 goes into 50: 8 times, and } 7 \times 8 = 48 \\
 & 40 & 50 - 48 = 4 \\
 & \underline{-36} & \text{7 goes into 40: 6 times, and } 7 \times 6 = 36 \\
 & 60 & 40 - 36 = 6 \\
 & \underline{-5\chi} & \text{7 goes into 60: } \chi \text{ times, and } 7 \times \chi = 5\chi \\
 & 1 & 60 - 5\chi = 1\text{, and we are now back to where we started}
\end{array}
$$

So, $\frac{1}{7} = 0.\overline{186\chi}_{12}$.

3 Negative Integer Bases

Similarly to positive bases, for a negative integer base $-b$ ($b \in \mathbb{Z}$ and $b > 0$), $D_b = [b]$.

We once again do not need a digit of b because if we had $(d_k)_b = b$, d_k has value

$$b(-b)^k = -(-b)(-b)^k = -(-b)^{k+1}.$$

This can be rewritten as

$$
\begin{aligned}
(-b)^{k+2} - (-b)^{k+2} - (-b)^{k+1} &= (-b)^{k+2} - (-b)(-b)^{k+1} - (-b)^{k+1} \\
&= (-b)^{k+2} + b(-b)^{k+1} - (-b)^{k+1} \\
&= (-b)^{k+2} + (b-1)(-b)^{k+1} \\
&= 1(-b)^{k+2} + (b-1)(-b)^{k+1} + 0b^k
\end{aligned}
$$

This means that $b = (1\ (b-1)\ 0)_{-b}$, and instead of expressing a number using b as d_k, we can have $d_{k+2} = 1$ and $d_{k+1} = b - 1$. For example, $10 = 190_{-10}$.

Extending this, the value of a digit $(d_k)_b = qb + r \geq b$ ($q, r \in \mathbb{N}, r < b$) would be

$$
\begin{aligned}
(qb + r)(-b)^k &= qb(-b)^k + r(-b)^k \\
&= -q(-b)^{k+1} + r(-b)^k \\
&= (-b)^{k+2} - (-b)^{k+2} - q(-b)^{k+1} + r(-b)^k \\
&= (-b)^{k+2} + b(-b)^{k+1} - q(-b)^{k+1} + r(-b)^k \\
&= 1(-b)^{k+2} + (b-q)(-b)^{k+1} + r(-b)^k
\end{aligned}
$$

So, $qb + r = (1\ (b-q)\ r)_b$, and we could have $d_{k+2} = 1$, $d_{k+1} = b - q$, and $d_k = r$, e.g. $34 = 174_{-10}$. But what if $b - q < 0$?

We do not need negative digits in negative bases because if $(d_k)_b = -n$ where $n \in \mathbb{N}$ and $n < b$, d_k has value

$$\begin{aligned}
-n(-b)^k &= (-b + b - n)(-b)^k \\
&= (-b)(-b)^k + (b - n)(-b)^k \\
&= 1(-b)^{k+1} + (b - n)(-b)^k
\end{aligned}$$

So, $-n = (1\ (b-n))_{-b}$, and instead of $d_k = -n$, we can have $d_{k+1} = 1$ and $d_k = b - n$ to express the same value, e.g. $-7 = 13_{-10}$. More generally, if $d_k = -(qb + r) \leq -b$, where $q, r \in Z, r < b$:

$$\begin{aligned}
-(qb + r)(-b)^k &= q(-b)^{k+1} - r(-b)^k \\
&= q(-b)^{k+1} + 1(-b)^{k+1} + (b - r)(-b)^k \\
&= (q + 1)^{k+1} + (b - r)(-b)^k
\end{aligned}$$

So, $d_k = -(qb + r)$ can be replaced with $((q + 1)\ (b - r))_{-b}$.

An interesting feature of negative bases is that they can represent negative numbers without using a negative sign, since they have place values that are negative. For example, $-13 = 27_{-10}$.

The counting process in negative bases is much different than that of positive bases. It is less straightforward. For example, in negadecimal, or base-(-10), counting from 1 is as follows:

$$1, 2, \ldots, 9, 190, 191, 192, \ldots, 199, 180, 181, \ldots$$

Negative bases also have what is called a *clearing algorithm*, which is something that helps us make arithmetic and conversion in negative bases easier. It allows us to temporarily ignore the restrictions on the digits that we can use. For a base $-b$, this is accomplished by adding b to any digit d_k and then adding 1 to d_{k+1}. This works because $1(-b)^{k+1} + b(-b)^k$ is always 0. So, we can also add and subtract multiples of 1 and b.

This means that we can temporarily have digits that are not in $D_{-b} = [b]$ and then "clear" them by cleverly adding and subtracting the polynomial $1(-b)^{k+1} + b(-b)^k$. We will discuss the clearing algorithm's application more in the next section.

We can also use this idea for positive integer bases, although it is not as useful because, per what we will later see, it would not make anything any easier: it would just change the order of steps for something like addition or multiplication. The clearing algorithm for a positive base b would be adding/subtracting $-b$ to d_k and then 1 to d_{k+1}.

3.1 Conversion

The power method that we discussed for positive bases does not work for negative bases because we cannot really determine the largest multiple of a power of the base that goes into a number when some of the place values are negative.

The remainder method, however, does not rely on the second base being positive since it uses remainders to find the digits. So it also works for negative bases, with one extra step:

5. If the base is negative, we may have gotten negative digits in the previous steps. If so, make them positive by borrowing a -1 from one place value away, that is increase the next digit by 1 and increase the current digit by b (base is $-b$).

This step is one of the uses of the clearing algorithm.

As an example that the remainder method works for negative bases too, let us convert 16 into base-(-3):

1. d_0 is the remainder of $\frac{16}{-3}$, which is 1.
2. d_1 is the remainder of $\frac{16-1}{(-3)^2} = \frac{15}{9}$, which is 6, divided by -3, which is -2.
3. d_2 is the remainder of $\frac{15-(-2)(-3)}{(-3)^3} = \frac{9}{-27}$, which is 9, divided by $(-3)^2 = 9$, which is 1.
4. So, we have $16 = 1(-2)1_{-3}$. But we do not want negative digits, so we can increase the 9s place value by 1 and increase the -3s place value by 3.
5. So, $16 = 211_{-3}$.

We can check this: $2(-3)^2 + 1(-3) + 1 = 18 - 3 + 1 = 16$.

However, instead of using remainders, we could also use only the clearing algorithm, which, for base-(-3), is adding multiples of 13_{-3} from a place value:

$$\begin{array}{rrrl} & & 16 & \text{is the number we are converting} \\ & -5 & -15 & = -5(1(-3)^1 + 3(-3)^0) = 0 \\ \underline{+2} & \underline{+6} & \underline{} & = 2(1(-3)^2 + 3(-3)^1) = 0 \\ 2 & 1 & 1 & \end{array}$$

Using only the clearing algorithm would take a long time for larger numbers, though.

To convert non-integer values, we would use division, as with positive bases.

3.2 Arithmetic

In addition for a negative integer base $-b$, when $m_k + n_k \geq b$, we carry a -1 over one place. This works because, as previously discussed, $b(-b)^k = -1(-b)^{k+1}$. For example, $(57 + 284)_{-10}$:

$$\begin{array}{r} {\scriptstyle -1\,-1} \\ 284 \\ \underline{+57} \\ 121 \end{array}$$

In subtraction, when $m_k - n_k = 0$, we borrow a -1 from 1 place away (we increase the digit in the next largest place by 1 and increase the current digit by b), because $1(-b)^{k+1} + b(-b)^k = 0$. However, subtraction is equivalent to adding the negative of a number. So, if we wanted to, instead of subtracting a number n_{-b} from m_{-b}, we could find the negative of n in base-$(-b)$ and then add m_{-b} and $-n_{-b}$.

In multiplication, when $m_k n_k = qb + r \geq b$ ($q, r \in \mathbb{N}$ and $q, r < b$), we carry a 1 over two places and a $b - q$ over one place. This is because, as seen previously, $qb + r = (1\ (b-q)\ r)_b$. This gives us a way to find the negative of a number in base-$(-b)$ without converting to base-b and then negating it and converting back. Instead, we can multiply by $(1\ (b-1))_{-b} = 1(-b) + (b-1) = -1$. For example if we wanted to find the negative of 152_{-10}, we could multiply by $19_{-10} = -1$:

$$\begin{array}{rl}
{\scriptstyle -4\,-1} & \\
152 & \\
\times\ \ 19 & \\
\hline
548 & = (152 \times 9)_{-10} \\
+\ \ 1520 & = (152 \times 10)_{-10} \\
\hline
0068 & \\
{\scriptstyle -1} &
\end{array}$$

So, $(-152 = 68)_{-10}$. Notice that when we move on to multiplying by the digit in the next biggest place value, we place a zero and then multiply by that digit, just like we do in positive number bases. This is because, no matter the base, multiplying by something that is the base times larger shifts the result to the left by one place.

Division is similar to that for non-decimal positive integers in that we go one digit at a time from the greatest place value finding how many of the divisor go into that part of the dividend and then subtracting to find the remainder. But it is different in that sometimes we will get negative values for a digit, which leads to "digits" that are more than one digit long that we have add together at the end, like the 18_{-10} in this example of $(427 \div 3)_{-10}$:

$$\begin{array}{rll}
 & 1 & \\
 & 189 & \\
3 & \overline{)427} & \\
 & \underline{-3} & \text{3 goes into 4: 1 time} \\
 & 12 & \\
 & \underline{-14} & \text{3 goes into } 12_{-10} = -8\text{: } -2 = 18_{-10} \text{ times} \\
 & 187 & \\
 & \underline{-187} & \text{3 goes into } 187_{-10} = 27\text{: 9 times} \\
 & 0 &
\end{array}$$

So, $(427 \div 3)_{-10} = 289_{-10}$. We can check this by doing $(3 \times 289)_{-10}$:

$$\begin{array}{r}
{\scriptstyle -2\,-2} \\
289 \\
\times\ \ 3 \\
\hline
427
\end{array}$$

There is one more thing. Instead of carrying a -1 over 1 place for addition like have done, there is an alternative of carrying a 1 over 2 places and a $b - q$ over 1 place, where q is the bs digit of the sum of digits in a column. But, this is really only useful when the sum in the next highest place is 0, since we do not want to add a -1 to it. Even in this case, though, we can use the borrowing that we did for subtraction of borrowing a 1 from 1 place away from the -1 to increase the -1 by

b, which ultimately leads to the same thing as what we would get by just carrying -1.

Also, if we do always use this method of carrying, it creates many roundabout scenarios. For example, the $(57 + 284)_{-10}$ that we did earlier:

$$\begin{array}{r}
\cdots \\
{\scriptstyle 1\,9} \\
{\scriptstyle 1\,9} \\
{\scriptstyle 1\,8} \\
{\scriptstyle 1\,9} \\
284 \\
\underline{+\ \ 57} \\
\ldots 00121
\end{array}$$

This carrying method can also create infinite loops of carrying, like with the example above and with $(1 + 19)_{-10}$:

$$\begin{array}{r}
\cdots \\
{\scriptstyle 1\,9} \\
{\scriptstyle 1\,9} \\
19 \\
\underline{+\ \ 1} \\
\ldots 000
\end{array}$$

4 Pure Imaginary Bases

Non-real bases have many interesting properties, such as their ability to represent all complex numbers in one part and with no sign.

For a pure imaginary base bi, instead of $D_{bi} = [b]$, $D_{bi} = [b^2]$. This is because every pure imaginary base bi has an effective negative base $(bi)^2 = -b^2$. The effective negative base determines the allowable digits: base-bi uses the same digits as base-$(-b^2)$.

All place values $(bi)^k$ where k is even are real. The effective negative base is the square of bi because all of the real place values will be powers of $(bi)^2 = -b^2$. This means that real part of any complex number is essentially expressed in base-$(-b^2)$. The only difference is that the real place values are dispersed among imaginary place values. So we need the same digits in base-bi as in base-$(-b^2)$ to be able to express all real numbers.

All place values $(bi)^k$ where k is odd are pure imaginary numbers. All such place values can be rewritten as $bi(bi)^{k-1} = bi(-b^2)^{\frac{k-1}{2}}$. So, all of the imaginary place values are the products of bi and an integer power of the effective negative base. So, to express any pure imaginary number yi in base-bi, we can first find what $\frac{yi}{bi} = \frac{y}{b}$, a real number, is in base-(bi) and then shift that to the left by one place value (essentially multiplying it by bi) to undo the division by bi. Therefore, we need the same digits as the effective negative base to be able to express all pure imaginary numbers. So,

$$\begin{aligned} a_{bi} &= (d_{2s}\ldots d_3 d_2 d_1 d_0.d_{-1}d_{-2}d_{-3}\ldots d_{2t})_{bi} \\ &= (d_{2s}\ldots d_2 d_0.d_{-2}\ldots d_{2t})_{-b^2} + bi\cdot(d_{2s-1}\ldots d_3 d_1.d_{-1}d_{-3}\ldots d_{2t+1})_{-b^2}. \end{aligned}$$

Together we need only the digits of the effective negative base to have a complete system.

One pure imaginary base is base-$2i$, or quater-imaginary. Its effective negative base is $-2^2 = -4$, and $D_{2i} = 0, 1, 2, 3$. Some of its place values are

$$\ldots (2i)^5 = 32i, (2i)^4 = 16, (2i)^3 = -8i, (2i)^2 = -4, (2i)^1 = 2i, (2i)^0 = 1\ldots$$

$1, -4, 16, \ldots$ are powers of -4 and therefore place values of base-(-4).
$2i, -8i, 32i, \ldots$ are products of $2i$ and powers of -4.

But, how would we represent any pure imaginary number hi with an odd coefficient h in base-$2i$ if all imaginary place values have even coefficients? If we could find a way to express i or $-i$ in base-$2i$, then we could express any such hi. Luckily, there is one: $0.2_{2i} = -i$, since

$$0.2_{2i} = 2(2i)^{-1} = \frac{2}{2i} = \frac{1}{i} = \frac{1\cdot i}{i\cdot i} = \frac{i}{-1} = -i.$$

So, for example, $i = 10.2_{2i}$ and $3i = 20.2_{2i}$.

Similarly, with any other pure imaginary base bi, d_{-1} can be used to express pure imaginary numbers that are not multiples of b.

As with any other bases, we can express fractional values. For example, $0.5 = \frac{i}{2i} = \frac{10.2_{2i}}{2i} = 1.02_{2i}$ (dividing by $2i$ shifts the numerator 10.2_{2i} one place value to the right).

Like negative bases, pure imaginary bases have a clearing algorithm. We can add/subtract multiples of $(1\ 0\ b^2)_{bi}$ since $1(bi)^2 + 0(bi) + b^2 = -b^2 + b^2 = 0$.

4.1 Conversion

To convert a number $x + yi$ from base-10 into an imaginary base bi:

1. Take the real part x and convert it into base-$(-b^2)$ using the remainder method or a clearing algorithm (or, if the numbers are small, we can use intuition on some level) to get $(\ldots d_2 d_1 d_0.d_{-1}d_{-2}\ldots)_{-b^2}$, which are the digits for the real place values.
2. Take the imaginary part y, divide it by 2 and convert that into base-$(-b^2)$ using the remainder method to get $(\ldots c_2 c_1 c_0.c_{-1}c_{-2}\ldots)_{-b^2}$, which are the digits for the imaginary place values.
3. We combine these to get $\ldots c_2 d_2 c_1 d_1 c_0 d_0.c_{-1}d_{-1}c_{-2}d_{-2}\ldots$, which is $(x + yi)_{bi}$.

For example, let us convert $19 - 7i$ into base-$2i$. The effective negative base is -4:

1. 19 in base-(-4) is $103_{-4} = 1(-4)^2 + 3(-4)^0 = 16 + 3$.
2. $\frac{-7}{2} = -3.5$ in base-(-4) is $11.2_{-4} = 1(-4)^1 + 1(-4)^0 + 2(-4)^{-1} = -4 + 1 + 2(-\frac{1}{4})$.
3. Put these together to get 11013.2_{2i}.

Just to check, this is $16 - 8i + 2i + 3 - i = 19 - 7i$.

4.2 Arithmetic

Arithmetic is very similar in pure imaginary bases as it is in negative bases because of its possession of an effective negative base. So, we can treat arithmetic in a pure imaginary base bi as two separate arithmetic processes in base-$(-b^2)$, one in real place values and one in imaginary place values.

So, for addition, we can carry a -1 over two places, doubling the number of places we carry over for negative bases so that we "skip" over the place values not similar to the current one in terms of being real or imaginary.

In subtraction, we borrow a -1 from 2 places away, or, as with negative bases, we can find the negative of the subtrahend in base-bi and then add that to the minuend.

In multiplication, when $m_k n_k = qb + r \geq b$, we carry a 1 over four places and a $b - q$ over two places.

Division by a real number in a pure imaginary base is similar to that in a negative base, but, with non-real bases, we must determine both how many of the divisor go into the real part of some part of a number and how many go into the imaginary part. For example, $(4 + 3i) \div 2 = 2 + i$ R 1. Here is an example of $(33 \div 2)_{2i}$:

$$
\begin{array}{rl}
1\,0.2 & \\
11.2 & \\
1 & \\
2\,\overline{)\,33.0} & \\
\underline{-2} & \text{2 goes into 3: 1 time, and } 2 \times 1 = 2 \\
13 & \\
\underline{-12} & \text{2 goes into } 13_{2i} = 3 + 2i\text{: } 11.2_{2i} = 1 + i \text{ times, and } (2 \times 11.2 = 12)_{2i} \\
10 & \\
\underline{-10} & \text{2 goes into } 10_{2i} = 2i\text{: } 10.2_{2i} = i \text{ times, and } (2 \times 10.2 = 10)_{2i} \\
0 &
\end{array}
$$

So, $(33 \div 2 = 22.22)_{2i}$. We can check this in base-10:

$$
(2 \times 22.22)_{2i} = 2 \times (2(2i) + 2(1) + 2(-i) + 2(-\tfrac{1}{2})) = 2 \times (3i + 1.5) = 6i + 3, \text{ and}
$$
$$
33_{2i} = 3(2i) + 3 = 6i + 3
$$

If we want to divide by a pure imaginary number, we can divide after multiplying both the divisor and dividend by bi to make the divisor a real number.

To divide by a complex number, we can use a strategy that we use in base-10 to divide by complex numbers. We can multiply both the divisor and dividend by the complex conjugate of the divisor so that the divisor becomes a real number. For example, to do

$$
(110023.2 \div 13)_{2i} = (19 - 9i) \div (3 + 2i),
$$

we can first find the complex conjugate of 13_{2i}:

$$
(\overline{13} = (-1)\ 3 = 1033)_{2i}
$$

Here we are negating the imaginary part of 13_{2i} (the 1 in the $2i$ place value) and then borrowing -1 from 2 places away to increase the digit -1 by 4, just like we would do for subtraction in base-$2i$. We can also think of this as using base-$2i$'s clearing algorithm.

Next, we multiply the divisor by its conjugate:

$$\begin{array}{r} {\scriptstyle -2-2} \\ 1033 \\ \underline{\times\quad 13} \\ 11211 \\ \underline{+\quad 10330} \\ 10101 \\ {\scriptstyle -1-1} \end{array}$$

To quickly check this in base-10, 13_{2i} times its conjugate is $(3 + 2i)(3 - 2i) = 9 + 4 = 13$, and $10101_{2i} = 16 - 4 + 1 = 13$.

We then multiply the dividend by the divisor's conjugate, which yields

$$(110023.2 \times 1033 = 10230303.2)_{2i}.$$

So, now instead of doing $(11003.2 \div 13)_{2i}$, we can do $(10230303.2 \div 10101)_{2i}$ (note that the work for the involved multiplications will not be shown, and assume all arithmetic and numbers are in base-$2i$):

$$\begin{array}{rl}
2 & \\
1 & \\
02 & \\
102 & \\
10101\,|\,\overline{10230303.2} & \\
\underline{-1030302} & \text{10101 goes into 102303: 10.2 times, and } 10101 \times 10.2 = 103030.2 \\
3232 & \\
\underline{-20202} & \text{10101 goes into 3232: 0.2 times, and } 10101 \times 0.2 = 2020.2 \\
12222 & \\
\underline{+\quad 3} & \text{adding the 3 from the ones place of the original dividend} \\
12121 & \\
\underline{-10101} & \text{10101 goes into 12121: 1 time} \\
20202 & \\
\underline{-20202} & \text{10101 goes into 20202: 2 times, and } 10101 \times 2 = 20202 \\
0 &
\end{array}$$

So, after all of that work, we have that the answer is $(10\ (2 + 0)\ (2 + 1)\ 2)_{2i} = 1023.2_{2i}$!

We can check if 1023.2_{2i} is the correct answer to $(11003.2 \div 13)_{2i}$ in base-10:

$$\begin{aligned}
1103.2_{2i} &= 19 - 9i \\
13_{2i} &= 3 + 2i \\
1023.2_{2i} &= 3 - 5i \\
\frac{19-9i}{3+2i} &= \frac{(19-9i)(3-2i)}{(3+2i)(3-2i)} = \frac{39-65i}{13} = 3 - 5i
\end{aligned}$$

One last thing: since we use a very similar method of carrying in pure imaginary bases as in negative bases, the same points discussed about the alternative method of carrying applies to pure imaginary bases: it is helpful in some situations but can lead to infinite loops of carrying.

5 Complex Bases

Complex bases can be extremely confusing to our decimal-oriented minds, even more so than negative and pure imaginary bases.

It has been proven that any complex base system with a base of the form $-n \pm i$ where $n \in \mathbb{N}$, and only complex base systems with bases of this form, can uniquely express every Gaussian integer using $D = [n^2 + 1]$ (Kátai and Szabó, 1979).

One such base is base-$(-1 + i)$, one of the simplest complex bases, which uses $D = [1^2 + 1] = [2] = \{0, 1\}$.

Like negative and pure imaginary bases, complex bases have clearing algorithms. The clearing algorithm of a base $-n \pm i$ is adding or subtracting multiples of $n^2 + 1$ from any digit d_k, then $2n$ from d_{k+1}, and 1 from d_{k+2}. This works because, after expanding and combining like terms, we can see that the polynomial

$$1(-n \pm i)^{k+2} + 2n(-n \pm i)^{k+1} + (n^2+1)(-n \pm i)^k = (-n \pm i)^k(1(-n \pm i)^2 + 2n(-n \pm i) + (n^2+1))$$

is always equal to 0, i.e.

$$(1\ (2n)\ (n^2+1))_{-n \pm i} = 0.$$

5.1 Conversion

Any Gaussian integer $s + ti$ can be cleverly rewritten as

$$t(-n+i) + (s+nt), \text{ or } (t\ (s+nt))_{-n+i}$$

for a base $-n + i$, or as

$$-t(-n-i) + (s-nt) = (-t\ (s-nt))_{-n-i}$$

for a base $-n - i$.

We can then convert this form into base-$(-n \pm i)$ by using its clearing algorithm to obtain only digits that are in $D_{-n \pm i}$. Here is an example of converting $5 + 6i$ into base-$(-3 + i)$, in which we use the fact that $5 + 6i = 6(-3 + i) + (5 + 3 \times 6) = (6\ (23))_{-3+i}$ and that the clearing algorithm for base-$(-3 + i)$ is adding multiples of $(1\ 6\ 10)_{-3+i}$:

$$\begin{array}{rrrrl} & & 6 & 23 & = 5+6i \\ & -2 & -12 & -20 & \\ +1 & +6 & +10 & & \\ \hline 1 & 4 & 4 & 3 & \end{array}$$

The is the result we want because $D_{-3+i} = [3^2] = [9]$, so $5 + 6i = 1443_{-3+i}$.

We can, of course, check this: $1443_{-3+i} = 1(-3+i)^3 + 4(-3+i)^2 + 4(-3+i) + 3 = (-18 + 26i) + 4(8 - 6i) + 4(-3+i) + 3 = -18 + 26i + 32 - 24i - 12 + 4i + 3 = 5 + 6i$.

There is no clear-cut way to count in complex bases without converting from base-10. Using the clearing algorithm (along with some addition), counting the integers in base-$(-1+i)$ starting at 1 is as the following:

$$\begin{array}{r} 1 \\ 1100 = 2 \\ 1101 \\ 111010000 = 4 \\ 111010001 \\ 111011100 = 6 \\ 111011101 \\ 111000000 = 8 \\ 111000001 \\ 111001100 = 10 \\ 111001101\ldots \end{array}$$

5.2 Arithmetic

If we have a base $-n \pm i$ and a digit $d_k = n^2 + 1 \notin D_{-n\pm i} = [n^2+1]$, we can replace it with $d_{k+3} = 1$, $d_{k+2} = 2n - 1$, $d_{k+1} = (n-1)^2$, and $d_k = 0$. This is because

$$n^2 + 1 = (1\ (2n-1)\ (n-1)^2\ 0)_{-n\pm i}.$$

We could verify this:

$$1(-n \pm i)^{k+3} + (2n-1)(-n \pm i)^{k+2} + (n-1)^2(-n \pm i)^{k+1} = (n^2+1)(n \pm i)^k$$

using algebra, but we will do so using the clearing algorithm for base-$(-n \pm i)$:

$$\begin{array}{rrrr} & & & n^2+1 \\ & -1 & -2n & -(n^2+1) \\ +1 & +2n & +(n^2+1) & \\ \hline 1 & 2n-1 & -2n+n^2+1 & 0 \end{array}$$

Therefore, $n^2 + 1 = (1\ (2n-1)\ (n-1)^2\ 0)_{-n\pm i}$, because $-2n + n^2 + 1 = (n-1)^2$.

So, for addition, when $m_k + n_k \geq n^2 + 1$, we carry a 1 over 3 places, a $2n - 1$ over 2 places, and a $(n-1)^2$ over 1 place. However, if the digits to the left are large enough, we can instead carry a -1 over 2 places and a $-2n$ over 1 place, leaving the current digit at 0. This is because $((-1)\ (-2n)\ 0)_{-n\pm i} = 0$ (which can be seen if we stopped one line earlier in our use of the clearing algorithm in the previous paragraph.

Similarly to pure imaginary and negative bases, if we only ever carry using $(1\ (2n-1)\ (n-1)^2\ 0)_{-n\pm i}$, and we never use $((-1)\ (-2n)\ 0)_{-n\pm i}$, there are situations in which we might encounter an infinite loop of carrying and/or borrowing values when doing arithmetic. For example, shown here is $(169 + 1)_{-3+i} = -1 + 1$ (Gilbert, 1984).

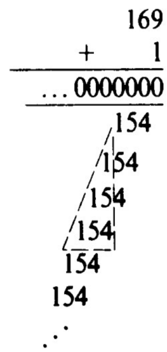


Figure 2: Infinite addition carrying loop

Here, since the first three columns sum to $10 = 3^2 + 1$, we repeatedly carry a 1 over 3 places, a $2(3) - 1 = 5$ over 2 places, and a $(3-1)^2 = 4$ over 1 place, which leads to an infinite string of zeros. Instead, we could carry a -1 over 2 places and a $-2(3) = -6$ over 1 place, which would immediately give us 0 as the answer.

For subtraction, when $m_k - n_k = -a < 0$ ($a \in \mathbb{N}$), we increase the digit two places higher by 1, increase the digit one place higher by $2n$, and the current digit by n^2+1, making it n^2+1-a. This is just using the clearing algorithm. If $2n > n^2$ (which is only for $n = 1$), we can do what we did for when a digit is $n^2 + 1$. If $n^2 + 1 - a < 0$, then we can repeat the process. Like with negative and pure imaginary bases, we can also avoid subtracting by multiplying the subtrahend by -1 and then adding the result to the minuend.

For multiplication, we can use a combination of polynomial multiplication and the clearing algorithm. The polynomial multiplication step emphasizes that the factors are polynomials in terms of the base $-n \pm i$, and so we multiply them how we would any other polynomials in x. Figure 3 shows the multiplication of 182_{-3+i} and 38_{-3+i} (Gilbert, 1984). The digits in this result are what we want because $D_{-3+i} = [9]$. So, $(182 \times 38 = 13546)_{-3+i}$.

We can approach division for complex bases very similarly to how we approached division for pure imaginary bases. We multiply the dividend and divisor to make sure that the divisor is real. Then, we divide by determining how many of the divisor go into both the real and imaginary parts of the digits as we work our way from the left to the right.

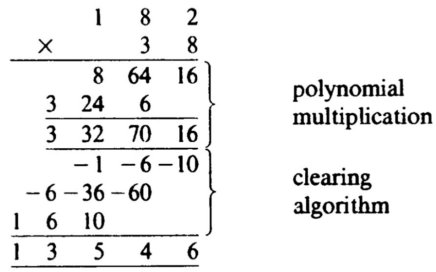


Figure 3: Complex base multiplication example

Here is the short example of $(133040 \div 3)_{-2+i}$ (assume all numbers without a subscript are in base-$(-2+i)$):

$$\begin{array}{rl} 1330 & = (1+2i)_{10} \\ 3 \,\overline{)\,133040} & = (3+6i)_{10} \\ \underline{-13304} & \text{3 goes into } 133040 = -3i_{10}\text{: } 133 = (-i)_{10} \text{ times, and } 3 \times 133 = 13304 \\ 0 & \end{array}$$

There are other ways of approaching division in complex bases. In one such approach, each complex base has a special set of remainders that it can have during division. This set of complex remainders also changes depending on the divisor and is difficult to find. If you wish to read more about this approach, I recommend reading William Gilbert's article (Gilbert, 1984, pp. 80-81).

6 Fractional and Irrational Bases

There are also real bases that are not integers.

6.1 Less Than One

The expression of any number in a base $\frac{1}{b}$ where $\frac{1}{b} < 1$ can be considered the reflection of that number's representation in base-b over the ones place value because $(\frac{1}{b})^n = b^{-n}$. So, any such base $\frac{1}{b}$ uses the same digits as base-b: $D_{\frac{1}{b}} = [b]$. For example, $18054.39_{10} = 934.5081_{\frac{1}{10}}$.

6.2 More Than One

For a base b where $b \notin \mathbb{Z}$ and $b > 1$, $D_b = [b]$. This is for very similar reasons for the limits on the digits of integer bases. If $(d_k)_b = \lceil b \rceil$, then the value of d_k is $\lceil b \rceil (b^k)$, which is more than b^{k+1} for all $b \notin \mathbb{Z}$. Therefore, some of this value could be expressed in the next largest place value with $d_{k+1} = 1$. This would decrease the value needed to be expressed in d_k from $\lceil b \rceil (b^k)$ to

$$\lceil b \rceil (b^k) - b^{k+1} = b^k(\lceil b \rceil - b) \leq b^k.$$

So, the maximum digit we would ever need is $\lfloor b \rfloor$.

There are irrational number bases, which can more easily express certain irrational numbers than rational bases. One of the simplest examples of this is base-$\sqrt{2}$, which uses the digits 0 and

1 because $\lfloor\sqrt{2}\rfloor = 1$. Like negative and pure imaginary bases, the place values alternate what kind of number they are. Every even place value is rational and every odd one is irrational. Hence, we could express every integer by using just the even place values, which would essentially just be using binary. As an example, $3 + \frac{\sqrt{2}}{2} = 101.1_{\sqrt{2}}$.

Another irrational base system is that uses the golden ratio as its base. Also sometimes called the *tau system*, or base-$(\frac{1+\sqrt{5}}{2})$, or base-ϕ, or base-τ, or maybe even phinary, it has some unique properties. This system uses the digits 0 and 1, as $\lfloor\phi\rfloor = 1$. Perhaps surprisingly, the system $(\phi, \{0,1\})$ is almost complete. Base-ϕ on its own has two unique representations for all integers. Since

$$\phi^n = \phi^{n-1} + \phi^{n-2},$$

the sequence 11_ϕ is equivalent to the sequence 100_ϕ, and so we will make the decision to always rewrite any sequence of digits 11 as a 100 for a number's standard representation.

The ability to rewrite two ones in a row as a single one in the next highest place value and vice versa is a very helpful tool when doing arithmetic in base-ϕ. For example, if we do $(101 + 11)_\phi$:

$$101_\phi + 11_\phi = 100_\phi + 1_\phi + 10_\phi + 1_\phi = 110_\phi + 1_\phi + 1_\phi = 1000_\phi + 1_\phi + 1_\phi$$

Now, we cannot add the two ones together to get a two or by carrying anything, so instead, we can rewrite one of them as 0.11_ϕ.

$$= 1000_\phi + 1_\phi + 0.11_\phi = 1001.11_\phi = 1010.01_\phi$$

So, $(101 + 11)_\phi = 1010.01_\phi$

If we list some of the powers of ϕ in terms of ϕ:

$$\begin{aligned}
\phi^3 &= 2\phi + 1 \\
\phi^2 &= \phi + 1 \\
\phi^1 &= \phi \\
\phi^0 &= 1 \\
\phi^{-1} &= \phi - 1 \\
\phi^{-2} &= -\phi + 2 \\
\phi^{-3} &= 2\phi - 3
\end{aligned}$$

It is easier to see how we would form integers.

To form a 2, we would need the ϕ terms of its digits to sum to 0 and constant terms to sum to 2. Luckily, $\phi^1 + \phi^{-2}$ does just this. So, $2 = 10.01_\phi$.

Similarly, $3 = \phi^2 + \phi^{-2} = 100.01_\phi$, $4 = \phi^2 + \phi^0 + \phi^{-2} = 101.01_\phi$, etc.

We can also use other irrational numbers, like π and e, as bases, but these systems would not be as complete or interesting as base-ϕ.

7 Aspects of a Good Positional System

First, let us establish that, for everyday use, positive integer bases are the best, as any other type of base would be more convoluted and difficult to deal with.

While there is no universal rating system for bases, there are two major factors that determine the usefulness of a number base. One is how large it is: we want number bases to be relatively small so that we do not have to use and interpret a plethora of different digits but also not too small so that we do not need too many digits to express individual numbers. The second is how many factors the base has.

7.1 Divisibility in Different Bases

Divisibility is very important because if a base b is divisible by an integer n, then $\frac{1}{n}$ can be easily expressed in base-b. For example, $7 \mid 14$ and $\frac{1}{7} = 0.2_{14}$, just like how $\frac{1}{5} = 0.2_{10}$. Also, it is much easier to tell divisibility by n in base-b than in other bases not divisible by n. For example, in base-8, any integer with $d_0 \in \{0, 4\}$ is divisible by 4.

Therefore, in general, we want bases to have as many factors as possible given its relative size. It is important that these factors are small and/or prime, since smaller prime numbers are more common in the factorizations of numbers than larger ones, and smaller factors prevent the base from becoming too large. So, a very important aspect of a base is its prime factorization. How well fractions are expressed in a base (we want radix expressions that are short and that terminate) is also important.

In any base b, there are useful divisibility rules for both $b+1$ and $b-1$, like those of 9 and 11 in base-10.

An integer a in base-b can be expressed with digits d_k as $d_n \dots d_2 d_1 d_0$ where n is a nonnegative integer. So,

$$\begin{aligned}
a &= d_n b^n + \cdots + d_2 b^2 + d_1 b + d_0 \\
&= (d_n + d_n(b^n - 1)) + \cdots + (d_2 + d_2(b^2 - 1)) + (d_1 + d_1(b-1)) + d_0 \\
&= (d_n(b^n - 1) + \cdots + d_2(b^2 - 1) + d_1(b-1)) + (d_n \cdots + d_2 + d_1 + d_0) \\
&= (d_n(b-1)(b^{n-1} + b^{n-2} + \cdots + b^2 + b + 1) + \cdots + d_2(b-1)(b+1) + d_1(b-1)) + (d_n \cdots + d_2 + d_1 + d_0) \\
&= (b-1)(d_n(b^{n-1} + b^{n-2} + \cdots + b^2 + b + 1) + \cdots + d_2(b+1) + d_1) + (d_n \cdots + d_2 + d_1 + d_0)
\end{aligned}$$

Therefore, a is divisible by $b-1$ if and only if $(d_n \cdots + d_2 + d_1 + d_0)$, or the sum of the digits, is divisible by $b-1$. This is the divisibility rule for 9 in base-10.

Now, for a divisibility rule for $b+1$:
Let n be an odd nonnegative integer.

$$\begin{aligned}
a &= d_n b^n + \cdots + d_2 b^2 + d_1 b + d_0 \\
&= d_n(b^n + 1) - d_n + \cdots + d_2(b^2 - 1) + d_2 + d_1(b+1) - d_1 + d_0 \\
&= (d_n(b^n + 1) + \cdots + d_2(b^2 - 1) + d_1(b+1)) + (-d_n + \cdots + d_2 - d_1 + d_0) \\
&= (d_n(b+1)(b^{n-1} - b^{n-2} + \cdots + b^2 - b + 1) + d_2(b+1)(b-1) + d_1(b+1)) + (-d_n + \cdots + d_2 - d_1 + d_0) \\
&= (b+1)(d_n(b^{n-1} - b^{n-2} + \cdots + b^2 - b + 1) + d_2(b-1) + d_1) + (-d_n + \cdots + d_2 - d_1 + d_0)
\end{aligned}$$

Therefore, a is divisible by $b + 1$ if and only if $(-d_n + \cdots + d_2 - d_1 + d_0)$ is divisible by $b + 1$. This is the divisibility rule that we have for 11 in base-10.

As a side note, these rules are switched for negative bases, that is, a is divisible by $b - 1$ in base-$(-b)$ if $(-d_n + \cdots + d_2 - d_1 + d_0)$ is divisible by $b - 1$, and a is divisible by $b + 1$ in base-$(-b)$ if $(d_n \cdots + d_2 + d_1 + d_0)$ is divisible by $b + 1$. For example, a number is divisible by 9 in negadecimal if the alternating sum of its digits is divisible by 9, and is divisible by 11 if the sum of its digit is divisible by 11. The reason for this is that every other place value in base-$(-b)$ is the additive inverse of the corresponding place value in base-b, so the alternating sum becomes the sum and the sum becomes the alternating sum.

A more general rule: if c is a factor of b, a number is divisible by c in base-b if d_0 is a multiple of c.

Like for 4 and 8 in base-10, there are useful divisibility rules in any base for all of the multiples of the base's factors.

Also, in any base, we can generate divisibility rules for primes that do not have a particularly nice relationship with the base, like that of 7 in base-10. We can also use the divisibility rules of two coprimes in combination for a way to tell divisibility by their product, like 6's rule in base-10 that uses divisibility by 2 and 3.

7.2 Advantages of Dozenal and Heximal

Although a societal shift to another number system would be very challenging and costly and not worth it, there are number bases, like base-12, also known as dozenal or duodecimal, and base-6, also know as heximal, seximal, or senary, that have advantages over our standardized base-10.

Although dozenal requires slightly more numerals than decimal (let χ be the digit for 10 and ϵ be the digit for 11 in dozenal), dozenal is considered the better of the two. This is thanks to the two-factor advantage of 12 over 10. Not counting themselves and 1, 10 has two factors and 12 has four.

However, heximal is also considered to better than base-10 because it requires four less numerals than base-10 but has the same number of factors as 10.

To compare these three bases, we will compare the representations of fractions in each of them:

Of all of them, heximal appears to have the simplest representations of these eight fractions. However, whether heximal or duodecimal is considered better depends on what aspects of a base we are prioritizing.

8 The Uses of Some Bases

8.1 Binary

Binary is used to store and transmit information in digital circuits and computers because of its bipolar nature. It only uses two digits, and each place value can represent an on-and-off switch,

	Decimal	Duodecimal	Heximal
1/2	0.5	0.6	0.3
1/3	$0.\overline{3}$	0.4	0.2
1/4	0.25	0.3	0.13
1/5	0.2	$0.\overline{2497}$	$0.\overline{1}$
1/6	$0.1\overline{6}$	0.2	0.1
1/7	$0.\overline{142857}$	$0.\overline{186\chi}$	$0.\overline{05}$
1/8	0.125	0.16	0.043
1/9	$0.\overline{1}$	0.14	0.04

Figure 4: Comparing fractions in different bases

with 0 typically representing off and 1 representing on. 0 results in a low, or false, voltage, while 1 results in a high, or true, voltage. Arithmetic in base-2 is also very simple. When adding in the same place value, a $0 + 0$ results in a 0, a $0 + 1$ results in a 1, and a $1 + 1$ results in a 1 in the place to the left and a 0 in the current place.

8.2 Hexadecimal

We also use hexadecimal (base-16) in computer science because it is a good middle ground between binary and decimal. Since 16 is a power of 2, computers can easily convert binary numbers into hexadecimal, and hexadecimal is easier for humans to read and interpret compared to binary because it more closely resembles decimal. Number representations in hexadecimal are much shorter than those in binary (4 times shorter, since $16 = 2^4$), and, to people, binary can appear like a meaningless string of 1s and 0s.

8.3 Quater-Imaginary

Computers use complex numbers in many different ways, and the arithmetic of complex numbers is important to several of the functions of a computer. Because of this, quater-imaginary base (base-$2i$) has the potential to be a more efficient number system than binary for computers. Quater-imaginary can represent negative numbers without a sign. Furthermore, it can represent all complex numbers, not just real and pure imaginary numbers, as just one part, whereas real bases need both a real and imaginary part. So, arithmetic with complex numbers can be done more efficiently in quater-imaginary than in binary, especially in regards to multiplication. In real bases, multiplying two complex numbers $a + bi$ and $c + di$ where $a, b, c, d \neq 0$ requires four multiplications and three additions:

$$(a + bi)(c + di) = ac + a(di) + c(bi) + (bi)(di)$$

In base-$2i$, this process only requires one multiplication.

9 References

"A Brief History of Numbers: How 0-9 Were Invented." Casio Education, Aug 2018. https://education.casio.co.uk/blog/a-brief-history-of-numbers.

"Additive, Multiplicative, and Ciphered Systems of Numeration."
https://www.hcusd2.org/vimages/shared/vnews/stories/53bd4762d3b99/4.1%20%20Additive,
%20Multiplicative,%20and%20Ciphered%20Systems%20of%20Numeration%20-%20filled%20in.pdf

Bergman, George. "A Number System with an Irrational Base." Mathematics Magazine, Vol. 31, No. 2. December, 1957.
https://math.berkeley.edu/ gbergman/papers/base_tau.pdf

"Ancient Chinese Mathematics." https://www.math.uci.edu/ ndonalds/math184/china.pdf.

"Fractional and Non-Integer Number Bases." Brilliant, retrieved June 14, 2023.
https://brilliant.org/wiki/fractional-number-base/.

Gilbert, William J. "Arithmetic in Complex Bases." University of Waterloo, March 1984.
https://www.math.uwaterloo.ca/ wgilbert/Research/ArithCxBases.pdf.

Gomony, Manil Dev, and Souradip Sarkar. "Quater-Imaginary Base for Complex Number Arithmetic Circuits." Hardware Platforms Research: Nokia Bell Labs, 2018.
https://past.date-conference.com/proceedings-archive/2018/pdf/0029.pdf.

Grussenmeyer, Kristina., & Arikan, Beytullah. "History of Zero."
http://webspace.ship.edu/msrenault/400%20Presentations/Presentation%203%20Zero.pdf

Hanson, Marilee. "Roman Numerals" English History, May 10, 2022.
https://englishhistory.net/ romans/roman-numerals/.

Jacobson, Roni. "60: Behind Every Second, Millenniums of History."
The New York Times, July 8, 2013.
https://www.nytimes.com/2013/07/09/science/60-behind-every-second-millenniums-of-history.html#: :text=In%20Babylonian%20astronomy%2C%20a%20year,of%2060%20from
%20the%20hour.

Katai, I., & Szabo, J. "Canonical number systems for complex integers." December 28, 1974.
https://web.northeastern.edu/seigen/11Magic/Binary/Complex/Canonical%20Number%20Systems
%20for%20Complex%20Integers%20%20Katai%20%20Szabo.pdf

MacTutor, November 2000. https://mathshistory.st-andrews.ac.uk/HistTopics/Zero/.

O'Connor, J. J., & Robertson, E. F. "Babylonian Numerals." Maths History, December 2000.
https://mathshistory.st-andrews.ac.uk/HistTopics/Babylonian_numerals/.

O'Connor, J. J., & Robertson, E. F. "The Arabic numeral system." January, 2001.
https://mathshistory.st-andrews.ac.uk/HistTopics/Arabic_numerals/

"Phi Numeral System." Phi numeral system - OeisWiki, May 9, 2021.
https://oeis.org/wiki/Phi_numeral_system.

Piché, Daniel G. "Complex Bases, Number Systems and Their Application to Fractal Wavelet Image Coding." University of Waterloo, 2002. https://www.collectionscanada.gc.ca/obj/s4/f2/dsk3/OWTU/TC-OWTU-243.pdf

"Radix notation." nLab. https://ncatlab.org/nlab/show/radix+notation.

Savas, Niko. "Why do we use hexadecimal?" Medium, Sep 23, 2016. https://medium.com/@savas/why-do-we-use-hexadecimal-d6d80b56f026.

Storr, Wayne. "Binary Numbers and the Binary Number System." Basic Electronics Tutorials, August 8, 2022. https://www.electronics-tutorials.ws/binary/bin_1.html.

How Often Do Powers of 2 Start with a 1?

Daniel Potievsky

1 Introduction

Suppose one were to start listing out powers of two. $2^1 = 2, 2^2 = 4, 2^3 = 8, 2^4 = 16, 2^5 = 32, 2^6 = 64, 2^7 = 128, 2^8 = 256, 2^9 = 512, 2^{10} = 1024, ...$

After listing the 10 first cases, we notice that 3 of them begin with a 1. What can be said of this pattern? How often do we encounter powers of 2 that start with a 1? Or we could rephrase this problem: given any arbitrary power of 2, what is the probability that this number starts with 1?

Let's consider the first n powers of 2: 2^1, 2^2, ..., 2^n. Suppose among these n powers, there are a_n that begin with a 1 when written down. That would mean that the probability that an arbitrarily chosen number from 2^1, 2^2, ..., 2^n starts with a 1 is $\frac{a_n}{n}$. This probability is dependent upon n, or the number of terms that we are considering. Thus, the limit

$$p_1 = \lim_{n\to\infty} \frac{a_n}{n}$$

(if it exists) will be the probability that an arbitrary power of 2 starts with a 1.

Now we turn to calculating this limit. Notice that among the two-digit numbers, the only power of 2 that starts with 1 is 16, and among the three-digit numbers, the only power of 2 that stats with 1 is 128. The same turns out to be true for four-digit numbers as well: 1024 is the only power of 2 that starts with 1. A question arises: does this pattern hold for all k-digit numbers?

Consider the smallest k-digit power of 2. This number must start with a 1. We can show this is true by contradiction.

Supposing the smallest k-digit power of 2 didn't start with a 1, we would be able to divide that number by 2, reaching another k-digit power of 2 that has a smaller leading digit. We can descend this way, and we will always reach a power of 2 that starts with 1. And we can also see that another k-digit power of 2 with a leading digit of 1 will not exist: the next power of 2 will have a leading digit of 2 or 3, and will continue likewise until we reach the $(k+1)$-digit power of 2 that has a leading digit of 1.

From here, it follows that if 2^n is an m-digit number, then among the numbers $2^1, 2^2, ..., 2^n$ there will be exactly $m-1$ that have a leading digit of 1. This implies that $a_n = m-1$.
But the claim that 2^n is an is an m-digit number can be rewritten as

$$10^{m-1} \leq 2^n < 10^m \Rightarrow m-1 \leq n\log 2 < m.$$

Recalling that $a_n = m - 1$, we obtain that $a_n \leq n \log 2 < a_n + 1$. From the left inequality, we get $\frac{a_n}{n} \leq \log 2$, and from the right inequality, we get that $\frac{a_n}{n} > \log 2 - \frac{1}{n}$. Combining this information, we see that $\log 2 - \frac{1}{n} < \frac{a_n}{n} \leq \log 2$.

Utilizing the Squeeze Theorem, we conclude that

$$p_1 = \lim_{n\to\infty} \frac{a_n}{n} = \log 2 \approx 0.30103...$$

This remarkable result shows that about 30% of all powers of 2 have a leading digit of 1.

2 Exercises

1. Let p_i represent the probability that an arbitrarily chosen power of 2 has a leading digit of i. Prove the following: $p_2 + p_3 = p_1$, $p_4 + p_5 = p_2$, $p_6 + p_7 = p_3$, and $p_8 + p_9 = p_4$.

2. Show that $p_4 = 1 - 3\log 2 \approx 0.096...$

3 The Generalization

A natural question arises after this problem: what is the probability that a power of an arbitrary natural number k has a leading digit of q?

Suppose the number k^n has a leading digit of q: $q \cdot 10^m \leq k^n < (q+1) \cdot 10^m$, where m is a natural number. Dividing this inequality by $q \cdot 10^m$ and taking the base-10 logarithm of both sides, we get that $0 \leq n \log k - \log q - m < \log \frac{q+1}{q}$. Because the base-10 logarithm is an increasing function, we have that for the upper bound, $\log \frac{q+1}{q} = \log 1 + \frac{1}{q} \leq \log 2 < \log 10 = 1$. Thus, by the inequality above, $n \log k - \log q - m$ is bounded between 0 and 1. And since m is a natural number, it follows that the fractional part of $n \log k - \log q$ is less than $\log \frac{q+1}{q}$.

A reminder that the fractional part of a number x is denoted $\{x\}$, and is defined as $x - \lfloor x \rfloor$. Conversely, if the fractional part of $n \log k - \log q$ is less than $\log \frac{q+1}{q}$, then k^n has a leading digit of n.

By showing this biconditional, we have illustrated the original problem is equivalent to the following: What is the probability that for an arbitrarily chosen number n, the inequality $\{n \log k - \log q\} < \log \frac{q+1}{q}$ holds. To solve this problem, we prove the following theorem.

3.1 Theorem on Fractional Parts

Let α, β be real numbers, where α is irrational, and I is some region length h contained within the interval [0, 1]. Consider the sequence $\alpha + \beta, 2\alpha + \beta, ..., n\alpha + \beta, ...$ Then the probability that an arbitrary term of this sequence has a fractional part that belongs to to I is h.

3.2 Proof

Let r be a natural number. The points $\frac{1}{r}, \frac{2}{r}, ..., \frac{r-1}{r}$ divide the interval [0, 1] into r segments, each of length $\frac{1}{r}$. We will now apply the combinatorial Pigeonhole Principle on these r subsections. Consider the numbers $\{\alpha\}$, $\{2\alpha\}$, ... , $\{(r+1)\alpha\}$. They all fall into the interval [0, 1]. But since there are $r + 1$ pigeons are r holes in this scenario, then there will at least exist two pigeons that fall into the same hole. That is, there will exist some p' and $p''(p' > p''$ without a loss of generality) such that the fractional parts of $p'\alpha$ and $p''\alpha$ differ by less than $\frac{1}{r}$. Thus, $|p'\alpha - p''\alpha - m| < \frac{1}{r}$ for some whole number m. Substituting $p = p' - p''$, we can say that $|p\alpha - m| = \varepsilon$, where $\varepsilon < \frac{1}{r}$. By the irrationality of α, we note that $\varepsilon \neq 0$.

Now, let s be the smallest natural number that exceeds $\frac{1}{\varepsilon}$, or $s - 1 \leq \frac{1}{\varepsilon} < s$. Examine s consecutive terms of the arithmetic sequence with a common difference of $p\alpha$: $\gamma + p\alpha, \gamma + 2p\alpha, ..., \gamma + sp\alpha$, where γ is some real number. Denote the fractional parts of these numbers as $x_1, x_2, ..., x_s$. By the equality $|p\alpha - m| = \varepsilon$, the numbers $x_1, x_2, ..., x_s$ differ by ε. And by the inequality $\frac{1}{s} < \epsilon \leq \frac{1}{s-1}$, we can conclude that the distance between x_1 and x_s is less than ε.

It follows from here that that number of points that fall into the region I of length h is between the values $\frac{h}{\varepsilon} - 1$ and $\frac{h}{\varepsilon} + 2$, that is, between $sh - 2$ and $sh + 2$ (noting that $h < 1$).

Now take ps consecutive terms of an arithmetic sequence $b_1, b_2, ...b_{ps}$ with a common difference of α, and let is write this sequence out as a table.

$$\begin{bmatrix} b_1 & b_{p+1} & b_{2p+1} & \dots & b_{(s-1)p+1} \\ b_2 & b_{p+2} & b_{2p+2} & \dots & b_{(s-1)p+2} \\ b_3 & b_{p+3} & b_{2p+3} & \dots & b_{(s-1)p+3} \\ \dots & \dots & \dots & \dots & \dots \\ b_p & b_{2p} & b_{3p} & \dots & b_{sp} \end{bmatrix}$$

Every row of this table forms a sequence of the form $\gamma + p\alpha$, as described in the previous paragraph. Since the number of rows equals p, then the number of terms from $b_1, b_2, ...b_{ps}$ that fall into the interval I is between $p(sh - 2)$ and $p(sh + 2)$.

Finally, let n be a natural number that exceeds rps. Dividing n by ps with remainder, we get $n = lps + q$, where $l \geq r$, $0 \leq q < ps$.

It follows that from the numbers $\alpha + \beta, 2\alpha + \beta, ..., n\alpha + \beta$, one can take ps consecutive terms of the arithmetic progression with common difference α l times, and q numbers will remain.

Therefore, $lp(sh - 2) \leq a_n \leq lp(sh + 2) + q$, where a_n is the number of terms from $\alpha + \beta, 2\alpha + \beta, ..., n\alpha + \beta$, the fractional parts of which lie in I. This inequality can be written as $(n - q)h - 2lp \leq a_n \leq (n - q)h + 2lp + q$ from where (by the inequalities $0 \leq h < 1$ and $q < ps$), we get $nh - 2lp - ps \leq a_n \leq nh + 2lp + ps$, that is $|\frac{a_n}{n} - h| \leq \frac{2lp+ps}{n} \leq \frac{2lp+ps}{lps} = \frac{2}{s} + \frac{1}{l} < \frac{3}{r}$ (since $l \geq r$ and $\frac{1}{s} < \varepsilon < \frac{1}{r}$. Thus, there exists a natural number N, namely $N = rps$, such that for $n > N$, $|\frac{a_n}{n} - h| < \frac{3}{r}$. Since r was chosen arbitrarily and can be made arbitrarily large, it follows that

$$\lim_{n\to\infty} \frac{a_n}{n} = h$$

which concludes the proof.

3.3 Resolution to the General Problem

We now may finish the general question: what is the probability that an arbitrarily chosen power of the natural number k has a leading digit of q?

Previously, we reduced this problem to the probability that for an arbitrarily chosen number n, the inequality $\{n \log k - \log q\} < \log \frac{q+1}{q}$ holds.

Note that if k itself is any power of 10 (1, 10, 100, ...), then all of its powers have a leading digit of 1, and the problem is trivial. But if k is not a power of 10, then $\alpha = \log k$ is an irrational number (as an exercise, prove this!).

From the Theorem on Fractional Parts, the probability that an arbitrarily chosen number from the sequence $n \log l - \log q (n = 1, 2, 3, ...)$ has a fractional part that belongs to [0, $\log \frac{q+1}{q}$] equals $\log \frac{q+1}{q}$. And this is the general solution! The probability that an arbitrarily chosen power of the natural number k has a leading digit of q is $\log \frac{q+1}{q}$.

Note that, surprisingly, the answer is independent of the base of exponentiation, k.

4 Additional Applications of the Theorem

4.1 Problem 1

Let α be an irrational number. Prove that there exists a natural number n such that $\cos n\alpha\pi > 0.999$.

We note that the inequality $\cos n\alpha\pi > 0.999$ is equivalent to $2m\pi - \arccos 0.999 < n\alpha\pi < 2m\pi + \arccos 0.999$ for some whole number m. Thus, $m < \frac{\alpha}{2} n + \frac{1}{2\pi} \arccos 0.999 < m + \frac{1}{\pi} \arccos 0.999$. In other words, the inequality $\cos n\alpha\pi > 0.999$ holds true if and only if the fractional part $\{\frac{\alpha}{2} n + \frac{1}{2\pi} \arccos 0.999\}$ belongs to the interval [0, $\frac{1}{\pi} \arccos 0.999$]. By the theorem on fractional parts, an arbitrarily chosen power of n follows this condition with a probability of $\frac{1}{\pi} \arccos 0.999 \approx 0.014$. Therefore, for a sufficiently large N, approximately 1.4% of the numbers $1, 2, ..., N$ satisfy the conditions of the problem.

4.2 Exercise 2

Find the probability that any arbitrary power of the number k starts with the combination of numbers $q_1 q_2 ... q_r$?

4.3 Exercise 3.1

Prove that for an arbitrary power of the number k, the probability that the j-th number from the left is q is equal to

$$\sum_{i=10^{j}-2}^{10^{j-1}-1} \log(1 + \frac{1}{q + 10i})$$

4.4 Exercise 3.2

Show that

$$\lim_{j \to \infty} \sum_{i=10^{j}-2}^{10^{j-1}-1} \log(1 + \frac{1}{q + 10i}) = \frac{1}{10}$$

Hint: First show that $\ln(1+x) < x$ for $x > 0$.

Then use the following: $\log(1+\frac{1}{10i)} - \log(1+\frac{1}{10i+q}) < \log(1+\frac{q}{100i(i+1)}) = \frac{1}{\ln 10}\ln(1+\frac{q}{100i(i+1)}) < \frac{1}{\ln 10} \cdot (1+\frac{q}{100i(i+1)}) = \frac{q}{100\ln 10}(\frac{1}{i-1} - \frac{1}{i})$.

Fibonacci: Getting Rich Using Math

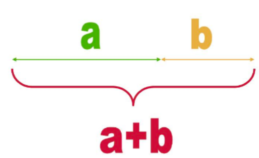
Danielle Rabiner

1 Introduction

Leonardo Bonacci, better known as Fibonacci, was an Italian-born mathematician. He is oftentimes regarded as one of the most influential and talented Western mathematicians of his time. Fibonacci traveled much of his youth, constantly interacting with merchants and traders. He met merchants from all along the Mediterranean coast, each with their own unique system of performing arithmetic. It was in Algeria where he discovered the Hindu-Arabic numeral system. In comparison to the Roman numerals he was used to, I II III IV, and so on, the Hindu-Arabic system was far more efficient for computations. The Hindu-Arabic numeral system is one that we use to this day, with 10 digits, 0 through 9. Fibonacci's love for numbers, combined with knowledge learned from the merchants he met in his travels inspired him to write *Liber Abaci*. *Liber Abaci* is widely recognized for its broadcasting of Hindu-Arabic numerals to western numerals. In the mathematical book, he discusses computations with this new numerical system, and proposes solutions to mathematical puzzles. This paper will discuss Fibonacci's influence on the Western world through his work in *Liber Abaci*, the connection between the Fibonacci sequence and the Golden Ratio, as well as real-world applications of the Fibonacci sequence.

2 Defining the Golden Ratio

If two quantities are in a Golden Ratio then their ratio is equal to the ratio of the larger of them to the sum. Let's take a look at the image below:



if segments a and b were to be in a Golden Ratio, $\phi = \frac{a}{b} = \frac{a+b}{a}$ (1). Let's see if we can represent (1) in a different form: $\frac{a}{b} = \frac{a}{a} + \frac{b}{a}$ (2). On the left side of (2) we have the ratio of $\frac{a}{b}$, which we have denoted ϕ. On the right side we have $\frac{b}{a}$, which is equal to the inverse of $\frac{a}{b}$, $\frac{a}{b}^{-1} = \phi^{-1} = \frac{1}{\phi}$. Lets plug this new found information into (2): $\phi = 1 + \frac{1}{\phi}$ (3). Now, let's try and solve (3). We can transform (3) into a quadratic equation. Multiply (3) by ϕ to get rid of it in the denominator under the condition that $\phi \neq 0$,

$$\phi = 1 + \frac{1}{\phi}$$
$$\phi^2 = \phi + 1$$
$$\phi^2 - \phi - 1 = 0$$

Let's call $\phi^2 - \phi - 1 = 0$ (4). We can now use the quadratic equation to solve (4). Using the quadratic formula,

$$\phi_{1,2} = \frac{1 \pm \sqrt{(-1)^2 - 4(1)(-1)}}{2(1)}$$
$$\phi_{1,2} = \frac{1 \pm \sqrt{1+4}}{2}$$

where $\phi_1 = \frac{1+\sqrt{5}}{2}$ (6) and $\phi_2 = \frac{1-\sqrt{5}}{2}$ (7). ϕ_1 is noted as the Golden Ratio and is ≈ 1.618. Evidently ϕ_2 is less than zero.

The Golden Ratio can also be found through a continued fraction. Let's consider the following continued fraction,

$$d = 1 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{1 + \cdots}}}$$

Let's label this (8) and take a look at the first few convergents of d:

$$d_0 = 1$$
$$d_2 = 1 + \frac{1}{1+1} = 1.5$$
$$d_3 = 1 + \frac{1}{1 + \frac{1}{1+1}} = 1 + \frac{1}{d_2} = 1.66667$$
$$d_4 = 1 + \frac{1}{1 + \frac{1}{1+\frac{1}{1+1}}} = 1 + \frac{1}{d_4} = 1.625$$

Notice the denominator of the next term is the value of the previous term. Here are the values of the partial sums up until d_{14}:

Proof of Continued Convergence	
d_0	1
d_1	1.5
d_2	1.666666667
d_3	1.6
d_4	1.625
d_5	1.615384615
d_6	1.619047619
d_7	1.617647059
d_8	1.618181818
d_9	1.617977528
d_{10}	1.618055556
d_{11}	1.618025751
d_{12}	1.618037135
d_{13}	1.618032787
d_{14}	1.618034448

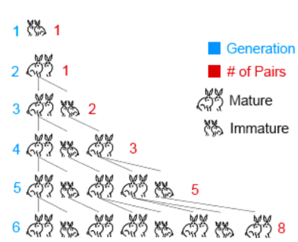
Around d_{10} is when you can see 1.618 emerge, or as we have previously discovered, the Golden Ratio, ϕ. The further we calculate and move toward d_∞ the evermore close we get to 1.618. In (7) notice since the fraction is infinite the lower part of the fraction is equivalent to the upper. Because of this, we can rewrite (7) as $d = 1 + \frac{1}{d}$. Which we can transform into a quadratic equation of the form $d^2 = d + 1$ or $d^2 - d - 1 = 0$ (9). By comparing (4) and (9), we have derived the quadratic equation which leads to the Golden Ratio in two different ways.

3 The Rabbit Island Problem

Your second section goes here. The real question is, how does any of this have relevance in our everyday life. The rabbit population may not be something that often crosses your mind, but this was something it was widely thought about by merchants in the 12th and 13th centuries. This brought about the problem of Rabbit Island. On pages 283 through 284 of *Liber Abaci*, Fibonacci ponders: In theory, if one started with a pair of rabbits, male and female, how many rabbits would they produce after one year? There were a few rules based on ideal circumstances to this puzzle:

1. No rabbit can die, or be consumed by a predator.
2. Each female rabbit must reproduce every month after her second month being alive.
3. Every time the rabbits reproduce, the female gives birth to a pair, one male and one female rabbit.

To help better visualize this problem, take a look at the diagram below:



For reference $f(n)$ represents the amount of pairs of rabbits at the beginning of any month, n. Additionally the function $g(n)$ marks the total number of rabbit pairs after n generations. One of the rules we must follow is that the rabbits must be mature, or above 2 months to mate. This means that in generation n, a rabbit born $n-2$ generations ago can produce offspring. We can model this as rabbits producing $= f(n-2)$. From this, the first time a new pair of rabbits will be produced is generation 3. Assuming the rabbits live indefinitely, there will always be $g(n-1)$ rabbits at any given time prior to a new generation.

We can combine the two expressions of finding independent aspects of the rabbit population into one equation that notates the entirety of the rabbit population.

In plain English, the total rabbit population is equal to the sum of the amount of rabbits able to produce with general rabbits. From this statement we can construct the general equation:

$$f(n) = f(n-1) + f(n-2)$$

Where $n > 2$ since the rabbits can only reproduce after month 2.

By now, you might recognize this equation as well as the sequence of rabbit pairs. The problem we just observed, the Rabbit Island, is what birthed the Fibonacci Sequence.

The Fibonacci sequence, to me, has always seemed fairly simple. If anything, it is something younger me thought I had created. Each term of the sequence can be noted as a sum of the previous two.

The sequence starts with the number $f(1) = 1$, the next term would be $f(2) = 1 + 0 = 1$, the next being $f(3) = 1 + 1 = 2$, the following term would be $f(4), f(4) = f(3) + f(2) = 2 + 1 = 3$. Do these numbers look familiar, hence the rabbit problem. We can generalize this as $f_n = f_{n-1} + f_{n-2}$ for $n > 1$.

Let's take a closer look at the ratio between two consecutive numbers in the Fibonacci sequence:

Term #	Fibonacci #	Ratio $\frac{f_{n+1}}{f_n}$
1	0	N/A
2	1	N/A
3	1	1
4	2	2
5	3	1.5
6	5	1.666666667
7	8	1.6
8	13	1.625
9	21	1.615384615
10	34	1.619047619
11	55	1.617647059
12	89	1.618181818
13	144	1.617977528

14	233	1.618055556
15	377	1.618025751
16	610	1.618037135
17	987	1.618032787
18	1597	1.618034448
19	2584	1.618033813
20	4181	1.618034056
21	6765	1.618033963
22	10946	1.618033999
23	17711	1.618033985
24	28657	1.61803399

It is evident that as the sequence progresses, the ratio approaches ϕ, the Golden Ratio. From this we can see that the ratio of two consecutive Fibonacci numbers is equal to the golden ratio:

$$\lim_{n\to\infty} \frac{f(n+1)}{f(n)} = \phi$$

4 Proof Through Induction

Let's see if we can generalize Fibbonaci's nth term, $f(n)$, using the Golden Ratio. To better understand this, let's look at (4): $\phi^2 - \phi - 1 = 0$. We can rewrite this equation as $\phi^2 = \phi + 1$ (10). From this we can find expressions for all powers of by mathematical induction. Mathematical induction works through proving the base case, n=1, and then applying that to $n = k$, and proving it must hold for all future cases, $n = k + 1$:

$$\begin{aligned}
\phi &= \phi \\
\phi^2 &= \phi + 1 \\
\phi^3 = \phi(\phi^2) \to \phi^3 = \phi(\phi+1) \to \phi^3 = \phi^2 + \phi \to \phi^3 = (\phi+1) + \phi \to \phi^3 &= 2\phi + 1 \\
\phi^4 = \phi^3 \times \phi \to \phi^4 = (2\phi+1)\phi \to \phi^4 = 2\phi^2 + \phi \to \phi^4 = 2(\phi+1) + \phi \to \phi^4 &= 3\phi + 2 \\
\phi^5 = \phi^4 \times \phi \to \phi^5 = (3\phi+2)\phi \to \phi^5 = 3\phi^2 + 2\phi \to \phi^5 = 3(\phi+1) + 2\phi \to \phi^5 &= 5\phi + 3 \\
\phi^6 = \phi^5 \times \phi \to \phi^6 = (5\phi+3)\phi - > \phi^6 = 5\phi^2 + 3\phi \to \phi^6 = 5(\phi+1) + 3\phi \to \phi^6 &= 8\phi + 5
\end{aligned}$$

And so forth. Hence we get $n = f_n + f_{n-1}$ (11), where f_n and f_{n-1} are consecutive numbers of the Fibonacci sequence.

We know from equations $\phi_1 = \frac{1+\sqrt{5}}{2}$ and $\phi_2 = \frac{1-\sqrt{5}}{2}$, ϕ_1 and ϕ_2 are the roots of the equation $\phi^2 = \phi + 1$. That means that they must satisfy $\phi^n = f_n\phi +_{n-1}$ (11). This is because it was derived inductively. Ergo, we can substitute each root into equation (11) one at a time. Which gives us the following: $\phi_1^n = f_n\phi_1 + f_{n-1}$ for ϕ_1(12), and $\phi_2^n = f_n\phi_2 + f_{n-1}$ for ϕ_2(13). Let's subtract (13) from (12):

$$\phi_1^n - \phi_2^n = (f_n\phi_1 + f_{n-1}) - (f_n\phi_2 + f_{n-1}) \ (14)$$

Let's simplify (14):

$$\phi_1^n - \phi_2^n = f_n(\phi_1 - \phi_2) + f_{n-1} - f_{n-1}$$
$$\phi_1^n - \phi_2^n = f_n(\phi_1 - \phi_2) \ (15)$$

Let's solve (15) for f_n:

$$f_n = \frac{\phi_1^n - \phi_2^n}{(\phi_1 - \phi_2}$$

Now using the actual values discovered in (6) and (7), we get the final formula for the nth term of Fibonacci sequence in closed form:

$$f_n = \frac{(\frac{1+\sqrt{5}}{2})^n - (\frac{1-\sqrt{5}}{2})^n}{\sqrt{5}}$$

For example, let's say we wanted to find the 12th term of the Fibonacci sequence. We could use the formula above to do so:

$$f_1 2 = \frac{(\frac{1+\sqrt{5}}{2})^1 2 - (\frac{1-\sqrt{5}}{2})^1 2}{\sqrt{5}}$$
$$f_1 2 = 144$$

This is in fact true.

5 Applications in Finance

This core idea seems elementary, so why is it so important? Fibonacci's understanding of rabbit populations, the golden ratio, and other discoveries in *Liber Abaci*, can be applied to a variety of things, both real world and mathematical. But this is not the only interesting observation that can be made about the Fibonacci sequence. When I was younger, I would sometimes come with my dad to work. I would stare at the huge computer screens, filled with strange charts and numbers. The world of finance has always been something I was interested in, but have never quite understood. Fibonacci's ratio can be used in finance to help traders identify when to stop losses, and target prices for an amicable price.

As previously shown,

$$\lim_{n\to\infty} \frac{f(n+1)}{f(n)} = 1.618 = \phi$$

But this is not the only ratio that converges to a constant. For example,

$$\lim_{n\to\infty} \frac{f(n+2)}{f(n)} = 2.618$$

$$\lim_{n\to\infty} \frac{f(n+3)}{f(n)} = 4.236$$

$$\lim_{n\to\infty} \frac{f(n+4)}{f(n)} = 6.854$$

The inverses of these first four limits of the Fibonacci sequence are called retracement levels in finance. People who work with this analysis are called technical analysts. Let's work out an example to better understand how they are actually used.

To find the retracement levels we must take the inverses of the four limits as follows:

Retracemnts Levels for Stocks	
ratio $\frac{f_n}{f_{n+1}}$	0.6181818182
ratio $\frac{f_n}{f_{n+2}}$	0.3823529412
ratio $\frac{f_n}{f_{n+3}}$	0.2363636364
ratio $\frac{f_n}{f_{n+4}}$	0.1459016393

For the purpose of this paper, let's focus on IBM stock. I have gone ahead and created a graph of the stock, as well as marked the retracement levels accordingly, and circled any main focus points that I will discuss later.

To help understand the graph:

$r_0 = 113.26$, which is 0% retracement level or the lowest value of the stock price.

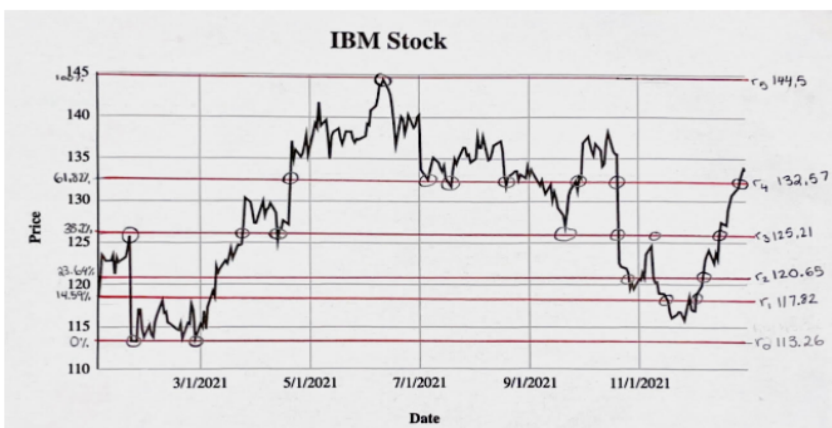
$r_5 = 144.5$, which is 100% retracement level or the highest value of the stock price.

$r_1 = r_0 + (r_5 - r_0) \times \frac{f_n}{f_{n+4}}$. By plugging in the actual numbers you get:

$r_1 = 113.26 + (144.5 - 113.26) \times 0.1459016393 = 117.82$

We can derive r_2 , r_3 , and r_4 this way. And they respectively are equal to:

$r_2 = 120.65$, $r_3 = 125.21$, $r_4 = 132.57$.



Let's consider the period of time between January 1, 2021 and May 1, 2021. As we can see the stock price in January tries to grow higher in value from r_1 , passing r_2, but finding resistance at level at r_3. Then between around February and March, it trades in the range of r_0 and r_1. What I find cool about this, is that it perfectly touches r_1 but finds resistance and never passes it. From around April to May, the stock price breaks r_3 and floats between r_3 and r_4 before eventually breaking r_4. After May 1, 2021, I notice the stock price precisely hit r_4 three times before breaking down. This would be a good time to sell. The main takeaway is: if it breaks up, buy. If it breaks down, sell.

Although the Fibonacci sequence was developed in the 13th century it is still extremely applicable and relevant today- this is what makes it so beautiful. Whether or not Fibonacci retracement works in relation to stock prices, or it is simply a psychological phenomenon, remains a curiosity of mine. The applications of Fibonacci are so much more complex than I could have ever imagined. And hey, maybe now I can predict some stock and become rich.

References

Bradley, Sean. “Generalized Fibonacci Sequences.” The Mathematics Teacher, vol. 93, no. 7, 2000, pp. 604–06. JSTOR, http://www.jstor.org/stable/27971512. Accessed 10 Jun. 2022.

Fabian Dablander. “The Fibonacci Sequence and Linear Algebra.” Fabian Dablander, 28 July 2019, https://fabiandablander.com/r/Fibonacci.html#: :text=The%20closed%2Dform%20expression %20of,%E2%88%9A52)n%5D.

fibandphi(AT)ronknott.com, Dr R Knott: “Why Do the Fibonacci Numbers Appear as the Number of Rabbits in the Field Each Month?” Why the Fibonacci Numbers Appear in the Rabbits Family Tree, https://r-knott.surrey.ac.uk/Fibonacci/rabserieswhy.html.

“Fibonacci Numbers.” Corporate Finance Institute, 27 July 2021, https://corporatefinanceinstitute.com/resources/knowledge/trading-investing/fibonacci-numbers/.

“Mathematical Treasure: Fibonacci’s Liber Abaci.” Mathematical Treasure: Fibonacci’s Liber Abaci — Mathematical Association of America, https://www.maa.org/press/periodicals/convergence/mathematical-treasure-fibonacci-s-liber-abaci.

Chaos Theory: Delightful Unpredictability

Shaon Anwar

1 Introduction

There is a scene in the movie Jurassic Park in which the fictional character Dr. Ian Malcolm attempts to explain the theory of chaos. He tells Dr. Settler that "it simply deals with predictability in complex systems. The shorthand is the butterfly effect, the butterfly flaps its wings in Central Park, you get rain in central Asia." Then, he illustrates further by dropping water droplets on her hand and saying that the flow of the water will change in unpredictable ways every distinct time "because tiny variations... the orientation of the tiny hairs on your hand, the blood cells and imperfections in the skin."

To a layperson, the concept of chaos conjures up images of randomness, changes, and unpredictability. In scientific circles, it means fluctuations in a deterministic system so sensitive to precise measurements that the output may appear completely random, yet with an exquisitely patterning structure. In this article, we will take a walk through this fascinating subfield of dynamic systems.

2 Advances in Chaos Theory and its Mathematics

Within the mathematical scene, progress is often made by intersecting parts of disciplines, revitalizing and building on precursors. Revolutionary mathematical ideas come from lots of hard work and collaboration with others. That's true of the evolution of chaos theory, too. The beginning of the story of chaos theory can be traced to the effort of finding a solution to the motion of the three-bodies, the problem set in a branch of physics following the gravitational relationships between celestial bodies. Newton who extensively studied the motion of the Earth around the sun was able to describe any system in which motion results from applied forces is a dynamical system, in the sense that its behavior evolves in time. Since this fundamental principle of dynamics establishes a relation between an acceleration and the applied forces, it is expressed by a differential equation which involves the second derivative of the position — the acceleration — and the position evolving under the influence of the forces on which it depends. Newton's solution to the problem of the motion of the Earth around the sun (law of gravitational attraction) naturally invited scientific attempts to extend Newtonian formulas to the sun, Earth and moon.

These efforts proved fruitless, since the three-body problem did not have a closed solution and small variations in orbital influences drastically changed movement. Ultimately, the seminal work of the French mathematician Henri Poincaré in the 1800s developed an insightful approach to the problem. The challenge lied in nonlinearity. Unlike linear systems that can be subdivided into smaller parts and combined back together (which allows for the simplification of complex problems), nonlinear systems do not fulfill the requirements of the superposition principle. The richness of behaviors that we encounter in our daily lives cannot be explained by a mathematical relationship of proportionality. For example, the spread of infectious diseases is heavily reliant on time and dependent on its initial variables, and thus in the long term, prediction is impossible. In the real world, it is difficult to find appropriate, predictable linear behavior for such natural processes. Poincaré saw that a system with even only three bodies would behave too unpredictably to be modeled using conventional methods. He opted for applying pictures and geometry to complex problems, thus introducing tools for dealing with dynamics. Thus, the mathematician's observations gave rise to the field of dynamic systems and offered a glimpse into the possibility of chaos.

The top floor of MIT's Green Building houses the Lorenz Center; it is named after Edward N. Lorenz who made a serendipitous discovery that subsequently altered the course of chaos theory and broke into popular culture. By the early 1960s, Lorenz had used a computer to construct a mathematical model of the weather, effectively combining the fields of mathematics and meteorology. His simulations used a set of differential equations which represented long-term changes in temperature, pressure, wind velocity, etc. Lorenz kept a continuous simulation and, as the story goes, one fateful day, he took a coffee break while it ran. Upon his return, he noticed that the weather patterns of the simulation changed drastically when compared to the previous ones. Lorenz initially thought there was an issue with the machine, but upon looking at the data, he realized that the differences stemmed from a simple rounding issue; Lorenz had inputted .506127 as .506. This small variation led to an output which did not uphold the previous results. The system was governed by deterministic laws: with finite systems of differential equations hence, with no randomness invading between one moment and the next. When the scientist narrowed the number of equations from twelve to three, Lorenz observed that the solutions kept oscillating, in an aperiodic state. This phenomenon, called sensitive dependence on initial conditions, rendered the solutions unpredictable. Lorenz illustrated this idea using an analogy of the mere flapping of butterfly's wings that can change the weather.

In his landmark work "Deterministic Nonperiodic Flow," Lorenz also showed that there was an underlying structure in chaos. While graphing data in three dimensions, he observed that plotting the trajectory of two nearby points would create ever-growing and widely separated regions, but iterating points off the regions would bring them closer. Such a complex system is called a "strange attractor," and dynamics discovered by Lorenz called the "Lorenz attractor."

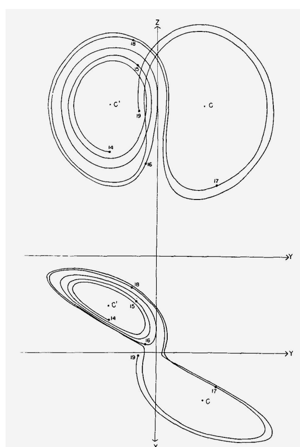


Fig. 2. Numerical solution of the convection equations. Projections on the X-Y-plane and the Y-Z-plane in phase space of the segment of the trajectory extending from iteration 1400 to iteration 1900. Numerals "14," "15," etc., denote positions at iterations 1400, 1500, etc. States of steady convection are denoted by C and C'.

Figure 1. Lorenz' original attractors

It bears mentioning here, that without the advent of computers, the discovery of the attractors would prove to be difficult; and without the work of two female mathematicians, the discovery of Lorenz's attractors would be perhaps impossible. Margaret Hamilton and Ellen Fetter, who programmed, plotted and supplemented Lorenz' research, helped reveal the mathematical signatures of this chaos (later, Hamilton's code became the structural foundation for many of NASA's future programs, including Skylab—the first American space station). Their contributions were indispensable in the field, and Lorenz acknowledged Fetter's help at the end of his original paper.

As a warm-up before taking on the function, let's examine a simple dynamic system that is not chaotic. In the given state s of the system at one moment in time, $f(s)$ is the state after one phase of time has passed. Following the behavior of a dynamic system, inputting the output back into the function will get us $f(f(s))$ which would be fed back in and so on. The dynamics of the system can be written as the following sequence: $s, f(s), f(f(s)), f(f(f(s)))\ldots$ As an example, let's consider the function $f(x) = 2x(1-x); 0 < x < 1$. Let's pick an arbitrary number for x from the given interval, and use 0.75. We get $f(0.75) = 0.375$, then $f(f(0.75)) = f(0.375) = 46875$, and then $f(f(f(0.75))) = f(0.46875) = 0.498047$, followed by $f(f(f(f(0.75)))) = f(0.498047) = 0.499992$. We see that with the initial state at 0.75, the system gets closer and closer to the state 0.5 with every step and passage of time (by the way, we can start with 0.25 for x, or any other value between 0 and 1 and the solution will still gravitate to 0.5).

Now, let's change the function to $g(x) = 4x(1-x)$ with x belonging to the interval 0 to 1. Then, $g(0.75) = 0.75, g(g(0.75)) = 0.75$, and continuing in this manner will reveal that all the states will remain fixed through all iterations. A dynamic state can also loop between two values, or some other finite set of points in any length of loop, or jump wildly without setting. More interestingly, two initial seed values of x within 0.000001 of one another will quickly form different trajectories, and the system will resemble nothing of the previous pattern. This manifests one of the defining features of chaos: sensitivity to initial conditions.

In the early 1960s, the American mathematician Stephen Smale discovered a simple and stable function that provided an explanation to the chaos of Poincare's three-body system. Smale's background in geometric analysis supplemented by his rigorous studies of literature on differential equations led to the hallmark of chaos: Smale's horseshoe function.

To further understand chaotic systems and Smale's horseshoes, let's subdivide our interval depending on whether the initial value is $x < 1/2$ or $x > 1/2$. After the orbit commences, let's note where the next iteration lands on. Subsequently, continue commencing orbits using the previous output as the new input, keeping track of which half of the horseshoe the output falls on. Each sequence of these recorded halves can be traced back to one specific initial input. Our function g, unlike our previous function f, does not express asymptotic behavior, and therefore maps to orbits of every random design. Now we are ready to expand our examples to Smale's horseshoe, used to plot functions with more dimensions. First, we can create a rectangle with vertices A, B, C, and D. We can then horizontally elongate and contract a new rectangle into the shape of a horseshoe.

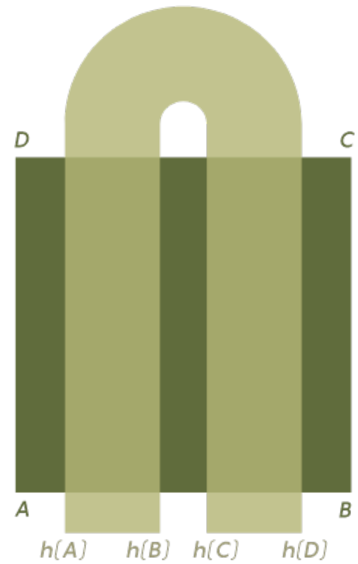


Figure 2. Smale's horseshoe mapped onto $ABCD$

Overlaying this horseshoe figure over rectangle $ABCD$ defines a function h which maps a point h into a point inside the horseshoe. Remapping these points by bending new horseshoes over and over demonstrates the phenomenon of chaos, where the outputs grow increasingly distinct and separate. Notably, the horseshoe is invertible, so mapping the inverse function of h yields a new, perpendicular horseshoe.

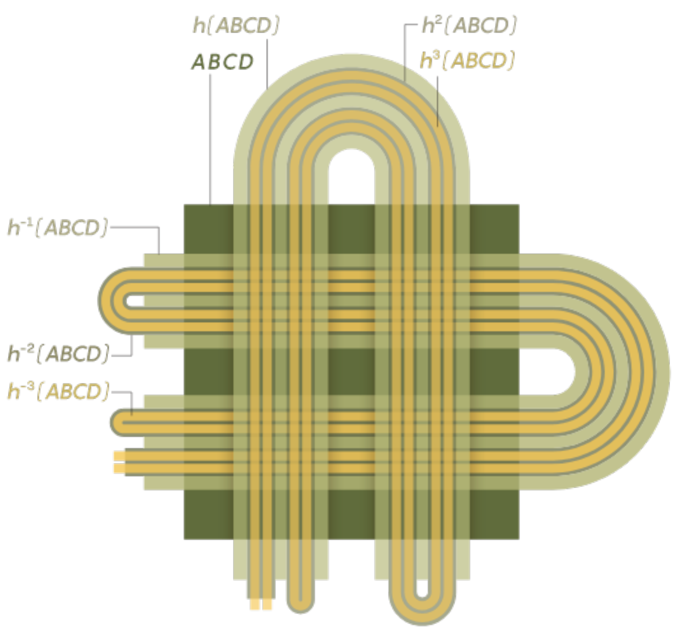
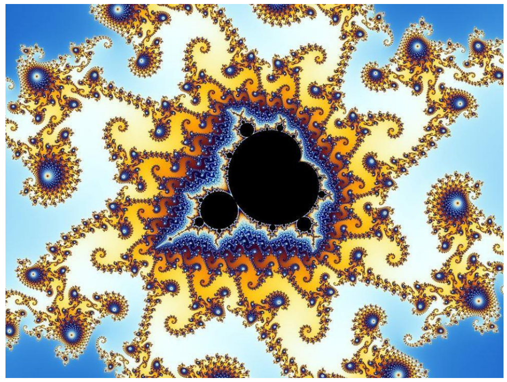


Figure 3. An example of points on a horseshoe being repeatedly mapped

The term *chaos* was introduced in a 1975 paper by James A. Yorke and his student T.Y. Li. The same year French-Polish mathematician Benoit Mandelbrot coined the term fractals as another class of "strange attractors," where smaller regions appear similar in structure to the whole thing.

(Both men received the 2003 Japan Prize in science and technology). Needless to say, fractals with their aesthetic appeal have added to popular fascination with nonlinear dynamics and chaos.



3 Conclusion

In the 1980s, movies such as *Jurassic Park* have helped to stimulate greater public interest in chaos theory. Although the fictional Dr. Malcom's efforts to explain the essence of chaos are rather clumsy, Professor Yorke of the University of Maryland in an interview remembers at least one graduate student in particular who "showed up from Australia because he liked the movie [Jurassic Park] and thought he should study chaos. And he wound up doing that." Perhaps this stretch into popular culture has lessened in the 21st century, but it does not mean that the influence of chaos theory has diminished. Nowadays, applications of the mathematics of chaos are deeply diverse and reach far beyond celestial mechanics. Studies on chaos dynamics extend to heart rhythms, laser physics, and the firing of neurons in our brain. On the philosophical level, chaos theory instructs us that we cannot predict what the future holds, but perhaps walking into chaos can be immensely enjoyable.

References

Cornell MAE. "MAE5790-1 Course introduction and overview." May 27, 2014, video, 1:16:30, https://www.youtube.com/watch?v=ycJEoqmQvwg&list=PLbN57C5Zdl6j_qJA-.

Dizikes, Peter. "When the Butterfly Effect Took Flight." Mit Technology Review, February 2011.

https://www.technologyreview.com/2011/02/22/196987/when-the-butterfly-effect-took-flight/.

Knill, Oliver. "Mathematics in Movies." Department of Mathematics, Harvard University, May

2023.
https://people.math.harvard.edu/ knill/mathmovies/.

Lorenz, Edward N. "Deterministic Nonperiodic Flow." Journal of the Atmospheric Sciences, March

1963.
https://journals.ametsoc.org/.

Richeson, David S. "How We Can Make Sense of Chaos." Quanta Magazine, March 2022.

https://www.quantamagazine.org/how-mathematicians-make-sense-of-chaos-%20%20%20%20
%20%2020220302/.
Sokol, Joshua. “The Hidden Heroines of Chaos.” Quanta Magazine, May 2019.

https://www.quantamagazine.org/the-hidden-heroines-of-chaos-20190520/.

Circles and Quadrilaterals

Patrick Was

1 Introduction

Squares, rectangles, trapezoids — all of these are the "special" quadrilaterals that we learn as elementary or middle school students. Most of them have distinct properties but there are sets of quadrilaterals that are considered more "specific" or "general" because they possess the properties of other quadrilaterals. Although this may seem obvious, this is also true for a more expansive range of quadrilaterals than the simple groups. This article contains some such quadrilaterals that have relations to circles, such as *cyclic quadrilaterals*, *tangential quadrilaterals*, *orthodiagonal quadrilaterals*, and the most specific of these, which contains a mixture of properties from each, *bicentric quadrilaterals*.

2 Cyclic Quadrilaterals

2.1 Definition

Definition (Cyclic Quadrilateral)**.** A convex quadrilateral circumscribed within a circle.

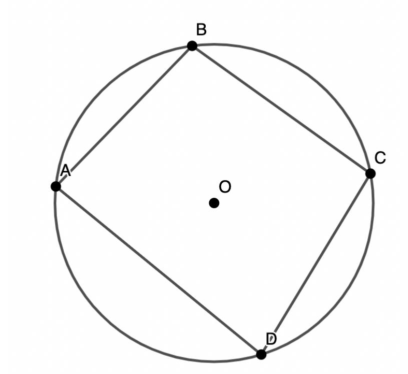


Figure 1: Cyclic quadrilateral

2.2 Properties

2.2.1 Supplementary Angles

In a cyclic quadrilateral, two opposite angles are supplementary.

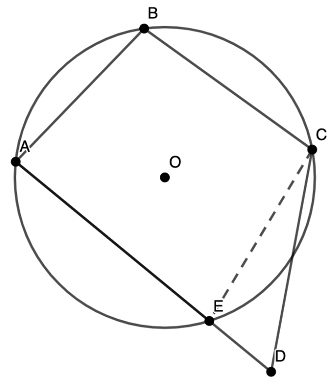


Figure 2: Supplementary angles

Consider quadrilateral $ABCD$. What we want to show is that the two opposite angles sum to $180°$. First, let the center of the circle be O. Let $\angle ABC$ be α and $\angle ADC$ be β. Constructing $\angle AOC$, we see that $\angle AOC = 2\alpha$ because the angle subtended by an arc of a circle at its center is twice the angle it subtends anywhere on the circle's circumference. Following the same reasoning,

$$\begin{aligned} 360° - \angle AOC &= 2\beta \\ \angle AOC + 360° - \angle AOC &= 360° \\ 2\alpha + 2\beta &= 360° \\ \alpha + \beta &= 180° \\ \angle ABC + \angle ADC &= 180° \end{aligned}$$

The other two angles must be supplementary for the same reasoning.

2.2.2 Ptolemy's Theorem

In a cyclic quadrilateral, the product of its sides is equal to the sum of the product of its two pairs of opposite sides. With cyclic quadrilateral $ABCD$, this means $AC \cdot BD = AB \cdot CD \cdot AD \cdot BC$.

Consider cyclic quadrilateral $ABCD$. AB is opposite to CD and AD is opposite to BC. Construct diagonals AC and BD. Because they subtend the same arcs, $\angle BAC = \angle BDC$ and $\angle ADB = \angle ACB$. Construct E on AC such that $\angle ABE = \angle CBD$. It follows that $\triangle ABE \sim \triangle DBC$ and $\triangle EBC \sim \triangle ABD$. Therefore, $\frac{AE}{AB} = \frac{DC}{DB}$ and $\frac{EC}{BC} = \frac{AD}{BD}$. This gives

$$AE \cdot DB = AB \cdot DC$$

and

$$EC \cdot BD = BC \cdot AD,$$

and because $AE + EC = AC$, we get $AC \cdot BD = AB \cdot CD + AD \cdot BC$.

2.2.3 Brahmagupta's Formula

The area S of a cyclic quadrilateral with sides a, b, c, d is given by $S = \sqrt{(s-a)(s-b)(s-c)(s-d)}$, where s is the semi-perimeter of the quadrilateral.

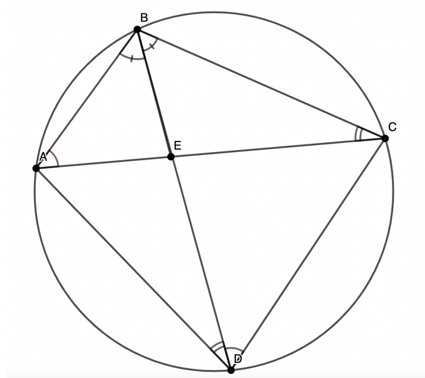


Figure 3: Ptolemy's Theorem

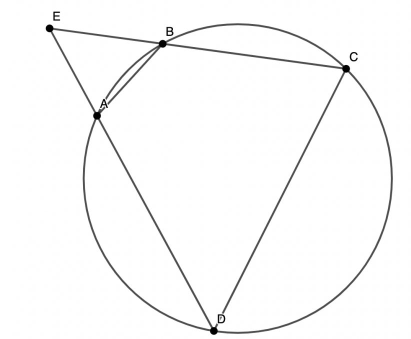


Figure 4: Brahmagupta's Formula

One can see that this is very similar to Heron's formula for a triangle. This observation will be used in the proof.

Consider quadrilateral $ABCD$, with $AB = a$, $BC = b$, $CD = c$, and $DA = d$. Extend the non-parallel opposite lines to a point E outside of the circumcircle. Denote $CE = e$ and $DE = f$. This proof assumes that these two sides are AD and BC. Heron's formula can now be used on $\triangle CDE$.

$$[CDE] = \sqrt{\frac{1}{16}(c+e+f)(-c+e+f)(c+e-f)(c-e+f)},$$

so

$$4[CDE] = \sqrt{(c+e+f)(-c+e+f)(c+e-f)(c-e+f)}.$$

We also have that $\angle BCD$ is supplementary to $\angle BAD$, which is supplementary to $\angle BAE$, meaning $\angle BCD \cong \angle BAE$. For similar reasoning, $\angle CDA \cong \angle ABE$. Therefore, $\triangle ABE \sim \triangle CDE$, meaning

$$\frac{[ABE]}{[CDE]} = \frac{a^2}{b^2},$$

and since $[ABE] = S = c^2 - a^2$,

$$\frac{S}{[CDE]} = \frac{c^2 - a^2}{a^2}.$$

Additionally, $\triangle ABE \sim \triangle CDE$, meaning that $\frac{e}{c} = \frac{f-d}{a}$ and $\frac{f}{c} = \frac{e-b}{a}$, which gives $e + f = c \cdot \frac{b+d}{c-a}$ and $e - f = c \cdot \frac{b-d}{c+a}$ also due to similar triangles. Substituting these into the original Heron's formula results in the four expressions:

$$2c \cdot \frac{s-a}{c-a}, 2c \cdot \frac{s-c}{c-a}, 2c \cdot \frac{s-d}{c+a}, 2c \cdot \frac{s-b}{c+a}.$$

Therefore, $[CDE] = \frac{c^2}{c^2-a^2}(\sqrt{(s-a)(s-b)(s-c)(s-d)}$, and

$$\begin{aligned} S &= [CDE] \cdot \frac{c^2-a^2}{a^2} \\ &= \frac{c^2-a^2}{a^2} \cdot \frac{c^2}{c^2-a^2} \cdot \sqrt{(s-a)(s-b)(s-c)(s-d)} \\ &= \sqrt{(s-a)(s-b)(s-c)(s-d)} \end{aligned}$$

3 Tangential Quadrilaterals

3.1 Definition

Definition (Tangential Quadrilateral)**.** A convex quadrilateral that contains an incircle that is tangent to all 4 sides of the quadrilateral.

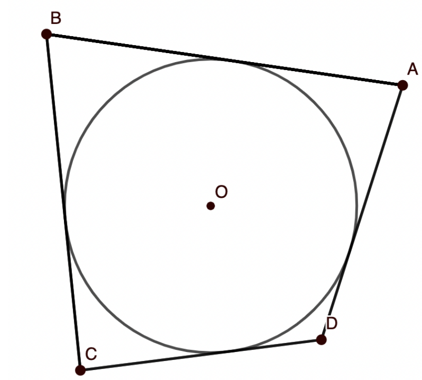


Figure 5: Tangential quadrilateral

3.2 Properties

3.2.1 Pitot's Theorem

The two sums of lengths of opposite sides are equal. In tangential quadrilateral $ABCD$, $AB + CD = AD + BC$.

The proof for this theorem is very quick. Since the sides are composed of tangent lines, let the points of tangency on segment AB be P, Q on BC, R on CD, and S on AD. It follows that $AP = AS$, $BP = BQ$, $DS = DR$, $CR = CQ$ Labeling AP as p, BP as q, DQ as r, and CR as s, From there, it can be seen that

$$\begin{aligned} AB &= AP + BP = p + q \\ CD &= CR + DR = r + s \\ BC &= BQ + CQ = q + s \\ AD &= AS + DS = p + r. \end{aligned}$$

Therefore, $p + q + r + s = q + s + p + r$, which implies that

$$AP + BP + CR + DR = BQ + CQ + AS + DS$$

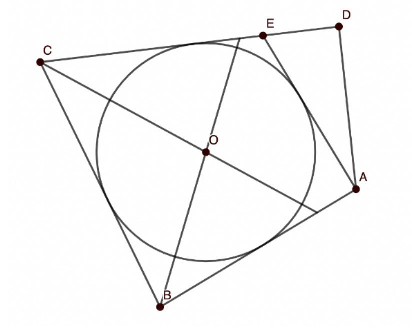


Figure 6: Pitot's Theorem

so $AB + CD = AD + BC$.

There is also a converse to this theorem: if there exists a convex quadrilateral $ABCD$ with $AB + CD = AD + BC$, then a semicircle tangent to all 4 sides can be inscribed within it.

Consider a convex quadrilateral $ABCD$. Instead of making a circle tangent to all 4 sides, which would be counterproductive to the purpose of the proof, construct a circle tangent to AB, BC, and CD using angle bisectors of $\angle B$ and $\angle C$. This can be done for the same reasoning as why an incircle can be constructed using angle bisectors. The angle bisectors will split the shape into triangles with equal length altitudes, which are the points of tangency.

The most standard proof for this is proof by contradiction.
If we assume that the circle is not tangent to AD, we can draw a tangent line through A to a point E lying on segment CD. By the previously proven Pitot's theorem, we have $AB + CE = AE + BC$. However, we also have $AB + CD = AD + BC$. Subtracting the two:

$$CD - CE = AD - AE.$$

The second is subtracted from the first to ensure that negative side lengths don't occur. We also have that

$$CD - CE = ED = AD - AE.$$

These are all part of $\triangle AED$, and because it is not true that the sides are greater than or equal to the sum of any other two, the triangle must be degenerate. Therefore, the original assumption was false, and it must be true that AD is tangent to $ABCD$. The circle is tangent to all 4 sides of $ABCD$.

It can also easily be seen from this that $AB + CD$ and $AD + BC$ are both equal to the semi-perimeter. $p + q + r + s = S$, the semiperimeter.

3.2.2 Area

Area can be expressed as $A = rs$, where r is the inradius and s is the semiperimeter. The reasoning behind this claim is nearly identical to that of a triangle with an incircle, especially since the areas can be expressed in the same way (but with a different semiperimeter value).

Consider O, the center of the incircle of quadrilateral $ABCD$. Segments can be drawn to the nearest vertices. This will result in 4 triangles: $\triangle AOD$, $\triangle BOA$, $\triangle COB$, and $\triangle DOC$. All of these have an altitude of r, meaning that their total areas combined can be expressed as $\frac{1}{2}ra + \frac{1}{2}rb + \frac{1}{2}rc + \frac{1}{2}rd$, where a, b, c, and d are sides of the quadrilateral. This can be rewritten as $\frac{1}{2}r(a + b + c + d)$, or rs, where s is the semiperimeter. Because the triangles partitioned the quadrilateral, we get $A = rs$

4 Orthodiagonal Quadrilaterals

4.1 Definition

Definition (Orthodiagonal Quadrilateral)**.** A convex quadrilateral whose diagonals intersect at right angles.

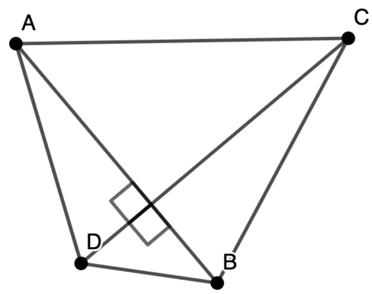


Figure 7: Orthodiagonal quadrilateral

4.2 Properties

4.2.1 Sum of Squares of Diagonals

A convex quadrilateral $ABCD$ is orthodiagonal iff $AB^2 + CD^2 = BC^2 + DA^2$.

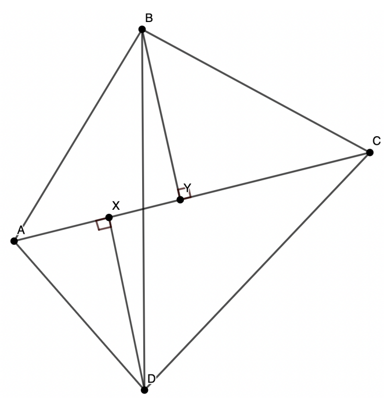


Figure 8: Property of orthodiagonal quadrilateral

Construct convex quadrilateral $ABCD$ and its diagonals, AC and BD. Drop altitudes from points D and B to AC, calling them x and y respectively. The figure can then be partitioned into 4 right triangles with the Pythagorean properties:

$$AY^2 + BY^2 = AB^2$$
$$BY^2 + CY^2 = BC^2$$
$$CX^2 + DX^2 = CD^2$$
$$AX^2 + DX^2 = DA^2$$

We can notice that some of these equations share terms, such as BY^2 and DX^2. Eliminating these from each other, we get

$$AB^2 - BC^2 + CD^2 - DA^2 = AY^2 - CY^2 + CX^2 - AX^2$$

The right hand side can be rearranged to:

$$\begin{aligned}
AY^2 - AX^2 + CX^2 - CY^2 &= (AY + AX)(AY - AX) + (CX + CY)(CX - CY) \\
&= (AY + AX)XY + (CX + CY)XY \\
&= (AX + CX + AY + CY)XY \\
&= 2(AC)XY.
\end{aligned}$$

We want the distance between x and y to be 0 so that the perpendiculars intersect at the intersection of the diagonals. Setting XY to 0:

$$AB^2 - BC^2 + CD^2 - DA^2 = 0,$$

so $AB^2 + CD^2 = BC^2 + DA^2$.

4.2.2 Sum of Angles of Diagonals

Another property is a convex quadrilateral $ABCD$ is orthodiagonal iff $\angle OAB + \angle OBA + \angle OCD + \angle ODC = 180^{\cdot}$, where O is the intersection of the diagonals.

The sum $\angle OAB + \angle OBA + \angle OCD + \angle ODC = 2\pi - 2\theta$, where θ is the angle between the diagonals (hence, the angles excluded from the theorem). Because we want theta to be equal to $90^{\cdot}$, $2\theta = 180^{\cdot}$, and therefore, $\angle OAB + \angle OBA + \angle OCD + \angle ODC = 180^{\cdot}$

There are a lot of properties to orthodiagonal quadrilaterals, but most of them revolve around similar reasoning to theorem 1 and are of the same form. The cirumradii, medians of triangles within the quadrilateral, and reciprocals of the altitudes are all related in the same proportion.

5 Bicentric Quadrilaterals

5.1 Definition

Definition (Bicentric Quadrilateral)**.** A convex quadrilateral that contains both a circumcircle and an incircle; satisfies both properties of cyclic and tangential quadrilaterals.

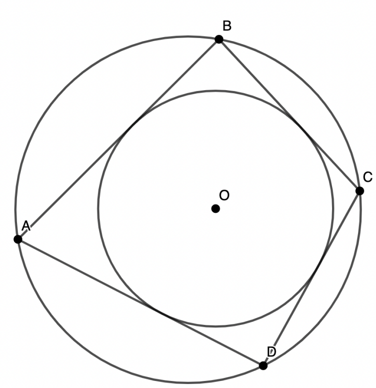


Figure 9: Bicentric quadrilateral

5.2 Properties

5.2.1 Area

A bicentric quadrilateral has area $\sqrt{abcd}$, where a, b, c, and d are its sides.

Consider a bicentric quadrilateral. Since, by definition, it is a quadrilateral that contains properties of both cyclic and tangential quadrilaterals, Brahmagupta's and Pitot's theorems can simply be algebraically combined to find the area of a bicentric quadrilateral. $S = \sqrt{(s-a)(s-b)(s-c)(s-d)}$, so

$$16S^2 = (-a+b+c+d)(a-b+c+d)(a+b-c+d)(a+b+c-d).$$

We have that $a + c = b + d$, so

$$\begin{aligned} a &= -c+b+d \\ b &= a+c-d \\ c &= -a+b+d \\ d &= a+c-b \end{aligned}$$

Plugging in a for -a, b for -b and so on,

$$16S^2 = (2a)(2b)(2c)(2d),$$

so $S = \sqrt{abcd}$.

6 Special quadrilaterals

Some of the standard groups of shapes are special cases of the ones listed above. *Squares*, because they have all of the properties that satisfy cyclic, tangential, and even orthodiagonal quadrilaterals, are bicentric. It can easily be seen that the area formula indeed does satisfy a square, as $A = s^2$ where $s =$ side and $A = \sqrt{s^4}$ are equivalent.

Given that all of the following kinds of quadrilaterals satisfy the property that their angles are supplementary, rectangles, isosceles trapezoids, and kites with two right angles are cyclic.

Although rhombuses aren't cyclic, they are tangential because of their paired sides. Kites are also tangential, orthodiagonal, and cyclic, making them bicentric.

7 Resources

"Brahmagupta's Formula and Theorem."
https://www.cut-the-knot.org/Generalization/Brahmagupta.shtml.

"Cyclic Quadrilateral." Byju's.
https://byjus.com/maths/cyclic-quadrilateral/.

Joseffson, Martin. "Characterizations of Bicentric Quadrilaterals." Forum Geometricorum.
https://forumgeom.fau.edu/FG2010volume10/FG201019.pdf.

Josefsson, Martin. "Characterizations or Orthodiagonal Quadrilaterals." Forum Geometricorum.
https://forumgeom.fau.edu/FG2012volume12/FG201202.pdf.

"Pitot's Theorem." Brilliant, retrieved June 14, 2023.

https://brilliant.org/wiki/pitots-theorem/.

"Ptolemy's Theorem." Brilliant, retrieved June 14, 2023. https://brilliant.org/wiki/ptolemys-theorem/.

Generative AI: Unleashing Creativity and Redefining Artistic Expression

Anastasia Lee

1 Introduction

Generative Artificial Intelligence (AI) has emerged as a transformative force in the world of creativity and art. With advancements in deep learning and neural network architectures, generative AI models like ChatGPT have garnered significant attention. In this overview, we delve into the technical intricacies of generative AI, exploring the capabilities of ChatGPT and its impact on the realm of AI art. From understanding the underlying technology to examining real-world applications, we unravel the potential of generative AI in fostering creative expression.

2 Understanding Generative AI

Generative AI refers to the branch of AI that focuses on creating new content, such as images, text, music, and more, that is not explicitly present in the training data. It utilizes deep learning techniques, primarily generative models, to produce coherent and contextually relevant outputs. These models learn patterns and structures from vast datasets and generate novel content by leveraging their learned knowledge.

3 ChatGPT: An Overview of Conversational AI

ChatGPT, developed by OpenAI, is a prominent example of a generative AI model designed for conversational interactions. It employs the GPT (Generative Pre-trained Transformer) architecture, a transformer-based neural network that excels in understanding and generating human-like text. Trained on diverse datasets, ChatGPT captures the relationships between words using self-attention mechanisms, allowing it to produce coherent and contextually relevant responses to user prompts. This conversational nature of ChatGPT paves the way for engaging and interactive AI-driven conversations, making it a valuable tool in AI art and various other domains.

3.1 Architecture

At the core of ChatGPT lies the Transformer architecture, a type of neural network that has revolutionized natural language processing tasks. Transformers leverage the power of self-attention mechanisms to capture relationships between words in a given sequence. Unlike traditional recurrent neural networks (RNNs) that process sequences sequentially, transformers can process words in parallel, resulting in more efficient and contextually aware language modeling.

3.2 Pre-training and Fine-tuning

ChatGPT undergoes a two-step training process: pre-training and fine-tuning. During pre-training, the model is exposed to a massive corpus of publicly available text data from the Internet. It learns to predict the next word in a sentence, gaining an understanding of grammar, context, and a wide range of linguistic patterns. This pre-training phase helps ChatGPT develop a solid foundation for language understanding.

After pre-training, the model goes through fine-tuning, which involves training it on more specific, curated datasets. Fine-tuning allows ChatGPT to adapt to specific tasks or domains and be more responsive to prompts from users.

3.3 Large-Scale Training Data

The effectiveness of ChatGPT is also attributed to the large-scale training data it is exposed to. The model is trained on a diverse range of text sources, including books, articles, websites, and other publicly available text from the Internet. This extensive training data helps ChatGPT capture a broad spectrum of linguistic patterns, contextual information, and world knowledge.

3.4 Attention Mechanisms

Attention mechanisms play a pivotal role in the Transformer architecture, enabling ChatGPT to attend to and weigh different parts of the input sequence when generating responses. Self-attention allows the model to assign importance to relevant words and consider their dependencies, capturing long-range relationships within the text. This attention mechanism contributes to the model's ability to generate coherent and contextually appropriate responses.

3.5 Iterative Generation

During inference, ChatGPT employs a technique called iterative generation to generate responses. Instead of generating the entire response in one pass, the model breaks it down into chunks and iteratively refines each part. This approach allows the model to create more coherent and contextually appropriate responses, providing a more engaging conversational experience.

4 The Creative Potential of Generative AI in Art

Generative AI has revolutionized the world of art, offering exciting possibilities for creative expression. AI art, also known as computational creativity, encompasses various forms of art generated or influenced by AI algorithms. ChatGPT and other generative models have opened up new avenues for artists to explore and experiment with innovative artistic techniques and styles.

4.1 Visual Art

Generative AI models can generate visually stunning artwork, create novel styles, and blend different artistic influences. Techniques like style transfer, where the style of one image is applied to another, enable the creation of unique and captivating visuals. AI models can generate original artworks based on text prompts or collaborate with human artists, providing inspiration and expanding the boundaries of traditional artistic techniques.

Input: An astronaut riding a horse in photorealistic style.

4.1.1 Deep Learning and Neural Networks

Deep learning and neural networks form the foundation of AI art. Neural networks, inspired by the structure and function of the human brain, learn patterns and relationships from vast datasets. Deep learning algorithms, composed of multiple layers of artificial neurons, analyze and process this data to generate new artistic content.

In the context of AI art, neural networks can be trained on large collections of existing artwork, enabling them to learn the unique styles, structures, and aesthetics of renowned artists. This knowledge is then utilized to generate new, AI-created artworks that mimic or blend different artistic styles.

4.1.2 Generative Adversarial Networks (GANs)

GANs have emerged as a powerful technology in AI art. A GAN consists of two neural networks: a generator and a discriminator. The generator generates synthetic content, such as images, while the discriminator tries to distinguish between real and generated content.

In the context of AI art, GANs can generate visually stunning and unique artworks. The generator learns to create new images or modify existing ones, while the discriminator helps ensure the generated content is of high quality and resembles the desired artistic style. GANs have been used to create AI-generated paintings, digital art, and even interactive installations, pushing the boundaries of traditional artistic techniques.

4.1.3 Style Transfer and Neural Style Transfer

Style transfer is a technique that enables the blending of artistic styles from one image onto another. Neural style transfer, a variation of style transfer, leverages deep neural networks to separate the content and style of an image. By extracting the content from one image and applying the style of another, AI models can generate visually captivating artworks that combine different artistic influences. This technology allows artists to experiment with new styles, blend aesthetics, and create visually striking compositions that fuse elements from various artistic sources.

4.1.4 Reinforcement Learning

Reinforcement learning, a type of machine learning, has also found applications in AI art. Reinforcement learning involves training an agent to interact with an environment and learn from feedback to maximize rewards. In the context of AI art, reinforcement learning can be used to iteratively refine and improve AI-generated artworks based on human feedback or predefined metrics. This iterative process allows the AI model to learn and adapt its artistic decisions, leading to the generation of aesthetically pleasing and innovative artworks.

4.2 Music

Generative AI has transformed music composition by enabling the creation of original pieces in diverse genres and styles. AI models can learn from vast musical libraries, capture patterns, and generate melodies, harmonies, and even entire compositions. This technology empowers musicians to experiment with new ideas, find inspiration, and collaborate with AI systems in the creative process.

4.3 Literature

AI algorithms have demonstrated remarkable abilities in generating text, including prose, poetry, and storytelling. Generative AI models like ChatGPT can be prompted with themes, characters, or styles to generate coherent narratives or contribute to collaborative storytelling projects. This opens up possibilities for engaging AI-assisted creative writing and exploring new literary frontiers.

5 Challenges and Ethical Considerations

While generative AI and AI art present immense opportunities, they also pose significant challenges and ethical considerations. Some key aspects to consider include:

1. Bias and Fairness: Generative models are trained on large datasets that may inadvertently capture biases present in society. It is crucial to address and mitigate biases to ensure fair representation in AI-generated content.

2. Authenticity and Attribution: Questions surrounding the authorship, ownership, and copyright of AI-generated art arise. Determining the contribution of AI systems and human artists in collaborative projects requires careful consideration and clear attribution guidelines.

3. Responsible Use: The potential for misuse of generative AI in creating misleading content or deepfake videos raises concerns. Implementing ethical guidelines and responsible use policies is essential to mitigate these risks.

6 Conclusion

Generative AI, exemplified by models like ChatGPT, has revolutionized the creative landscape by enabling interactive and collaborative AI-driven experiences. With its ability to generate text, visuals, music, and more, generative AI fuels innovation and expands the boundaries of artistic expression. While ethical considerations and challenges exist, responsible adoption of generative AI empowers artists, musicians, writers, and creators to unlock new levels of creativity and forge exciting collaborations between human imagination and AI-powered capabilities. As this technology continues to evolve, it promises to redefine the future of art, creativity, and human-AI interactions.

Note: This article was written by ChatGPT!

7 References

Gordon, Rachel. “AI System Makes Models like Dall-e 2 More Creative.” MIT News Massachusetts Institute of Technology, September 8, 2022. https://news.mit.edu/2022/ai-system-makes-models-like-dall-e-2-more-creative-0908.

https://chat.openai.com/

https://openai.com/dall-e-2/

OpenAI. “Introducing Chatgpt.” Introducing ChatGPT, November 30, 2022. https://openai.com/blog/chatgpt.

Newton's Sums

David Jiang

1 Introduction

Newton's sums, or Newton's identities, is a useful and efficient way to find the power sums of the roots of a polynomial without directly solving for the roots. Newton's sums is derived from Vieta's formulas, which relates the coefficients of the polynomial to its roots. For more information on Vieta's formulas, see this Brilliant article[1].

In this article, the following definitions will be used for simplicity. Do note that this notation is not universal, and if referenced in a formal proof, it should be followed with an explicit definition.

2 Notation

Let $f(x)$ be a polynomial, with $f \in \mathbb{Q}[x]$, of degree n and roots $r_1, r_2, \cdots, r_n$. Then,

Definition (Power Sum)**.** The k^{th} power sum P_k, where $k \in \mathbb{N}$, of the roots of $f(x)$ is the sum of each of the roots raised to the k^{th} power. In other words,

$$P_k = \sum_{i=0}^{n} a_i^k.$$

Definition (Elementary Symmetric Polynomial)**.** The j^{th} elementary symmetric polynomial S_j, with $j \in \mathbb{N}$ and $j \leq n$, of the roots of $f(x)$ is defined as

$$S_j = \sum_{1 \leq i_1 < i_2 < \cdots < i_j \leq n} r_{i_1} r_{i_2} \cdots r_{i_j}.$$

Experienced problem solvers will notice that these polynomials also show up in Vieta's formulas.

3 Motivation

Problem 1. *The polynomial $f(x) = x^2 - 3x + 2$ has roots r and s. What is P_n, for $n \in \mathbb{N}$?*

$n = 1$ is a straightforward application of Vieta's formulas. The sum of the roots is $r + s = 3$, so $P_1 = S_1 = 3$.

When $n = 2$, we can rewrite $r^2 + s^2$ in terms of S_1 and S_2. Note that

$$r^2 + s^2 = (r + s)^2 - 2rs,$$

[1] https://brilliant.org/wiki/vietas-formula/#:~~:text=x2%2Bbx%2Bc%E2%89%A1x2%E2%88%92(,q)x%2Bpq.&text=This%20is%20the%20so%Dcalled,to%20polynomials%20of%20higher%20degree.&text=x%202%20%E2%88%92%20(%20p%20%2B%20q,x%20%2B%20p%20q%20%3D%200.

and by Vieta's, this leads to $r^2+s^2=5$. Note that the roots of $f(x)$ are 1 and 2, and 1^2+2^2 indeed is 5.

Less obvious is how we express r^3+s^3 in terms of the symmetric sums. To get the cubic terms, we cube $r+s$, so

$$(r+s)^3 = r^3+3r^2s+3rs^2+s^3,$$

and the right side simplifies as $r^3+s^3+3rs(r+s)$. Thus $r^3+s^3=(r+s)^3-3rs(r+s)$, which equals 9.

As we continue raising S_1 to the nth power, there doesn't seem to be an easily recognizable pattern. For example,

$$\begin{aligned} r^4+s^4 &= (r+s)^4-(4r^3s+6r^2s^2+4rs^3) \\ &= (r+s)^4-2rs(r^2+s^2+3rs) \\ &= (r+s)^4-2rs\left((r+s)^2+rs\right) \\ &= (r+s)^4-2(r+s)^2rs-2(rs)^2 \end{aligned}$$

and by a similar method,

$$r^5+s^5=(r+s)^5-5(r+s)^3rs-5(r+s)(rs)^2.$$

In particular, the coefficients follow no nice formula, and as n gets larger, the expansion of P_n becomes longer.

However, notice that in the expansion of P_4, we solved r^2+s^2 in terms of S_1 and S_2, which we have already done previously. Instead of expressing P_n solely in terms of S_1 and S_2, we can express it recursively, in terms of both S and P.

In $n=3$, consider other methods to get the cubic terms. Rather than cubing $r+s$, we can take $(r^2+s^2)(r+s)$. Then,

$$(r^2+s^2)(r+s)=r^3+s^3+rs(r+s),$$

so $r^3+s^3=(r^2+s^2)(r+s)-(r+s)rs$. By definition, this is also equal to $P_2S_1-P_1S_2$. Similarly, when $n=4$, we take

$$(r^3+s^3)(r+s)=r^4+s^4+rs(r^2+s^2),$$

or $P_4=P_3S_1-P_2S_2$. Continuing, we can solve for $P_5=P_4S_1-P_3S_2$ and $P_6=P_5S_1-P_4S_2$. In general, it seems that $P_k=P_{k-1}S_1-P_{k-2}S_2$ for non-negative integers k.

4 Generalization

We can extend this to higher degree polynomials. When we find the power sums of a cubic polynomial with roots r, s, t, a similar pattern occurs. We get

$$\begin{aligned} r+s+t &= S_1 \\ r^2+s^2+t^2 &= P_1S_1-2S_2 \\ r^3+s^3+t^3 &= P_2S_1-P_1S_2+3S_3 \\ r^4+s^4+t^4 &= P_3S_1-P_2S_2+P_1S_3 \end{aligned}$$

and so on. After applying a similar method to 4th, 5th, and even 6th degree polynomials, we arrive at a recursive generalization of the power sums known as Newton's Sums.

Theorem (Newton's Sums)**.** *For a polynomial $f(x) = a_nx^n + a_{n-1}x^{n-1} + \ldots + a_1x + a_0$ with roots $r_1, r_2, \cdots, r_n$,*

$$P_k = P_{k-1}S_1 - P_{k-2}S_2 + P_{k-3}S_3 - \ldots + (-1)^{n+1}P_{k-n}S_n$$

when $k > n$ and

$$P_k = P_{k-1}S_1 - P_{k-2}S_2 + P_{k-3}S_3 - \ldots + (-1)^{k+1}kS_k$$

when $k \leq n$, where $k \in \mathbb{N}$.

Proof. Consider the case when $k > n$. Take each root r_i, where i is an integer between 1 and n. By definition, $f(r_i) = 0$, and dividing by a_n in the polynomial expansion yields

$$r_i^n - S_1r_i^{n-1} + S_2r_i^{n-2} - \ldots + (-1)^nS_n = 0.$$

Notice that this is a direct application of Vieta's formulas. Now, we multiply each equation by r_i^{k-n}, giving

$$r_i^k - S_1r_i^{k-1} + S_2r_i^{k-2} - \ldots + (-1)^nS_nr_i^{k-n} = 0.$$

Summing each equation given by each of the roots and simplifying, we get

$$P_k = S_1P_{k-1} - S_2P_{k-2} + \ldots + (-1)^{n+1}S_nP_{k-n}.$$

Next, we consider when $k \leq n$. Note that

$$f(x) = a_nx^n + a_{n-1}x^{n-1} + \ldots + a_0 = a_n(x - r_1)(x - r_2)\ldots(x - r_n),$$

and dividing by a_n yields

$$x^n - S_1x^{n-1} + S_2x^{n-2} + \ldots + (-1)^nS_n = (x - r_1)(x - r_2)\ldots(x - r_n).$$

Taking the derivative of both sides with respect to x,

$$nx^{n-1} - (n-1)S_1x^{n-2} + (n-2)S_2x^{n-3} + \ldots + (-1)^{n+1}S_{n-1} = \sum_{i=1}^{n} \frac{\frac{f(x)}{a_n}}{x - r_i}.$$

Recall that $\frac{f(x)}{a_n} = x^n - S_1x^{n-1} + S_2x^{n-2} - \ldots + (-1)^nS_n$. To simplify the right hand side of the equation, we use the following lemma.

Lemma 1. *If r is a root of $f(x)$ and if $\frac{f(x)}{a_n} = (x - r)P(x)$, then the coefficient of the x^m term of $P(x)$ is*

$$(-1)^{n-m-1}S_{n-m-1} + (-1)^{n-m-2}S_{n-m-2}r + (-1)^{n-m-3}S_{n-m-3}r^2 \ldots + r^{n-m-1}.$$

for non-negative integers $m < n$.

The lemma can be proven using induction and synthetic division, and it is left as an exercise to the reader. Using lemma 1 and summing over all roots r_i, we get the polynomial

$$nx^{n-1} + (P_1 - nS_1)x^{n-2} + (P_2 - P_1S_1 + nS_2)x^{n-3} + \ldots + (P_n - P_{n-1}S_1 + \ldots + (-1)^{n-1}nS_n).$$

Then, matching coefficients, we get

$$\begin{aligned}
-(n-1)S_1 &= P_1 - nS_1 \\
(n-2)S_2 &= P_2 - P_1S_1 + nS_2 \\
-(n-3)S_3 &= P_3 - P_2S_1 + P_1S_2 - nS_3 \\
&\ddots \\
(-1)^{n-1}S_{n-1} &= P_n - P_{n-1}S_1 + P + n - 2S_2 - \ldots + (-1)^{n-1}nS_n
\end{aligned}$$

After solving for the highest P_k, we get the desired formulas and we are done. □

5 Problems

Let's walk through a basic example.

Problem 2 (2003 HMMT February Algebra #7)**.** *Let a, b, c be the three roots of $p(x) = x^3 + x^2 - 333x - 1001$. Find $a^3 + b^3 + c^3$.*

Solution. This is a direct application of Newton's sums. By the coefficients of $p(x)$, we know that $S_1 = -1$, $S_2 = -333$, and $S_3 = 1001$. Clearly, $P_1 = S_1 = -1$, and $P_2 = P_1S_1 - 2S_2$, which is equal to $(-1)^2 - (2)(-333) = 667$. Thus, we can use the following equation

$$P_3 = P_2S_1 - P_1S_2 + 3P_3,$$

which is $(667)(-1) - (-1)(-333) + (3)(1001) = \boxed{2003.}$ □

For more practice, here is a handful of applications of Newton's sums, not necessarily in increasing order of difficulty.

5.1 Practice Problems

Problem 3 (2008 HMMT February Algebra #1)**.** *Positive real numbers x, y satisfy the equations $x^2 + y^2 = 1$ and $x^4 + y^4 = \frac{17}{18}$. Find xy.*

Problem 4 (2019 AMC 12A #17)**.** *Let s_k denote the sum of the kth powers of the roots of the polynomial $x^3 - 5x^2 + 8x - 13$. In particular, $s_0 = 3$, $s_1 = 5$, and $s_2 = 9$. Let a, b, and c be real numbers such that $s_{k+1} = a\,s_k + b\,s_{k-1} + c\,s_{k-2}$ for $k = 2, 3, \ldots.$ What is $a + b + c$?*

Problem 5 (2010 CHMMC Individual #5)**.** *Let $x = \frac{3-\sqrt{5}}{2}$. Compute the exact value of $x^8 + \frac{1}{x^8}$.*

Problem 6 (2019 Purple Comet High School #28)**.** *There are positive integers m and n such that $m^2 - n = 32$ and $\sqrt[5]{m + \sqrt{n}} + \sqrt[5]{m - \sqrt{n}}$ is a real root of the polynomial $x^5 - 10x^3 + 20x - 40$. Find $m + n$.*

Problem 7 (1985 IMO Longlist #88)**.** *Determine the range of $w(w + x)(w + y)(w + z)$, where x, y, z, and w are real numbers such that*

$$x + y + z + w = x^7 + y^7 + z^7 + w^7 = 0.$$

Analytics in a New Sports Age

Tejas Siddaramaiah

1 Introduction

As basketball has reached new heights in popularity around the world, so has the desire to win and the need to have an edge on other teams have also increased. Sports used to just be about averages and had never anything more complicated beyond that. However, now advanced analytics have begun taking a stronger influence in how basketball is played. In this article, I will talk about some of the analytical models used in the National Basketball Association to evaluate individual players in their each team's game strategies.

2 Basketball

2.1 Basic Statistics

The most commonly quoted statistics quoted by casual fans are points per game, assists per game, and rebounds per game, which are all averages. Accuracy is measured by field goal percentage and free throw percentage, which are both percentages of makes over attempts. These statistics can be misleading as they don't tell the whole story of the value of a player to their team.

2.2 Player Efficiency Rating

Player Efficiency Rating, better known as PER, is a statistic created by John Hollinger to measure a player's per minute productivity. The unadjusted value is calculated in the following formula. Most of the

```
uPER = (1 / MP) *
     [ 3P
     + (2/3) * AST
     + (2 - factor * (team_AST / team_FG)) * FG
     + (FT *0.5 * (1 + (1 - (team_AST / team_FG)) + (2/3) * (team_AST / team_FG)))
     - VOP * TOV
     - VOP * DRB% * (FGA - FG)
     - VOP * 0.44 * (0.44 + (0.56 * DRB%)) * (FTA - FT)
     + VOP * (1 - DRB%) * (TRB - ORB)
     + VOP * DRB% * ORB
     + VOP * STL
     + VOP * DRB% * BLK
     - PF * ((lg_FT / lg_PF) - 0.44 * (lg_FTA / lg_PF) * VOP) ]
```

Figure 1: PER formula

```
factor = (2 / 3) - (0.5 * (lg_AST / lg_FG)) / (2 * (lg_FG / lg_FT))
VOP    = lg_PTS / (lg_FGA - lg_ORB + lg_TOV + 0.44 * lg_FTA)
DRB%   = (lg_TRB - lg_ORB) / lg_TRB
```

Figure 2: Meanings of terms

terms are contractions for basic statistics used in the league, like assists and minutes played. Some more definitions (by order of appearance):

- team = player's team
- VOP = value of possession
- DRB% = defensive rebound percentage
- = lg = league

The 1/MP (1/Minutes Played) is used to scale the PER based on minutes played. The player's positive contributions, like three pointers and assists, are added while the negative contributions, like turnovers and personal fouls, are subtracted.

The unadjusted PER is then adjusted by multiplying a pace adjustment, which is league pace divided by team pace. Since different teams play at different paces, a player's production also changes from team to team, thus the need for a pace adjustment. This is to scale player's PER to show a player's contribution to the team rather than just a result of the team's system of playing.

The adjusted PER is finally multiplied by (15/league average PER) to set the league average PER to 15 for better comprehension by fans. A PER above 15 means better contributions than an average player while below 15 means worse.

2.3 Win Shares

Win shares are an estimation of the amount of wins a player contributed to their team's season. Win shares are a sum of offensive win shares and defensive win shares.

Offensive win shares are calculated by marginal offense divided by marginal points per win and defensive win shares are calculated by marginal defense divided by marginal points per win, so win shares are calculated by

$$\frac{\text{marginal offense} + \text{marginal defense}}{\text{marginal points per win}}.$$

Marginal offense is calculated by the formula

$$\text{points produced} - (0.92)(\text{league points per possession})(\text{offensive possessions}).$$

This is essentially the difference in points produced by the player and by an average player. Marginal defense is calculated by the formula:

$$(1.08)\left(\frac{\text{player minutes}}{\text{team minutes}}\right)(\text{team defensive possessions})\left(\text{league points per possession} - \frac{\text{defensive rating}}{100}\right).$$

This is essentially the difference in points allowed by an average player and the player. Marginal points per win is calculated by the formula

$$(0.32)(\text{league points per game})\left(\frac{\text{team pace}}{\text{league pace}}\right).$$

This is essentially the factor to standardize win shares for each team.

The sum of the win shares of all the members of a team should sum to the number of wins the team had. There is an average margin of error of 2.74 wins.

2.4 Conclusion

Sports, like basketball are becoming more ingrained with math and statistics. The two examples that I presented are a few of the many different analytics that a team uses to evaluate its players. If a player displays an above average PER and positive win shares, they are deemed conducive to winning basketball. These statistics can also be used to compare with other players from other teams to influence transactions in the league. Since the NBA is a multi-billion dollar organization, these transactions can be worth millions of dollars to teams. This real-world application of statistics and math show us its value.

References

"Calculating PER." Basketball Reference, Sports Reference, https://www.basketball-reference.com/about/per.html.

"NBA Win Shares." Basketball Reference, Sports Reference, https://www.basketball-reference.com/about/ws.html.

Stuyvesant Alumni Perspectives: Milan Haiman, Class of 2019

Anastasia Lee

In his senior year as a math major at MIT, Milan Haiman (Class of 2019) shares his memories of Stuy, his love for combinatorics, and his varied math experiences, including winning a Gold medal (8th place) for Hungary in the 2019 International Mathematics Olympiad (IMO) in the United Kingdom.

1 The biggest factor in Milan's success in competing in and teaching math? Going to Stuy

Although Milan did not participate in math competitions before high school, he attended New York Math Circle from fifth grade through middle school. One of his NYMC teachers was Mr. Kats, who would eventually also teach one of Milan's calculus and two of his Math Team classes and become one of his favorite teachers. During middle school, Milan also attended MathPath and in high school, he participated in MathILy, Canada/USA Mathcamp, Math-M-Addicts, and MOP (Mathematics Olympiad Program), but he was most impacted by his four years at Stuy. He particularly enjoyed training and competing with the Math Team, where he served as a captain.

"I made lots of friends on the Math Team," he emphasized. "If I hadn't done that, then I don't think I would have liked math as much and done as much math in high school and probably would not have been as interested in college, etc." Although Milan's specific interests in math have changed since high school, he remarked, “You can keep changing how you think about [math], but as long as there’s people whom you like working with, I think that’s probably the most important thing.” Milan especially enjoyed all the New York City Math Team trips to competitions such as ARML (American Regional Mathematics League) and NYSML (New York State Mathematics League), and as an alumnus, he reunited with the teams and coaches when they traveled to Cambridge, Massachusetts to compete in HMMT (Harvard-MIT Mathematics Tournament).

As a senior Math Team captain at Stuy, Milan also gained valuable teaching experience, leading Math Team classes every morning and organizing practices for an entire year. “I got to choose whatever I wanted to teach,” he recalled fondly. Later, he would apply this experience in his college and summer employment. “Overall, I think I really like teaching,” he shared. “I definitely want to continue teaching in grad school.”

2 Before, during, and after the IMO

When asked for a fun fact about himself, Milan replied that he is half Indian and half Hungarian. But the whole Milan represented Hungary in the 2019 International Mathematical Olympiad, which was held in the United Kingdom in the city of Bath. He described some of his training before the global competition, saying, “The summer before my senior year, I went to some camps in Hungary.” Then in the winter, he attended a joint Hungary-UK math camp for 20 British students and 20 Hungarian students, which included interesting lectures and several practice Olympiad exams.

Milan then traveled for 10 hours from New York to Bucharest to represent Hungary at RMM (Romanian Master of Mathematics). He thought it was interesting that the travel time for the three other students who represented Hungary was longer than his. “[Bucharest] is in the South of Romania and Budapest is in the north of Hungary, so it’s around 14 hours on the train,” he explained. His adventure included sleeping in the same room as his three teammates on very squeaky beds and searching for vegetarian food options for himself. As for the contest itself, Milan recounted an interesting turn of events on Day 1 of the two-day competition. “Because of an unexpected incident, the Day 2 problems had to be changed,” Milan described. “So the problems on the second day were not as good. I later heard about what was supposed to be on that day, and it seemed much more interesting.”

After one additional training camp in Hungary, Milan headed for Bath, which is about a two-hour drive from London, to compete in the IMO. He cherished meeting students from all over the world and appreciated that it was well-organized, including the availability of “really nice fruits.” He also had the opportunity to visit famous historical monuments, recollecting, “After the contest days of the IMO, the students get to go on excursions while the papers are being graded. So, I got to see Stonehenge, which was pretty cool! It was quite big and I have no idea how people made it so long ago.” During another trip, Milan and his fellow competitors enjoyed “see[ing] the actual baths” in Bath.

Milan ultimately earned a Gold medal, scoring 40 out of 42 points, ranking an impressive 8th place in the world. When asked at what point he realized that he had the potential to perform so well on an international level, he replied, “I think probably after the IMO was over.” He confided, “I also think I was quite lucky with the problems on the IMO. My best subject was combinatorics and then after that was geometry, but mostly because I knew how to bash the geometry problems, not solve them how you are supposed to do. The way the IMO works is there's three problems on each day, so six total, and then each day [the problems] increase in difficulty. And there's a mix of subjects: algebra, number theory, combinatorics, and geometry. But there's obviously going to be some subjects for the harder problems and some for the easier problems. So the first problems [on each day], #1 and #4, which are the two easiest, were algebra and number theory and #2 and #3 were geometry and combinatorics, and then #5 and #6 were combinatorics and geometry. So I still think I was pretty lucky, and if the tests were different, then I probably wouldn't have done as well."

Milan continued his involvement with Olympiad math after high school. The summer after junior year of college, he worked as a counselor at MOP, teaching mostly on subjects in combinatorics but also on topics in geometry, like inversion. But he reflected, "Maybe the thing I'm happiest about is having a problem on the IMO." Below is Problem 3 of the 2020 IMO which Milan wrote (except he originally framed the problem using coins rather than pebbles):

There are $4n$ *pebbles of weights* $1, 2, 3, \ldots, 4n$*. Each pebble is colored in one of* n *colors and there are four pebbles of each color. Show that we can arrange the pebbles into two piles so that the following two conditions are both satisfied:*

- The total weights of both piles are the same.
- Each pile contains two pebbles of each color.

"There are all the students who go to the IMO, and the IMO problems are preserved forever, so lots of people will look at them," he explained. "So having a nice problem that I liked and having that shared with everyone, I think that's pretty cool."

3 Living, learning, researching, and teaching at MIT

When asked about what makes an MIT education unique, Milan asserts, "The main thing that makes MIT different from other colleges is how the dorms work. At some colleges, I think the culture is that you don't really spend much time in your dorm, and you hang out in other places, but at MIT, each dorm has different [groups of] people, so you have potentially similar kinds of people in the same dorm. You more or less pick which one you want to be in, so there's some self-selection. The dorm I'm in is called Next House; I've made a lot of friends here. Also, the dorms aren't divided [among] class years. When I was a freshman, the person across the hall from me was a junior, and the person next to me was a senior, so you're just mixed in with all the different years, and you can ask for lots of advice from them. I think most colleges will have a freshman dorm. There's probably advantages to both [systems], but I think definitely one of the reasons I wanted to go to MIT was how the dorms work."

In addition to completing his undergraduate coursework, Milan is specifically interested in researching "extremal combinatorics." He has already begun pursuing this passion, most notably through Summer Program in Undergraduate Research (SPUR) at MIT. His research, with his mentor Yannick Yao, entitled "Irreducibility of Generalized Permutahedra, Supermodular Functions, and Balanced Multisets" won the Hartley Rogers Jr. Prize in 2022. According to the Roger's Prize website, Milan's paper "makes a serious contribution to the study of irreducible generalized permutohedra, establishing a double exponential upper bound for the number of such objects in dimension n." From the Duluth Research Experiences for Undergraduates (REU) 2021 summer program, Milan's research was entitled "The Dimension of Divisibility Orders and Multiset Posets."

In his earliest undergraduate research from 2020 to 2021, Milan was supervised by Professor Yufei Zhao while researching "Graphs with high second eigenvalue multiplicity."

Milan also found several opportunities to teach math during his college years. He volunteered to lecture on Catalan numbers at MIT Splash during his freshman and senior years of college, up to 80 students at one time. Milan also worked as a counselor the summer after freshman year college at Canada/USA Mathcamp and as an instructor (with another MIT student) for the Ghana IMO camp in January of his sophomore year of college, both of which were limited to online participation during the pandemic. Being employed as a teaching assistant for a large class called "Probability and Random Variables" during senior year of college allowed Milan to teach weekly recitations and hold office hours. "As a grad student, I'm sure there will be lots of opportunities to do [more] teaching," he predicted.

4 A few tips for current Stuy students

- Use your time wisely.

  "I know a lot of Stuy students will lose sleep for small improvements on their grades. I definitely don't think that's worth it. Getting sleep is a good use of time!"

- Think about what you're doing.

  "Don't just do it because that's what you were doing before and that's what everyone else is doing. You should enjoy what you're doing."

- When writing a math problem, make it something you want to think about just for the sake of thinking about it.

  "So I probably have stronger opinions than the average person about what makes a good problem. I think it should have a nice statement, and it should have a nice solution. The solution should really be about coming up with an idea. And then once you've come up with that idea, you should be able to solve the problem easily. It shouldn't be about looking at the problem, having a general sense of what's going to happen, but then having to do a bunch of calculations to figure out the solution. When I read the problem, I want it to be something I would actually want to think about."

- If you're in Mr. Chew's senior Math Team class, understand isosceles tetrahedrons.

  "He'll want you to learn a trick to deal with 3D geometry."

Stuyvesant Math and Computer Science Achievements

2022-2023

2022 AP Computer Science A Exam
Skai Nzeuton and Chun Yeung "Frank" Wong were two of only 369 in the world to earn a perfect score.

2022 Congressional App Challenge
Randy Sim was a winner of this contest, designed to support STEM and computer coding education for middle and high school students within the 9th Congressional District.

American Regions Mathematics League (ARML) Competition
NYC Math Team's "Tin Man" team finished in 5th place overall.

Harvard-MIT Mathematics Tournament (HMMT)
NYC Math Team's "Tin Man" team placed 7th in the Team Round. Daniel Potievsky finished in 5th place on the Combinatorics Round.

Princeton University Mathematics Competition (PUMaC)
NYC Math Team's "Tin Man" and "Scarecrow" teams finished in 4th place and 10th place overall.

New York State Mathematics League (NYSML)
NYC Math Team's teams, "Tin Man," "Scarecrow," "Lollipop Guild," and "Lion" won 1st place, 2nd place, 3rd place, and 5th place overall.

Philadelphia Classic (PClassic)
Nicholas Tarsis, Yuhao "Ben" Pan, Chun Yeung "Frank" Wong, and Chongtian "Mark" Ma placed 2nd in the advanced division.

High School Contest in Mathematical Modeling (HiMCM)
Rebecca Bao, Theodore Landa, Eli White, and Eric Zheng's outstanding performance earned an invitation to participate in the International Mathematical Modeling Challenge.

NYC Terra STEM Fair
Daniel Liu finished 2nd place in the Software and Robotics Division of the NYC Terra STEM Fair for his research project entitled, "Comparison of Convolutional Neural Network Architectures in Training A Machine Learning Classifier for Urban Home Sounds."

Student Forecasting Tournament
Stuyvesant teams "Wille the Pooh" (Christopher Louie, Christopher Shen, Ryan Radwan and Andy Huang) and "Limit DNE" (Rebecca Bao, Bingde Jiang, Edwin Lin, and Gloria Yip)

finished 1st and 2nd place.

USA Junior Mathematical Olympiad (USAJMO)/USA Mathematical Olympiad (USAMO)
John Gupta-She won a USAMO Silver Medal, Josiah Moltz and Joseph Othman won USAMO Bronze Medals, and Paul Gutkovich received a USAMO Honorable Mention. Ian Buchanan was named a USAJMO Winner.

USAMO Qualifiers
John Gupta-She (12th)
Paul Gutkovich (12th)
Josiah Moltz (12th)
Joseph Othman (12th)

USAJMO Qualifiers
Ian Buchanan (10th) Winner

American Mathematics Competition (AMC) 12A
John Gupta-She and Josiah Moltz were School Winners with scores of 126.

American Mathematics Competition (AMC) 12B
Joseph Othman and Paul Gutkovich were School Winners with scores of 136.5.

American Mathematics Competition (AMC) 10A
Steven Lou was School Winner with a score of 136.5.

American Mathematics Competition (AMC) 10B
Steven Lou and Leonna Wang were School Winners with scores of 115.5.

American Mathematics Competition (AMC) 12A Distinguished Honor Roll
John Gupta-She, Paul Gutkovich, Julia Kozak, Andrew Li, Edwin Lin, Mikayla Lin, Haokai Ma, Josiah Moltz, William Opich, Joseph Othman, Aditya Pahuja, Jacob Paltrowitz, Daniel Potievsky, Nikita Safronov, Stanley Tung, Chun Yeung Wong, Kathleen Zhang, Elie Zheng

American Mathematics Competition (AMC) 12B Distinguished Honor Roll
Lingjie Fang, Paul Gutkovich, John Gupta-She, Joseph Jeon, Julia Kozak, Jacob Kirmayer, Andrew Li, Haokai Ma, Josiah Moltz, Joseph Othman, Jacob Paltrowitz, Noam Pasman, Daniel Potievsky, Ruiwen Tang, Nicholas Tarsis, Stanley Tung, Chun Yeung Wong, Kathleen Zhang, Qinwen Zheng

American Mathematics Competition (AMC) 10A Distinguished Honor Roll
Ian Buchanan, Andy Jiang, Sophia Jin, Audrey Jing, Shuichi Kinoshita, Steven Lou, Chongtian Ma, Victoria Reguyal, Leonna Wang, Enmei Yang, Connor Yau, Jackie Zeng, Brandon Zhang, Ashley Zhu

American Mathematics Competition (AMC) 10B Distinguished Honor Roll
Krish Bhandari, Ian Buchanan, Andy Jiang, Sophia Jin, Audrey Jing, Shuichi Kinoshita, Steven

Lou, Chongtian Ma, Leonna Wang, Connor Yau, Mina Yeh, Jackie Zeng, Brandon Zhang

90 American Invitational Mathematics Examination (AIME) Qualifiers
Krish Bhandari, Ian Buchanan, Patrick Cao, Finn Charest, Vincent Chen, Joshua Choi, Richard Delgado, Maya Doron-Repa, Konstantin Dubovskiy, Lee Eisenberg, Lingjie Fang, Rin Fukuoka, Stuti Girdhar, John Gupta-She, Paul Gutkovich, Zhengxi He, Charles Hua, Stanley Huang, Aaron Hui, Joseph Jeon, Henry Ji, Andy Jiang, Bingde Jiang, David Jiang, Sophia Jin, Audrey Jing, Alyssa Kang, Matthew Karp, Shuichi Kinoshita, Jacob Kirmayer, Julia Kozak, Georgy Krasnov, Keerthan Kumarappan, Anastasia Lee, Mary Lee, Michael Lee, Ryan Lee, Andrew Li, Aiden Lim, Curt Lin, Edwin Lin, Ethan Lin, Isaac Lin, Mikayla Lin, Ethan Liu, Katherine Liu, Steven Lou, Christopher Lu, Chongtian Ma, Dylan Ma, Haokai Ma, Aiden Mizhen, Josiah Moltz, William Opich, Joseph Othman, Aditya Pahuja, Jacob Paltrowitz, Yuhao Pan, Ryan Junhyuk Park, Noam Pasman, Ryan Peng, Daniel Potievsky, Medha Prasad, Daphne Qin, Victoria Reguyal, Nikita Safronov, Shadman Shaharia, Ethan Sharma, Charles Shen, Chenkai Shen, Charles Tang, Ruiwen Tang, William Tang, Nicholas Tarsis, Stanley Tung, Leonna Wang, Maya Wassercug, Chun Yeung Wong, Mengxuan Wu, Enmei Yang, Connor Yau, Cyrus Yau, Mina Yeh, Jackie Zeng, Brandon Zhang, Calvin Zhang, Kathleen Zhang, Raymond Zhang, Elie Zheng, Qinwen Zheng, Royce Zheng, Ashley Zhu

Mathematics and Computer Science Department Senior Awards
The Dr. Richard Rothenberg Memorial Award for Highest Honors in Mathematics: Joseph Othman

The Richard B. Geller Memorial Scholarship for Mathematics: Julia Kozak, Josiah Moltz

Henrietta O. Midonick Award: Kathleen Zhang

David J. Pichler Memorial Scholarship: Tasnim Chowdhury

The Okean Family Memorial Award for Scholarship in Pure or Applied Mathematics: John Gupta-She, Paul Gutkovich, Lin Wang

Irene Finkel Memorial Award for Excellence in Mathematics: Maya Doron-Repa, Jacob Paltrowitz

Highest Honors in Computer Science: Ruiwen Tang

High Honors in Computer Science: Vernon Hughes, Yuk Kwan Lau, Lauren Lee, Karen Shekyan, Chun Yeung Wong

Honors in Computer Science: Anthony Sun, Maya Doron-Repa, Samantha Hua, Brian Li, Vincent Lin, Josiah Moltz, Gabriel Thompson, William Vongphanith

Golden Ducky for Service to the Computer Science Community: Kate Alvarez, Brianna Tieu, Eduardo Lozano, Samantha Hua, Lauren Lee, Yuhao Pan, Andrey Sokolov, Ruiwen Tang, Nicholas Tarsis, Gabriel Thompson, William Vongphanith, Lin Wang, Chun Yeung Wong, Theodore Yochum